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Technical Report No. 73

Prefilled Syringe User Requirements for Biotechnology Applications

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Technical Report No. 73

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1.0 Introduction

Since the mid-2000's, a large number of subcutaneously injected biotechnology drug products have been packaged in prefilled syringes (PFSs) because of significant benefits compared to vials or ampules. PFSs reduce medical dosing errors and the risk of microbial contamination by decreasing the number of necessary manipulations prior to injection, and they improve compliance due to ease of use. In addition to the patient benefits, PFSs often have lower overfill regulatory requirements than vials, thereby reducing product waste.

1.1 Purpose/Scope

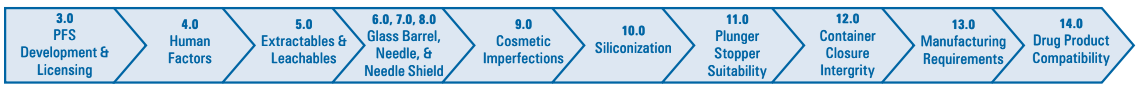
The purpose of this technical report is to present a discussion of requirements for the 1 mL long glass PFS for biotechnology applications (**Figure 1.1-1**). Topics covered are material selection and evaluation for suitability, syringe preparation and handling (including human factors), and drug product compatibility (physical and chemical) with the syringe materials and mode of delivery. Plastic syringes and ancillary devices, such as autoinjectors, are not within scope.

The terms "end user," "customer," and "user" are used interchangeably throughout this report. However, the definitions of these terms change according to the topic under discussion as the report progresses through the product life cycle of the PFS. For example, during development, the end user or target population may be the target patient, while the customer or user during production and delivery may be the supplier or manufacturer. Additionally, while this technical report provides more specific information related to the U.S. and European regions, requirements for different regulatory agency jurisdictions may vary, and applicants and manufacturers should be aware of the proper qualification and validation expectations for the intended market.

While the focus of this report is on 1 mL long glass PFSs for biotechnology (large molecule) drug products, the concepts presented may be applied to other products and applications (e.g., small molecule drug products, luer, luer-lock, and plastic syringes). Every attempt has been made to provide comprehensive content, but some particular techniques and technologies may not be included.

The task force that prepared this report consisted of members representing global companies and representatives from regulatory bodies to ensure a consensus on the methods, terminology, and practices described. Experts in PFSs and PFS manufacturing performed technical reviews.

The information in this technical report appears in the general order of operations for developing and manufacturing 1 mL long glass PFSs. Throughout the report, the chevron diagram (below) serves as a guide for the reader and reflects the focus of each chapter.



The glossary contains a list of definitions and acronyms used in this report. When available, same or similar definitions to those included in FDA or ICH guidelines are used.

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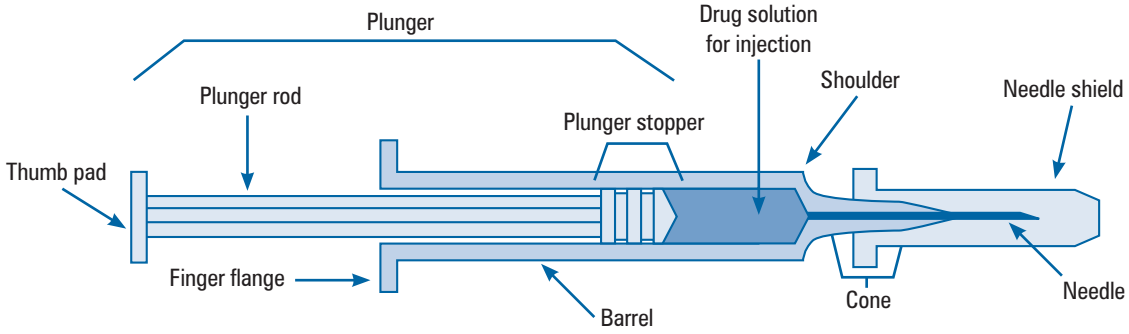

Figure 1.1-1 Schematic drawing of a PFS

A typical 1 mL long staked needle syringe consists of a Type I borosilicate glass barrel. For a staked needle syringe, a stainless-steel needle is affixed to the syringe cone with adhesive, and the needle is embedded in an elastomeric needle shield. The inside of the syringe is siliconized to allow the plunger to be easily depressed during administration of the injectable ***(1,2)***.

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2.0 Glossary of Terms

Backstop (Finger Plate or Thumb Plate)
Feature that enhances the area to hold the syringe and is usually designed to avoid accidental removal of the plunger from the back. By design, it may also serve as a flange extender to facilitate holding of the syringe during injection.

Break-loose Force
Energy required to initiate plunger movement within the syringe barrel upon injection.

Component, Primary
Element of the assembled prefilled syringe (needle, plunger stopper and tip closure, or adhesive) directly in contact with the drug.

Component, Secondary
Element of the assembled prefilled syringe (plunger rod, backstop, or safety system) that interacts with the primary components and provides functionality to the delivery system.

Contextual Inquiry
Ethnographic research method used to observe and analyze behaviors in actual end-use contexts (actual environments and use scenarios).

Elastomer
Thermoplastic material formulation (that may or may not contain rubber/natural latex) derived from elastic polymer; often used interchangeably with the term "rubber."

Extractable
Organic or inorganic chemical entity that is forced out of container closure system materials and components under laboratory experimental conditions *(3)*.

Extrusion Force (Propagation Force, Glide Force, Syringeability)
Filled syringe delivery force (that does not include break-loose force) quantified as the highest non-break-loose force to complete the injection stroke.

Formative Usability Evaluation
Observed actual or simulated use of early prototypes to help reliably identify product concept-specific, use-related hazards that may have been missed by other methods.

Glide Force
Force in Newtons (N) required for plunger movement within an empty syringe.

Hold-Up Volume (Residual Volume, Nonexpellable Volume, Dead Volume)
Amount of fluid remaining in the syringe when the plunger has reached the end of travel within the barrel.

Leachable
Organic or inorganic chemical entity that migrates from pharmaceutical container closure system components into a drug product formulation *(3)*.

Needle (Cannula)
A thin, hollow, metal tube commonly used with a syringe to inject substances into the body.

Needle-Safety Device
A safety feature or mechanism that effectively reduces the risk of an exposure incident (e.g., accidental needlestick) *(4)*.

Needle Shield (Needle Cover)
Rigid or flexible polymeric substance used to seal and protect the needle.

Nest (Tub Nest)
Rigid plastic support placed into a tub to keep syringes upright with sufficient separation to allow for easy manipulation by manual or automated fill-line systems.

Plunger
The combined components of the plunger rod and plunger stopper.

Plunger Rod
Portion of the plunger assembly which provides a thumb pad for depressing the plunger and is attached to the plunger stopper inside the syringe barrel.

Plunger Stopper (Piston, Stopper, Plunger, Gasket, Plug, Bung, Closure)
Elastomeric component, which acts as a seal to prevent product leakage and also as the portion of the syringe which forces the expulsion of the contents of the syringe when depressed.

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Plunger Stopper-Barrel Interface
Circumferential contact points between the ribs of the plunger stopper and the inner diameter of the syringe barrel.

Prefillable
Syringes and associated components considered as starting material for the filling and assembling process of a prefilled syringe.

Prefilled
Syringe assembly after being filled with pharmaceutical product and being closed.

Prefilled Syringe
A syringe that has been prefilled to contain a specific dose of medication.

Ready for Filling
Prepared for loading with the pharmaceutical product.

Rondo Tray
Plastic packaging with a horizontal design used to accommodate syringes during the manufacturing process.

Schlieren
Optical inhomogeneities in transparent material not necessarily visible to the human eye. In the context of this technical report, Schlieren (from German *Schliere*, meaning "streak") are due to localized variations in silicone oil thicknesses.

Staked Needle
Nonremovable, fixed needle attached to syringe body/cone with adhesive.

Steribag
Sealing bag made at least partly of steam-permeable material to wrap and seal containers for sterilization (e.g., tub-nest configurations).

Suitability (Machinability)
Physical interactions of components with equipment and environment.

Summative/Validation Testing
Evaluative testing providing evidence to show that representative users in the expected use conditions can perform all essential and critical tasks required for safe and effective use of the device.

Syringeability
The qualities required for delivery of the syringe contents at a given injection rate via a needle of a predetermined gauge and length that enables optimal injection conditions for the drug product, taking into account the product's physical properties, flow rate, injection force, and needle gauge.

Thumb Pad
Disk at end of plunger rod.

Tub
Plastic box that contains a vertically oriented syringe nest and is sealed with a lid.

Use-related Risk Analysis
A systematic assessment of all of the user steps involved in using the device with particular consideration of the potential use errors, their associated clinical consequences along with the risk-mitigation strategies, and the method of validating the risk-mitigation strategies.

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2.1 Abbreviations

AAS Atomic absorption spectroscopy

AET Analytical evaluation threshold

API Active pharmaceutical ingredient

AQL Acceptable quality level

BLA Biologic license application

C/C Container closure

CCI Container closure integrity

DMF Drug master file

EPDM Ethylenepropylenediene monomer

FMEA Failure mode and effects analysis

FTIR Fourier transform infrared spectroscopy

GC Gas chromatography

HFE Human factors engineering

HPLC High-performance liquid chromatography

ICM-MS Inductively coupled plasma-mass spectrometry

ICP-OES Inductively coupled plasma optical emission spectroscopy

ISE Injection site reaction

LPP Low-pressure plasma

MDD Medical Devices Directive

NDA New drug application

PFS Prefilled syringe

RABS Restricted access barrier system

RNS Rigid needle shield

TPV Thermoplastic vulcanizate

VHP Vaporized hydrogen peroxide

WFI Water for injection

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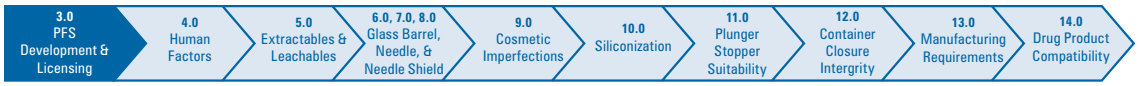
3.0 Prefilled Syringe Development and Licensing



This section discusses considerations for planning the development of the PFS and eventual marketing approval.

A developer of a PFS must define its purpose, evaluate the supplier, and adhere to regulatory and marketing requirements. For example, in the United States, a finished PFS is considered a combination product and regulated by the Center for Drug Evaluation and Research (CDER), the Center for Biologics Evaluation and Research (CBER), or the Center for Device and Radiological Health (CDRH) of the U.S. FDA, depending on its primary/principal mode of action (PMOA). Based on the same principle of PMOA, products in the European Union (EU) that incorporate or administer a drug may be regulated as either medical devices or medicinal products as, the term combination product per se does not exist yet in EU regulations. A container/closure (C/C) system, such as a PFS used in the packaging of a medicinal product for human use, is approved as part of the medicinal product application.

3.1 High-Level Overview of Key System Functions

In order to have a comprehensive understanding of the PFS up to the end of its shelf life, it is necessary to consider the ability of the PFS to provide an integral C/C system and to consider the interaction of the related subsystems. These considerations should support the goal of treating the patient, obtaining market approval for a drug, and producing and supplying the drug delivery system. These three goals holistically comprise the key system functions of the PFS life cycle.

3.1.1 Key System Function: Treating the Patient

The ultimate purpose of the PFS subsystem is to contain the drug and administer the expected dose within the expected timeframe to the patient. Administration of the drug delivery system is achieved through various avenues, such as the following:

- self-administration with a plunger rod PFS or with additional device accessories
- healthcare provider administration with or without safety systems

Subsequently, to ensure proper administration of the drug, it is necessary to address multiple factors when treating the patient. Factors include the following:

- drug dose volume, viscosity, and temperature
- duration of injection
- targeted tissue and route of administration
- injection technique (pinch, angle, or region)

3.1.2 Key System Function: Enabling Market Approval

The license holder must generate a comprehensive data package to ensure the suitability of the PFS for the intended use as prescribed by various regulatory authorities, such as the FDA and European Medicines Agency, and in accordance with applicable guidance documents to receive regulatory approval for the drug product *(5)*.

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3.1.3 Key System Function: Enabling Production and Distribution of the Delivery System

To produce and distribute the combined drug and delivery system, the system should be:

- compatible with the filling and inspection process, including the plunging process (e.g., Ventube™ or vacuum)
- compatible with the assembly process (plunger rod insertion, labeling, and secondary packaging)
- suitable for delivery to end users, including factors such as cold chain requirements, transport conditions (shock, vibration, and pressure), and protection provided by the secondary and tertiary packaging *(6–8)*

Table 3.1.3-1 below provides a high-level view of the interactions between subsystem components or stakeholders in relation to key system functions.

Table 3.1.3-1 Overview of interactions between key system functions & components in the PFS life cycle

Key System Functions	Contributors and Stakeholders					
	Drug	PFS subsystem	Fill/Finish manufacturing equipment	Ancillary devices	Distribution	Patient
Treating the Patient						
Allows the transfer of the expected dose within the expected time frame in the targeted tissue	•	•		•		•
Contains drug	•	•	•			
Obtaining approval for drug						
Adequately protect the dosage form	•	•	•	•	•	
Composed of materials that are considered safe	•	•			•	
Producing and delivering the drug delivery system						
Be compatible with the filling and inspection process	•	•	•			
Be compatible with the assembly process		•	•	•		
Be suitable for delivery to end user				•	•	•

Notes on Contributors and Stakeholders

1. "Drug" refers to the medicine delivered in the PFS.
2. The PFS subsystem includes a packaged glass barrel with an attached needle that is protected with a needle shield and a plunger stopper.
3. "Manufacturing equipment" refers to the different machines used to process the subsystem to produce a full drug delivery system. The tasks of this equipment might include filling, stoppering, inspecting, or preparing secondary packaging for a product.
4. Ancillary devices include autoinjectors, safety devices, plunger rods, backstops, and other assistive devices to help the user inject the product.
5. "Distribution" refers to the outbound distribution channel from the license holder to the patient.
6. The patient is the person who ultimately receives the treatment.

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3.2 Regulatory Considerations

The following sections provide considerations for manufacturer, supplier, instructions for healthcare provider, regulatory submission/eCTD content format, standards, and technical references applicable to the PFS.

3.2.1 Manufacturer Considerations

A drug product manufacturer must follow the marketing requirements in the country where the product is marketed. In the United States, a PFS is considered to be a combination product as defined by 21 CFR 3.2(e). These combination products are assigned to the lead center based on the PMOA ***(9,10)***. Both CDER and CBER provide guidance on PFS drugs and biologics.

Drug manufacturers must comply with cGMPs to ensure that the PFS product is safe and efficacious. Each constituent part of the PFS product is subject to its applicable cGMP requirements. The device constituent is subject to the quality system regulations for devices as described in 21 CFR 820, whereas the drug or biological product constituent is subject to the regulations under 21 CFR 211 ***(11,12)***. For biological products, the regulations described in 21 CFR 610 also apply ***(13)***. The cGMP requirements for combination products are codified under the final rule in 21 CFR 4, which provides guidelines on how to achieve compliance under one operating principle with additional provisions under another ***(14)***. These include, 1) demonstration of compliance with all cGMP regulations applicable to each of the constituent parts included in the combination product, or 2) demonstration of compliance with either the drug cGMP or the quality system regulations for devices, and with provisions that are not shared by both regulations. If the PFS product complies with the drug regulations, then it must also demonstrate compliance with requirements for management responsibility, design controls, purchasing controls, and corrective and preventative actions in the quality system regulations. Similarly, regulations pertaining to testing and approval of C/C, expiry, stability, and reserve samples must be followed for drug/biologic product when the quality system regulations are followed. Once again, the requirements in 21 CFR 610 are applicable to biological products.

Products that incorporate a drug or administer a drug may be regulated in the European Union as either medical devices or medicinal products, depending on the principal intended function of the product and the method by which this function is achieved. PFSs are categorized as devices for administering a medicinal product when the device and the medicinal product form a single integral product designed for exclusive use in the given combination, and the device is not reusable or refillable. These products are covered by the regulations for medicinal products, such as Directive 2001/83/EC ***(15)***. However, relevant essential requirements in Annex 1 of Medical Devices Directive (MDD) 93/42/EEC apply to the safety and performance-related features of the device. Guidelines related to drug delivery products regulated as medicinal products, such as PFSs, are provided in medical devices guidance document MEDDEV 2.1/3, Revision 3 ***(16,17)***.

Following standards like ISO 13485, Medical Devices – Quality Management Systems – Requirements for Regulatory Purposes, and ISO 14971, Medical Devices –Application of Risk Management to Medical Devices ***(18,19)*** will allow companies to demonstrate compliance with Annex 1 of the MDD.

3.2.2 Supplier Considerations for Filing

Suppliers of PFS components can include information related to the manufacturing of the syringe, needle, and accessory components in the drug master file (DMF) for applications submitted to the FDA in the United States. The manufacturer's supplier agreement should define the roles and responsibilities of each party. In other countries, such information may be submitted in the marketing

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application dossier to the health authorities. Additionally, filing requirements for different regulatory agency jurisdictions may vary, and applicants should be aware of the proper procedures for the intended market.

Elements to be included in the DMF include an overall description of the C/C system, packaging component diagram, dimensional specifications, and information on the manufacturing and processing facility (washing, siliconization, sterilization, and environmental controls), component suppliers [of glass barrels, ceramic paint, plunger stoppers, needles, silicone, needle adhesive, needle shields, plunger rods, flange extenders, or backstops], suitability (protection from light, compatibility, and sterility), and component safety (description of component qualification). The quality controls implemented during syringe barrel manufacture, lubrication, needle assembly, sterilization, and transport are essential for assessment of the information supplied in the DMF. In addition, the inspection of incoming materials or components, in-process inspection, and final release procedures provide vital information on the quality of the components. The DMF should include information on the stability of syringe components without any product, and a list of regulatory requirements fulfilled for the components and revalidation plan. The DMF and the supply agreement should also include a mechanism for communication of quality issues and customer notification.

Elastomeric closure components for PFS systems (e.g., tip caps and plunger stoppers) must meet current U.S. and European regulatory requirements for compendial compliance. Component suppliers should ensure that adequate information is provided to the health authorities for proper review of sponsors' drug applications, including elastomer formulation ingredients and appropriate certifications (e.g., BSE/TSE, dry natural rubber latex, and animal-derived raw materials), compendial testing results for physicochemical properties, and biological reactivity assessment ***(21–23).*** In the United States, suppliers typically elect to compile this information in a Type III DMF.

In some instances, post-processing of elastomeric components (e.g., washing or sterilization) may be performed by a third party, such as the component supplier. In such cases, adequate supporting data should be made available to enable proper agency review of these processes. This information can be contained within a Type V DMF in the United States and should include supporting information on process validation, process controls, testing, and packaging qualification (for sterilization processes).

As DMF owners, suppliers can provide sponsors with letters of authorization to applicable DMFs for use in sponsor applications. Per 21 CFR 314.420(c), suppliers in the United States are required to notify their letter of authorization holders when significant changes are made to the DMFs. Suppliers should also provide further support to sponsor applications by responding to agency questions regarding information in their DMFs in the United States. Refer to 21 CFR 4 for further guidance on compliance with design control expectations.

3.2.3 Healthcare Provider Instruction for use of PFSs

The license holder must provide healthcare providers with instructions for use and disposal of PFSs. Needlestick injuries have the potential to expose PSF users, such as healthcare providers, to bloodborne pathogens. The Occupational Safety and Health Administration (OSHA) and Centers for Disease Control and Prevention (CDC) guidelines describe the requirements and practices in the United States to prevent such injuries ***(24,25).*** The disposal requirements for PFSs may vary between countries, and suggested storage conditions for the PFSs should be followed.

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3.2.4 Regulatory Submission Content for PFS Combination Products

In the United States, the marketing application for PFS combination products could be a new drug application (NDA), or a biologic license application (BLA), depending on whether the product is defined as a drug or biologic. Proprietary information about the device can be submitted in a master file and in letters of authorization to the master file provided in the application. Important considerations for evaluating human factors affecting the use of the PFSs include the intended patient population for the proposed indication and the condition of use, and whether the PFS is self-administered or administered by a healthcare provider. **(Note:** Product-specific submission requirements are not discussed here. Only elements that are important in the context of a combination product are presented in this section.**)**

The application should contain a description of the dosage form, composition, primary C/C system, and any ancillary component/plunger rod information that may be attached during packaging and labeling. The manufacturing and processing facility information should be included. The description of all primary packaging components (syringe barrel with staked needle, adhesive used to bond the needle to the glass tip of the barrel, needle shield, and plunger stopper) should include drawings, specifications, quality details, and supplier information. A description of the manufacturing process, including lubrication of the primary packaging components and material/components inspection, in-process controls, and release operations should be included in the supplier's DMF. In addition, if syringe markings and fill lines are present, additional details on the validation of the marking process are required; these details may be included in the application or by reference to the supplier's DMF. A summary containing details of the sterilization method and validation of the sterilization process providing a sterility assurance level of 10^{-6} should be included. The integrity and compatibility of the C/C system should be evaluated. The potential for leachables and particulates due to interaction of the syringe components with the product should be assessed, the impact of shipping on the C/C system should also be assessed.

PFSs have the potential to introduce new impurities or components, such as adhesive, tungsten, and silicone, which can impact product quality. The impact of such components should be carefully assessed. Syringe barrels and plunger stoppers are typically siliconized for use in PFS applications. A thorough assessment of the impact of component variability on product quality and stability is performed, but the approach is very company specific depending on the manufacturing techniques, component selection, and drug solution formulation. In any case, stability data to demonstrate that the commercial C/C system does not affect product stability is required. The effects of moisture and light are often addressed as part of stability testing. Temperature is a general concern given that most protein drugs are quite labile. Moisture is a concern for lyophilized dosage forms in syringes (or vials). The secondary packing description should clearly explain how the syringe is protected during transport.

The final finished product after filling and assembling should be tested for performance. Testing of the PFS includes testing of delivered (extractable) volume, break-loose and glide forces, and CCI. In addition, a range of physicochemical and microbiological parameters should be tested. Draft guidance was issued by the FDA in January 2013 on the type of marketing submission that may be required for post-approval changes to a combination product in the United States as defined in 21 CFR 3.2(e) ***(26).***

In Europe, the marketing application could be a medicinal product application. The marketing application dossier includes information related to the PFS.

At the time of this writing, the FDA and European Commission have not published guidance on the electronic version of the common technical document (CTD) for devices that are part of a combination product. A template for ICH CTD Module 3 is attached (refer to **Appendix I**) as a guide for providing information about a drug delivery device that is part of a combination product in a BLA/NDA. An example of a completed CTD Module 3 of a combination product application is provided in **Appendix II**.

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3.2.5 Standards and Technical References Applicable to PFS

A C/C system (constituent parts of a combination product), such as a PFS used in the packaging of a human drug or biologic, is approved as part of the drug application. A successful licensure application requires multiple tests and studies according to various industrial and compendial standards to demonstrate the suitability of the selected C/C system, which can be validated for its intended use based on the target product profile. Suitability is defined in the United States to cover four aspects: protection, safety, compatibility, and performance, according to the FDA ***(12).*** Different regulatory bodies may recognize different standards. **Appendix III** lists the standards, guidance, compendial methods, and technical reports to assist with the development of a PFS combination product. The FDA recognizes consensus standards, and EU harmonized standards are available online ***(16,27).*** The various testing standards help ensure regulatory compliance. Other PDA technical reports provide useful information related to sterilization, CCI test methods, and glass defects ***(28–33).***

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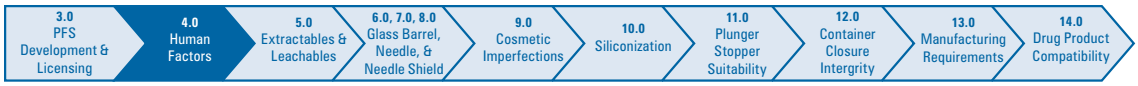
4.0 PFS Human Factors Engineering



Human factors engineering (HFE) considers a product's design, intended uses, users, and use environments, which can affect the safe and effective use of a product ***(34,35).*** The intent of HFE is to improve the quality of a product's user interface design to eliminate or minimize errors during use.

This section describes best practices related to the application of HFE and use-related risk management to the development of PFSs, including:

- key principles of HFE as they apply to PFSs (intended to supplement and not to replace the current health authority guidance and approaches described within recognized consensus standards)
- key considerations in understanding the use of PFSs
- typical areas of usability issues (known issues) to consider in the development of PFSs

4.1 Current Standards and Guidance

Regulatory bodies may request evidence of acceptable human factors summative/validation testing results and supporting processes during premarket application or submission reviews. The failure of manufacturers to apply a compliant and adequate HFE process can lead to serious consequences, including product approval delays/nonapprovals, negative audit and inspection findings, clinical and market complaints, and ultimately an unsafe, ineffective, and unsuccessful product. Available HFE standards and guidances include those listed in **Table 4.1-1.**

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Table 4.1-1 Human Factors Engineering-related Standards and Guidances

Standard/Guidance ID	Title
ANSI/AAMI/IEC/EN 62366-1:2015	Medical devices – Application of usability engineering to medical devices
AAMI/ANSI HE75:2009	Human Factors Engineering – Design of Medical Devices
ISO 14971:2007/(R) 2012	Medical Devices – Application of risk management to medical devices
IEC 60601-1-6:2010	Medical electrical equipment – Part 1-6: General requirements for basic safety and essential performance – Collateral standard: Usability
IEC 60601-1-8 Edition 2.1 2012-11	Medical electrical equipment – Part 1-8: General requirements for basic safety and essential performance – Collateral Standard: General requirements, tests and guidance for alarm systems in medical electrical equipment and medical electrical systems
IEC 60601-1-11:2010	Medical electrical equipment – Part 1-11: General requirements for basic safety and essential performance – Collateral Standard: Requirements for medical electrical equipment and medical electrical systems used in the home healthcare environment
FDA/CDRH document # 1497	Medical Device Use – Safety: Incorporating Human Factors Engineering into Risk Management (2000): http://www.fda.gov/medicaldevices/deviceregulationandguidance/guidancedocuments/ucm094460.htm
FDA/CDRH document # 1128	Guidance on Medical Device Patient Labeling (2001): http://www.fda.gov/medicaldevices/deviceregulationandguidance/guidancedocuments/ucm070782.htm
FDA/CDRH document # 1757	Applying Human Factors and Usability Engineering to Optimize Medical Devices (2011, draft*): http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/GuidanceDocuments/ucm259748.htm
FDA/CDRH document # 1750	Design Considerations for Devices Intended for Home Use (2014): http://www.fda.gov/medicaldevices/deviceregulationandguidance/guidancedocuments/ucm331675.htm
FDA/CDRH docket # HFA-305	Safety Considerations for Product Design to Minimize Medication Errors (2012 draft*): http://www.fda.gov/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/ucm331808.htm
21 CFR 820.30, Subparts A-D	Human Factors Implications of the New GMP Rule Overall Requirements of the New Quality System Regulation http://www.fda.gov/medicaldevices/deviceregulationandguidance/humanfactors/ucm119215.htm
EMA/606103/2014	EMA Guidance – April 2015, Good practice guide on risk minimization and prevention of medication errors. Pharmacovigilance Risk Assessment Committee (PRAC). (http://www.ema.europa.eu/docs/en_GB/document_library/Regulatory_and_procedural_guideline/2015/04/WC500185538.pdf)

Note: The FDA draft guidance documents referenced above marked with * are not intended for implementation until finalized and are subject to change. The final versions will represent FDA's final position on those areas.

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4.2 Human Factors Evaluation

The application of HFE throughout the product development process can optimize product design and minimize use-related failures and difficulties. The need to perform a human factors summative/validation study and include human factors data in premarket submissions can be determined based on whether there are potential use errors that could cause harm. For PFSs, there are known use errors that can result in delayed therapy (overdosing, underdosing, needle-stick injuries) and cause harm, such as reduced efficacy, infection, etc. A human factors summative/validation study can demonstrate that representative users can use the device under simulated use conditions without producing patterns of failures that could result in negative clinical impact to patients or injury to device users (refer to **Section 4.8**).

4.3 Device-Use Characterization

The HFE process begins with characterizing device use. This includes consideration of user populations, use environment, and intended use scenarios as discussed below.

4.3.1 User Populations

Common users of PFSs are healthcare providers, lay caregivers, and patients. User groups often have unique characteristics that are capable of affecting the user's ability to learn and use the product safely and effectively. The generally accepted term for a distinct group of users with unique characteristics in HFE is "user population."

Within each user population, individual user characteristics may vary. Key areas of differences between users or user populations include knowledge, skills, abilities, experience, cognitive functioning, perceptual acuity, and manual dexterity. Furthermore, user attitudes, emotions, and expectations may also affect the use of the device. Other factors that typically describe variations between user populations and user groups include gender, age, disease-related impairments, previous device experience and literacy, and education levels.

The design of the device should be evaluated with individual users who are representative of the intended users. This approach helps ensure that designs are optimized to support the needs of intended users to the extent possible.

Key considerations for users of injection devices in a home-use environment include the following:

- **Injection-related anxiety/phobia:** anxieties and phobias can make the task of self-injection/injecting another individual difficult and may cause users to withdraw the needle before the injection is complete.
- **Disease-related impairments:** the same disease that the injection is intended to treat can impair the ability of users to inject the product by reducing hand and finger strength, impairing sensory abilities, and reducing manual coordination and dexterity.
- **Negative transfer of learning between different products:** users with prior experience with injection devices that operate in a specific way may incorrectly use a device that operates differently due to the assumption that the devices operate in the same way.

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4.3.2 Use Environment

The use environment can affect the user's ability to use a device safely and effectively. Specific considerations for HFE may include the following:

- lighting (e.g., ambient, direct, and reflective light) that might affect the user's ability to see the syringe components, markings, and accessories
- thermal factors (e.g., air temperature, humidity, heat, moisture, and air flow) that might affect how well the user can grip, hold, and inject the syringe
- auditory factors (e.g., background noise and alarms) that might cause distractions and an inability to hear feedback from the syringe (if present)
- workload (e.g., distraction and interruptions) that may cause difficulty, especially for users with cognitive, sensory, and/or manual dexterity impairment

4.3.3 Intended Use Scenarios

Intended use scenarios describe how the user is expected to use the device. A device may also have multiple reasonably foreseeable use scenarios that may differ by user group. For example, a healthcare provider may be required to select the appropriate product for a patient; the same healthcare provider may need to visually identify and select a PFS among PFSs containing different drug strengths or different drugs. Alternatively, it may be necessary for the patient to possess the ability to self-administer the drug in the selected PFS.

Use scenarios are determined in part by the intended use of the product, and may include such factors as dosing regimen (e.g., variable dosing or fix dosing), dose frequency (e.g., once a day or once a month), type of medication (e.g., emergency versus routine), extent of training (e.g., naïve or experienced user), and/or clinical supervision prior to or during use. Foreseeable scenarios involving misuse or use error (including stress, performing the tasks within a specific time limit, or the need to wear gloves when using the PFS) should be identified and considered part of the application of human factors' techniques during design of the user interface.

4.4 Defining User Needs

After characterization of device use and prior to initiation of the design process, manufacturers should investigate and define end-user needs. End-user needs can be documented as user requirements, and as such, validated (i.e., are testable).

Multiple sources exist for defining user needs, such as the following:

- user group-specific needs (**Section 4.3.1**)
- use environment (**Section 4.3.2**)
- intended use (**Section 4.3.3**)
- potential hazardous situations that may arise (**Section 4.6.1**)
- reviews of relevant published material (**Section 4.6.1.1**)
- empirical research conducted by interviews or observations of potential users (**Section 4.6.1.2**)
- known use-related problems with the same or similar products
- health authority guidance(s) and consensus standards (**Section 4.1**)

User requirements may be further refined as new information is evaluated during device development.

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4.5 Design Requirements

To enable the product design process, user requirements are translated into engineering design requirements. Design requirements are comprehensive, clear, quantifiable, and verifiable. For example, a user requirement might state, "The user needs to be able to remove the needle cap." This might correspond to a specific design input requirement specifying the use of a suitable cap removal force or cap/syringe geometry. Design requirements related to the user-device interface may be derived from a range of sources, including anthropometric size and strength data or product-specific industry guidelines and standards.

Once a design prototype has been developed, requirements may be further refined through a process of iterative usability evaluation and risk analysis. In most cases, such evaluations should include early formative testing with users (**Section 4.7**) in which the suitability of the design input requirement may be assessed and adapted as necessary.

4.6 Risk Management

Use-related risk management is an integral part of the HFE process in the development of medical devices and combination products. This includes use-related hazard identification, risk analysis, identification and implementation of mitigation strategies. For further information on risk evaluation, analysis, and management refer to *PDA Technical Report No. 54: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations* ***(36).***

4.6.1 Hazard Identification

A hazard is a potential source of harm. Use errors and difficulties related to the product/user-interface design can lead to hazardous situations. All known and foreseeable use-related hazards associated with products are typically identified and documented within the use-related risk analysis (see **Section 4.6.2).**

Early in the design process, it is best practice to identify and investigate potential hazards based on a systematic review of all known information related to the type of product under development. This information is typically acquired through literature searches, empirical research, and analytical methods. These sources are described in the following sections.

4.6.1.1 Literature Search

Conduct a literature search to investigate known use-related issues with previous models or similar products. Typical sources of information include the following:

- scientific literature (e.g., study findings published in HFE and clinical journals)
- health authority complaints databases and reports (e.g., Manufacturer and User Facility Device Experience [MAUDE] and Medical Device Reporting [MDR] databases and adverse reporting data files)
- health authority alerts and/or warnings
- postmarketing product complaints for previous variants or similar devices
- clinical experience data, including clinical complaint records
- specialist support organizations (e.g., disease-specific patient support groups and associations)
- published standards and guidance
- previous formative and summative/validation studies of the device or similar device(s)
- support organization websites, social media, and blogs

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4.6.1.2 Empirical Research

Empirical research is an approach to identify and specify user needs, identify potential use-related hazards of the device, and analyze behaviors in end-use contexts. Empirical research methods may include the following:

- **Contextual inquiry:** this ethnographic research method is used to observe and analyze behaviors in actual end-use contexts (actual environments and use scenarios). Contextual inquiry can reveal important use-related behaviors in actual use that may not be present in a controlled or artificial test environment (e.g., observing the use of existing syringes to identify use-related difficulties).
- **User or expert interviews and focus groups:** eliciting input from users through open-ended and/or semi-structured interviews or conducting focus groups is a valuable means of attaining user insight. Since users often differ significantly in personal characteristics and use-related behaviors, a diversity of end users should be included.
- **Formative HFE:** simulated use of early prototypes are effective for identifying product use-related hazards and assessing the adequacy of implemented risk controls in addition to areas for further improvement. **Section 4.7** provides further details on this topic.

4.6.1.3 Analytical Use Error Identification Methods

Analytical methods are useful in assessing user interfaces and key components of use-related risk analysis. Analytical methods involve the systematic description, decomposition, and assessment of tasks often used in identifying hazardous situations. Many analytical methods are available in different domains and areas of focus. Refer to Stanton et al. for a comprehensive list of available methods ***(19,37–39)***.

A common analytical method effective for assessing a user interface is Hierarchical Task Analysis. This method systematically decomposes the steps necessary to attain the overall goal of the user into sub-goals and ultimately to the lowest-level tasks deemed appropriate for analysis. Hierarchical Task Analysis suggests how to organize related tasks, identifies sequence of specific tasks, and generates a clear structure for risk analysis. Once use-related hazards have been comprehensively identified, the risks associated with those hazards are assessed.

4.6.2 Use-Related Risk Analysis

Employ a use-related risk analysis to systematically assess all of the steps involved in using the device (e.g., based on a task analysis), the errors that users might commit or the tasks users might fail to perform, potential negative clinical consequences of use errors and task failures, risk-mitigation strategies to reduce any moderate or high risks to acceptable levels, and the method of validating the risk-mitigation strategies ***(40)***.

The potential use errors can be assessed for severity, occurrence, and detectability as described below ***(36,41)***.

- Severity rating defines the potential impact of the hazardous situation on the user. The severity rating is determined based on product knowledge, customer interface, previous usability test results, and product complaints. Refer to ISO 14971 (2012) for comprehensive examples. When more than one hazard is identified, the most severe rating should be used with respect to a specific task. ISO 14971 (2012) presents a description of each severity level and the rating assigned to each hazard identified in the FMEA ***(37)***. It is a recognized consensus standard applicable to medical devices ***(42)***.

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- Occurrence rating is an estimate of the frequency with which the identified harm occurs. It is determined based on product knowledge, previous usability test results, and product complaints. However, because it is difficult to determine accurately, estimates of severity should be used as the primary indicator of use error criticality (see **Section 4.6.2.1**).
- Detectability is the ability to detect, discover, or determine the existence, presence, or fact of a hazard. Detectability must be factored in the estimate of overall risks, together with the likelihood of occurrence and severity of harm. The risk control strategy may include processes that improve the detectability of a hazard ***(43)***.

While conducting the use-related risk analysis, it is important to consider the impact (potential harm) on the patient if a use error occurs during use of the device. For example, the instructions tell the user to place the product in a refrigerator, but the user leaves the product out at ambient temperature and then uses the product. In this scenario, the worst-case harm may be that the drug efficacy is affected, which could result in ineffective treatment. For this use error, the probability of occurrence is estimated as "occasional," and the severity of the error rated as "serious" by the medical representative on the team (in this scenario the definition of "serious" could be an error that results in injury or impairment requiring professional medical intervention).

After severity and occurrence are rated, risk mitigation strategies are developed to minimize identified use-related hazards. These may include the following, in order of priority:

1. Design the device to remove the hazard, thereby reducing the likelihood that the user will encounter the hazard, or reduce the consequences if hazard occurs
2. Make the user interface— including its operating logic—error resistant or error tolerant
3. Alert the user to the hazard through the user interface (e.g., labeling/IFU)
4. Develop written procedures and training that inform the user of the hazard, and provide steps to avoid the hazard

If unacceptable risks caused by use errors are identified through the risk assessment process, then the elements of the user interface that led to those use errors should be evaluated further to achieve acceptable risk levels by appropriate means, such as product redesign, to ensure that the changes made to eliminate or minimize hazards do not introduce new hazards ***(44)***.

Formative evaluations and human factors summative/validation testing are methods that demonstrate the effectiveness of the implemented risk mitigation strategies. Justification for residual risks may take into account whether to eliminate these risks, or if eliminating/reducing these risks is highly impractical and the residual risk is clearly outweighed by the benefits of the device.

4.6.2.1 Essential and Safety-Critical Task Identification

Employ the use-related risk analysis to identify essential and safety-critical tasks. Essential tasks are those that are necessary for successful use of the device for its intended purposes. Examples of typical essential tasks are removing a PFS from packaging, removing a needle shield or cap, inserting a needle into an injection site, and fully depressing the plunger. Safety-critical tasks are those that, if omitted or the user performs incorrectly, could have a negative clinical impact. A task may be both safety-critical and essential simultaneously. Examples of safety-critical tasks typically include full depression of the syringe plunger rod and activation of the safety needle-device mechanism. In addition, user understanding of all safety-critical warnings and caution statements in the device labeling is essential.

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Each task is assessed for device dependency based on interaction, if any, with the PFS:

- In device-dependent tasks, interaction with the device is required (e.g., removing a cap, depressing a plunger, or activating a needle-safety device). If a user successfully completes a task, then this is an indication that the success of the task completed is reasonably controlled by the design of the device. The performance of device-dependent tasks is important to ensure successful user performance as part of the summative/validation testing.
- Device-independent ("ancillary") tasks include preparation tasks and are broadly defined as those that support the successful operation of the device (e.g., washing hands or checking the expiration date). These are often highly dependent on compliance with instructions; validation may occur through user comprehension of the labeling materials (notably the instructions for use) in summative/validation testing. Labeling comprehension is an important determinant when assessing whether critical warnings and cautions have been appropriately captured and users understand the intent.

4.6.2.2 Common PFS Use-Related Hazards

Table 4.6.2.2-1 offers several examples of common use-related hazards and harm associated with PFS use. The table is not intended to be exhaustive, and certain hazards and harms may not apply to all situations. Once use-related hazards have been comprehensively identified, the risks associated with those hazards are assessed. It is the manufacturer's responsibility to identify, implement, and validate risk mitigation strategies that effectively minimize risk (refer to **Section 4.6.2** for the priority order of risk controls that should be implemented).

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Table 4.6.2.2-1 Examples of Common Use-Related Hazards and Harms Associated with PFS

Use error	Example of a hazard	Example of a harm(s)
Inability to open packaging	Inability to use device	Reduced efficacy
Inability to remove needle shield		
Incorrect removal of needle shield	Damaged needle	Injury
Incomplete plunger depression	Partial injection or no dose administered	Reduced efficacy
Premature removal of device from injection site		
Failure to engage or activate a needle-safety feature	Exposed used needle	Injury
Reuse of a syringe or needle	Contaminated syringe or needle	Infection
Inability to attach needle (if applicable to device type)	No dose administered	Reduced efficacy
Inability to set an accurate dose for devices requiring dose adjustment (if applicable to device type)	Inaccurate dose set	Overdose Reduced efficacy (Underdose)
Removal of plunger from syringe	Damaged device	Reduced efficacy
Failure to clean injection site prior to injection or contamination of injection site after cleaning (e.g., by blowing on or touching the site)	Unclean injection site	Infection
Touching by needle to other surface (e.g., of a table) prior to injection	Contaminated needle	
Failure to examine the product label	Administration of incorrect drug product or expired drug product	Injury Reduced efficacy
Failure to examine the syringe prior to use	Damaged device	Injury
Failure to check the expiration date	Expired product	Reduced efficacy and safety
Failure to wait for the PFS to warm up to room temperature	Cold injection	Discomfort or pain
Recapping of the needle	Handling of unprotected needle	Injury

4.7 Formative Usability Evaluation

After the user and design requirements have been defined and the risk management documentation process has been initiated, the user interface design is integrated into the product design and then implemented. Formative human factors evaluations assess the usability and use safety of the device, its packaging, labeling (including instructions for use), and training programs using product prototypes. The intent of formative usability studies is to optimize the user interface by "designing out" any usability problems or potential use-related hazards during the development of the product. The type and number of formative evaluations varies depending on the development effort and on the use-related risk analyses. The goal of this stage is to reduce use-related risks as much as is reasonably possible before the implementation of the design verification and summative/validation stage of development.

A written study protocol and report provides the necessary documentation for each formative human factors evaluation. Typical content for the protocol includes a description of the scope and objectives of the study, as well as a description of the participants, use environment, and prototype product. In addition, the formative study report provides a description and evaluation of any variants (e.g., different rigid needle shields or different versions of the instructions for use) along with any actionable

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conclusions from the study noted. Using the results and analyses from the formative study report, the use-related risk analysis is updated accordingly to ensure that any new risks and risk mitigations are appropriately captured.

4.8 Summative/Validation Testing

The summative/validation test is intended to be the culmination of all prior HFE analyses and formative testing. At this point, modifications to the user interface and product design are complete based on the results of iterative testing. The product is ready for validation testing.

With the appropriate design, execution, and acceptable results, the human factors summative/validation test provides evidence that the device is safe and effective for intended use. This decision is based on the following two considerations:

1) a representative sample of users satisfactorily completing all essential and critical tasks under expected use conditions, and

2) there are no additional patterns of use errors resulting from design deficiencies.

If this study is the first HFE effort (e.g., no formative evaluations were performed), or if the formative evaluations were not sufficiently robust, the summative/validation testing often uncovers user interface design problems that may result in use errors. When this happens, these new errors are evaluated to determine if additional user interface modifications are necessary. If the device is modified as a result of observed use errors, then the modified device is revalidated to demonstrate that the modifications effectively address the observed use errors and do not introduce new use errors. Since modifications to devices can be made during scale-up, the validation/summative testing is conducted using production equivalent devices (final production units or demonstrable equivalents). The participants and test environment is expected to be representative of intended user populations and use environment. If there will be more than one user population (e.g., caregivers, patients, and/or healthcare professionals), then the summative/validation study should be performed with $n \geq 15$ in each distinct user group ***(45)***.

Formative usability evaluations may be conducted using empirical methods (e.g., user studies or expert reviews), analytical methods (e.g., hierarchical task analysis), or simulated-use studies. Possible formative studies include evaluations of different prototype device designs, instructional and/or training material, and safety-critical and essential tasks. Typically, these studies are conducted under conditions of simulated use (e.g., injection into a pad), and are facilitated by a human factors specialist to ensure that use-related issues are captured and appropriate root-cause analyses are performed. According to human factors best practices, a formative human factors study typically includes a minimum of five to eight participants ***(38)***.

For products intended for submission to the U.S. FDA, study participants are typically representative of the intended user population (or populations) in the United States (e.g., able to read and speak English and reside in the United States). The tasks selected for testing are typically derived from the results of a comprehensive assessment of use-related hazards and analysis of risks that consider all functions of the device. The tasks are prioritized to reflect the relative importance of the task and the severity of the potential impact of omission or use error on the safety of the device and the user. Training provided in the study is typically representative of the end-user training for actual use. If it is foreseeable that actual users might not or will not consistently receive training, then an untrained group should be included in the summative/validation study to represent the anticipated population of untrained users. To account for "training decay," time is allowed to elapse between the training and

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the test session. This decay time is reflective of the decay time in actual use. For devices that are used daily, a (simulated) training decay period between training and testing of at least 30 minutes is usually appropriate. For devices that are used at longer intervals, an ideal representative training decay would be at least one day or longer.

The testing is designed to capture all objective (observational/performance) data and collect subjective data (narrative responses to open-ended questions) after completion of simulated use tasks. Subjective data include responses to questions about the use of the device overall, critical tasks, any observed use errors, and use difficulties. A key purpose of the subjective data is to obtain an explanation from the user's perspective on the cause of use errors or use difficulties noted during simulated use in addition to the user's overall use experience.

For the purposes of assessing performance in summative/validation testing, a use error is generally defined as action or lack of action that was different from that expected by the manufacturer and can lead to patient harm (i.e., overdose, underdose, or incorrect therapy). Performance success criteria are typically predefined prior to the summative/validation testing, and the study moderator ensures that the use errors are accurately recorded as representing the errors that could cause harm during actual use. The study results should then be used to refine the use-related risk analysis to determine the overall acceptability of use-related risk and determine if further modifications are needed to the device design, device labeling (including instructions for use), and/or the training (if applicable) to reduce the use-related risks to acceptable levels.

The human factors summative/validation study report typically includes the following elements:

- rationale for the study design, including the data collected, the number and description of distinct user populations and test participants (who was studied), and critical and essential tasks (what tasks/ intended uses were assessed)
- materials used within the study (what was tested)
- summary of the test methods used (how was the study performed)
- summary of both objective task performance and subjective results obtained from the study
- detailed root-cause analysis for all use errors and difficulties that occurred
- conclusion about safe and effective use

4.9 Labeling

Labeling includes device labels, instructions for use, information provided on the device, package inserts, package labels, and any accompanying informational materials. Labeling is considered part of the product user interface. As with the device design, labeling development starts with characterization of the user and use environment. It is important to consider the physical and cognitive abilities of users, as well as the limitations of the environment during label development, especially for self-administered devices. The workload or stress level of the user population should be a consideration because these can limit users' ability to understand and remember complex, multistep instructions.

Consider using several best-practice principles when developing instructional materials to ensure that the layout, content, and organization support safe and effective use of the device ***(46, 47)***. Use short, declarative, active-voice sentences to describe specific tasks for users. Enhance user instructions further with additional graphics or diagrams.

The use of formative evaluations can ensure that product labeling is ready for validation. As previously noted in **Section 4.7**, this can include expert reviews as well as testing with representative users.

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Warnings and other risk mitigations implemented as part of labeling can also be evaluated. Finally, labeling represents a part of the user interface and is included in summative/validation testing to confirm safe and effective use. Information such as warnings and precautions are considered critical tasks and tested through assessments of users' ability to read and understand the information.

4.10 Changes to Currently Marketed Products

Manufacturers may change a currently marketed PFS product for a variety of reasons. These may include use problems identified in postmarket reports or customer complaints, or changes due to other reasons, such as changes related to design or manufacturing, or new indicators related to the use of a "platform" device (e.g., new patient population, new user groups, new use environments, etc.).

An HFE impact assessment is developed and documented to evaluate product changes, describe each change, provide a rationale for making the change, and include an updated use-related risk analysis focusing on how those changes have impacted the user tasks and the risk profile ***(48).*** The assessment will indicate if revalidation of the risk-based decision is necessary. To inform the decision further, the history of commercial product complaints associated with use problems is typically included in the use-related risk analysis. If the change eliminates the possibility of the use error without affecting other aspects of use, then summative/validation testing may not be necessary with appropriate justification. If the HFE impact assessment or the use-related risk analysis identifies new use errors or hazardous situations that could lead to harm, then the summative/validation testing is needed and designed to focus on all aspects of the user interaction affected by the changes.

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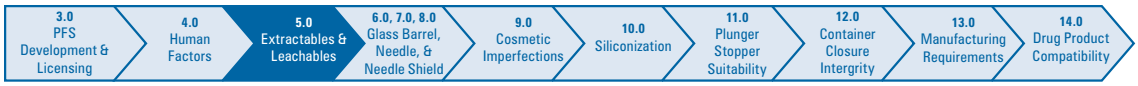
5.0 Extractables and Leachables Evaluation



The characterization of extractables and leachables presents an important component in drug product development. Extractables are compounds that are forced out of a respective material or component under laboratory experimental conditions (i.e., high extraction power of the solvent, increased extraction temperature, but shorter exposure time than during normal use). Leachables are those compounds that actually migrate from the respective material or component (e.g., the pharmaceutical C/C system) into the drug product formulation during its intended storage and use conditions. Typically, leachables represent a subset of the amount of extractables ***(3)***. The following questions are essential to consider in the course of characterization:

- Which compounds can migrate out of the C/C materials (extractables)?
- Which compounds and how much of each compound do in fact migrate into the drug product solution (leachables)?
- What is the impact of these compounds on patient safety?
- Do any of these compounds have a direct critical impact on the quality of the drug product?

Resolution of these questions requires knowledge of materials science, the component manufacturing process, analytical chemistry, and drug product manufacturing processes. The quantity and frequency of dosing are critical factors required for a thorough, expert review of the quality and safety implications, and requires a toxicologist to provide input into the evaluation.

This section describes considerations for evaluating extractables and leachables that may migrate from a PFS. Although this section may provide useful information to the pharmaceutical manufacturer, this technical report is not a regulatory guidance document, and manufacturers should be prepared to justify their evaluation approach to regulatory reviewers/inspectors.

5.1 Production of Syringe Barrels

The choice of the glass raw material is the first driver for controlling potential leachables in drug products. The pharmaceutical industry uses Borosilicate Type 1 glass tubes as raw material for manufacturing syringe barrels. Type 1 glass has the highest hydrolytic resistance and chemical stability of available glass formulations, and therefore limits migration of glass component raw materials into the drug product. Glass quality for hydrolytic resistance must meet the testing requirements in the relevant pharmacopeias (e.g., European Pharmacopoeia, USP, or Japanese Pharmacopoeia).

Delamination (the scaling off of very thin, visible-to-subvisible glass flakes) of glass surfaces contacting the product has been reported for some products packaged in glass vials, whereas delamination has not been reported in glass syringes so far. However, this possibility cannot be fully excluded during the storage of certain formulations. USP <1660> Evaluation of the Inner Surface Durability of Glass Containers addresses delamination and contributing factors in more detail ***(49)***.

The syringe barrel manufacturing process consists of four main steps, each of which can generate or introduce compounds that may later become extractables and/or leachables. Descriptions of each step are included in the relevant sections of this report. However, in this section, only a summary of the steps is included to describe the possible sources of extractable and leachable substances.

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5.1.1 Cutting Step (Horizontal Forming)

In the first step of the syringe barrel manufacturing process, glass tubing is cut to the appropriate size. The cut tubing pieces are then washed to remove glass particles and transferred to a forming machine. During this step, the glass does not undergo any treatment that could affect its physicochemical stability. This step has little impact on the generation of extractable or leachable substances.

5.1.2 Forming Step (Horizontal/Vertical Forming)

In the second step, the syringe tip and flange are sequentially formed by local heating of the glass cut/tube. Annealing and forming steps can locally modify the surface composition of the glass, and consequently, influence the leachable chemical profile. Some manufacturers may use tooling lubricants during the manufacturing process, which could subsequently be viewed as a process leachable.

Tungsten pins are commonly used to form the tip channel. Tungsten residuals have received attention due to some cases of product incompatibility reported in the literature ***(50).*** Such sensitivity must be evaluated through testing and compatibility studies (e.g., spiking studies as appropriate). Residual tungsten levels may be controlled or minimized during the formation step by use of process parameters and washing, and possibly eliminated through use of substitute pin materials if a drug has significant sensitivity to tungsten. In addition to sensitivity to tungsten, some drugs may possibly exhibit sensitivity to other metals. Therefore, if pins made out of substitute metals—such as iridium or platinum—are employed, then give consideration to employing the same control or minimization processes as those used for tungsten to reduce possible interactions between the drug and pin residues on the inner surface of the syringe. It is known that in areas where formation occurs (such as the tip area), a higher concentration of pin-derived compounds may occur than in the areas that are not in direct contact with heat (i.e., hot flames) ***(51).*** Once formed, the syringe barrel passes through an annealing oven to release stress in the glass; then the needle is affixed.

5.1.3 Needle Assembly Step

In the third step, the glass barrel and needle are assembled. Some manufacturing operations conduct this step in line with the formation process. The needle is mechanically placed into the tip to a predefined depth; then adhesive is added and polymerized by ultraviolet (UV) light exposure.

The adhesive used to bond the needle into the glass syringe is a possible source of extractables and leachables, such as photoinitiator byproducts, residual monomers, and adhesion promoters ***(52, 53).*** After needle assembly, the syringes are packed into trays or other support/containers for further processing.

5.1.4 Washing, Siliconization, Needle-Shield Positioning, and Packaging

The syringe supplier, pharmaceutical company, or contract fillers for bulk syringes perform the final step of the syringe barrel manufacturing process, which includes washing, siliconization, and sterilization. In this step, the bulk syringes are packaged into trays and cartoned. After the supplier completes the washing, siliconization, needle-shield positioning, and sterilization processes, the syringes are designated as ready-for-filling. Refer to **Section 10.0** for a full description of the washing, siliconization, and needle shield positioning steps.

Whether processed by the supplier or the pharmaceutical company, the syringes are washed in a controlled environment with hot WFI. After the syringes are dried and lubricated with medical-grade silicone oil, the needle shield is assembled. The washing process significantly reduces residual levels of process impurities.

Silicone oil is not formally considered to be extractable or leachable, but it is known to emulsify or migrate as very small droplets into liquid drug formulations. It is possible to analytically assess silicone

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oil migration using particulate counting systems that can discriminate silicone oil droplets from other particulate impurities (e.g., microflow imaging), or quantified (e.g., atomic absorption spectroscopy). Compatibility of the drug molecule and the formulation with the siliconized syringe should be evaluated as part of long-term compatibility testing.

After siliconization, the syringe barrels are nested (tip downward) and visually inspected. The nest is placed into a tub and packaged prior to sterilization. Refer to **Section 11.8** for additional details related to secondary packaging in the packaging process.

5.2 Ethylene Oxide and Gamma Irradiation Sterilization and Decontamination Processes

As described in **Section 13.0**, several sterile barrier systems in one or two bag(s) are used in the final packaging of the washed and siliconized syringes (e.g., cardboard or plastic cases), depending on the customer requirements. These packages are placed on a pallet and undergo a sterilization cycle or a suitable alternative process. The needle shield elastomer formulations are designed to allow sufficient gas penetration to achieve gas sterilization (e.g., ethylene oxide), especially at the material interfaces. The drawback of this property is the retention via absorption of ethylene oxide (EO) and its potential desorption over time. Ethylene oxide and ethylene chlorohydrin residuals must comply with ISO 10993-7, Biological Evaluation of Medical Devices, and with the limits of 1 μg/mL residual EO and 50 μg/mL ethylene chlorohydrin per container established in EMA CPMP/QWP 159/01 by the European Agency for the Evaluation of Medicinal Products (*Note: for guidance on limitations of EO in medical products*) ***(54–56).*** These are acceptance criteria for the batch release of empty syringes. Given that process residuals—such as EO—may impact the drug product quality, it is likely that product-specific limits will need to be evaluated.

Clean, lubricated plunger stoppers packaged in nests or in bulk are frequently sterilized by gamma irradiation in plastic bags at validated doses. Moist heat may also be used for sterilization depending upon the requirements of the pharmaceutical company. Suppliers can use various packaging configurations for plunger stoppers, including systems designed to connect to isolators.

The sterilization process may affect extractables and leachables (e.g., depletion of volatile compounds by autoclaving, generation of new compounds by radiolysis during irradiation). Therefore, the method used to sterilize the component for evaluation is important and should represent the intended conditions of use.

5.3 Sources of Extractable and Leachable Substances

Possible extractables and leachables from a 1 mL long ready-to-fill syringe are chemicals introduced by various components and processes, including:

- glass tubing composition
- glass converting process
- injection needle
- adhesive used for needle bonding
- elastomeric components
- labeling inks or adhesives
- residual ethylene oxide
- other process residuals

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The primary sources of packaging component extractables and leachables for 1 mL long staked needle PFSs are glass and elastomeric components (plunger stopper and needle shield).

Sections 8.0 and **11.0** respectively, provide thorough discussions of user requirements for needle shield and plunger stopper selection. In this section, discussion is limited to the characteristics that may affect extractable and leachable substances.

5.3.1 Elastomeric Components

The majority of elastomeric components are made from rubber materials that are composed of several different ingredients and are subjected to chemical cross-linking (curing) in a process that takes place at an elevated temperature (thermoset rubber). The primary ingredients in pharmaceutical rubber are the elastomer (high-molecular-weight polymer), filler (inorganic mineral-based material), pigment (inorganic or organic), and curing system for cross-linking. Other frequently used ingredients are antioxidants, plasticizers, waxes, and accelerators that help accelerate the curing reaction. All of these ingredients may already contain additives or generate impurities during the curing reaction. Examples of potential extractables are polymer stabilizers, antioxidants and their degradation products, low-molecular-weight fractions of elastomers and plasticizers, metal ions, accelerators, and side products generated during the curing reaction.

Some thermoplastic rubber formulations used in manufacturing needle shields do not require a curing step; thus, extractables related to curing are avoided. However, the extractables not related to curing remain. Plunger stopper formulations typically use a bromobutyl or chlorobutyl polymer as the elastomer. Plunger stoppers typically have a higher level of chemical cleanliness than needle-shield formulations. This is because needle-shield formulations typically made of polyisoprene or styrene-butadiene rubber have more complicated curing systems. The design of these systems (polyisoprene or styrene-butadiene rubber) enhances permeability when used with EO or moist heat sterilization. Another design parameter to investigate is whether using polymeric coatings that act as barriers between the plunger stopper and the drug product solution can further reduce the levels of potential leachables from the plunger stopper.

Suppliers of elastomeric components can provide packages of extractables information ranging from potential extractables lists to qualitative and quantitative determinations of extractables after extraction with model solvents under model conditions. Leachables can undergo hydrolysis or other reactions (thereby forming secondary leachables), or can react with the drug substance. Extractables data packages may contain physicochemical information only or may be complemented by toxicological data on extractables, all of which should be reviewed by internal experts. The extractable information provided by the supplier may supplement the pharmaceutical company's evaluation but cannot replace drug compatibility studies for which the pharmaceutical company remains responsible.

5.3.2 Glass Components

Extractables related to glass are primarily metallic ions from mineral materials in the glass raw materials. The glass typically used for syringes consists mainly of silicon oxide, aluminum oxide, and boron oxide as network-forming components (e.g., borosilicate glasses) spiked with network-modifying components, such as sodium or potassium oxides. High concentrations of leachable glass components, such as sodium or potassium, may cause the pH in unbuffered solution formulations to shift, and thus become critical and affect the quality and stability of the drug product, and/or the stability of the glass surface. Some glass also contains calcium, barium, magnesium, titanium, iron, or cerium to provide certain characteristics, such as color, strength, or formability (melting temperature). Potential contaminators during glass production or forming are refining agents, such as arsenic oxide, anti-

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mony oxide, alkali halides, and alkaline earth halides. During syringe formation, tungsten contamination may occur when using tungsten pins to form the syringe cone (see **Sections 5.1.2** and **14.6.2.1**). Trace elements may also be present as contaminants from the added compounds listed above.

Within siliconized syringes, silicone may be present as polydimethylsiloxanes (polymeric organosilicon). Silicone and tungsten species have the potential to interact with protein-based active pharmaceutical ingredients, and therefore need to be assessed ***(50)***.

5.4 Extraction Conditions and Sample Preparation

An important step for the evaluation of potential leachable substances is the choice of adequate extraction techniques and conditions. The current practice is to generate separate extraction profiles on each PFS device component (e.g., rubber stopper, syringe barrel, needle shield, and staked needle adhesive), and the assembled PFS device system. The extractables profiles of each PFS device component help to identify issues related to individual components and the source of the extractables. An alternative approach involves extraction on the assembled syringe system. By exposing solvents only to the product contact surfaces of the PFS device, this technique provides a truly accurate extraction profile representative of the PFS device system.

A risk-based approach should be used to ensure detection of all critical substances that might migrate into a pharmaceutical formulation from the packaging material. Various extraction strategies may be used to force the migration of substances out of the packaging material, including different solvents, sample preparation, accelerated temperatures, and time. The most commonly accepted practices include a series of studies: the controlled extraction study, simulated extraction study, and leachable study. These studies are useful for defining the set of target leachables needed to develop the appropriate quantitative analytical methods.

Before being tested, components must be prepared (washed, siliconized, and/or sterilized) in a manner representative of expected usage. In addition, the extraction time should be optimized at least once for each basic material, such as for the elastomeric rubber or adhesive. USP <381> Elastomeric Closures for Injections ***(21)*** details such evaluation parameters. **Tables 5.4-1** and **5.4-2** outline considerations for proper extractables study design. Complete information on how to conduct these studies is available in other guidances and publications ***(21,22,49–52)***.

Table 5.4-1 Extraction study types

	Study Type		
	Controlled extraction	**Simulated Extraction**	**Leachables**
Extraction medium	Model solvents, including organic solvents and/or aqueous/organic mixtures	Solvent is representative of the intended formulation	Intended drug product or a buffer without the active component
Temperature/ Storage time	Exaggerated high temperatures	Similar to accelerated stability study	Typical stability study values

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Table 5.4-2 Potential extraction study considerations

Materials to Evaluate	Syringe glass Silicone Needle Needle adhesive Plunger stopper and needle shield elastomers
Extraction Techniques	Reflux Soxhlet extraction Ultrasonication Maceration (soaking)
Parameters to Modify	Time Temperature Solvent composition: • pH • Polarity • Ionic strength • Surfactant concentration • Aqueous/organic solvent mixture ratio

5.5 Analytical Methodology

To compile a complete list of all extractables of the PFS components—and thus of potential leachables in drug or placebo solutions—a wide spectrum of different analytical techniques is recommended. A combination of two or more of the techniques outlined in **Table 5.5-1** may be considered to acquire adequate qualitative and quantitative extraction profiles. The selection and evaluation of the particular method or combination of techniques should be based upon the specific application and materials used in the PFS manufacture and processing.

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Table 5.5-1 Analytical techniques for extractables and leachables evaluation

Technique	Target analyte(s)	Features	Notes
GC (Gas chromatography)	Complex mixtures of volatile or semivolatile organic compounds	Can identify and quantify single components Can analyze volatile organic compounds using headspace-GC/MS on neat sample material	Absolute confirmation of an analyte possible via spectrum and retention time comparisons Easily confirmed with standard compound if commercially available
HPLC-MS (High performance liquid chromatography-Mass spectroscopy)	Nonvolatile (and some semivolatile) organic compounds	Qualitative and quantitative determination	UV/Vis detection is advisable and available in most commercial instruments Commercial MS databases are not readily available
Ion chromatography	Anionic and cationic compounds	Can quantify sulfates, nitrates, or acetate/formates	Can analyze compounds in aqueous extracts and solutions
ICP-MS (Inductively coupled plasma-Mass spectroscopy)	Elements, trace metals	Allows multi-elemental (60 to 80 elements) analysis with semiquantitation in less than a minute	Can analyze elements with low sensitivity or spectral disturbance
ICP-OES (Inductively coupled plasma-Optical emission spectrometry)	Elements, trace metals	Can be used in addition to ICP-MS to detect most cations	May also be performed on aqueous extracts Subject to multiple spectral interferences
AAS (Atomic absorption spectroscopy)	Inorganic and organic-bound silicon	Sensitive Can differentiate inorganic and organic silicon depending on sample preparation	Elements with low sensitivity or spectral disturbance can be analyzed Can analyze several elements, but only sequentially Affected by chemical and physical matrix interference
FTIR (Fourier transform infrared spectroscopy)	Polymers and some additives	Can be performed directly on the neat material using attenuated total reflectance-infrared spectroscopy	Qualitative and quantitative determination of silicone in hexane or toluene extracts is possible ***(51,52)*** Identification of elastomeric material by pyrolysis and infrared spectroscopy is an option ***(57)***

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5.6 Data Evaluation

For extractables studies, a semiquantitative evaluation of substances compared to suitable internal reference substances or relative response factors is considered appropriate. Even so, for each analytical technique, method suitability tests should be conducted during the analysis using a suitable mix of test substances (e.g., stabilizers, antioxidants, filling materials, and element standards). Even for screening studies, an analysis of appropriate spiking samples is recommended to confirm method performance in the given sample matrix.

For evaluation of leachable substances, (e.g., long-term exposure studies), methods should be developed and validated according to suitable guidelines or compendia, such as ICH Q2 (R1) ***(58).*** The current recommendation is to pay particular attention to product and method interference, especially during leachable determinations.

Additional leachable substances may also be identified that were not detected during extractables studies. This may be due to an interaction of leached substances with components of the pharmaceutical matrix (e.g., oxidation/reduction processes, degradation, or formation of drug-extractable complexes). The structure elucidation of such interaction products can sometimes be difficult. Since the compound structure of these substances is sometimes difficult to identify, a threshold approach (**Section 5.7**) should be considered to balance resource requirements in a risk-based approach to compound identification. The use of mass spectrometry (MS) instrumentation with high resolution, different ionization techniques (e.g., chemical or electrospray ionization), or capability to perform multiple MS experiments is sometimes helpful for identification challenges. For example, signals with low molecular masses are often connected to oxidation of polymer additives, such as antioxidants [e.g., BHT or Tris (2,4-di-tert-butylphenyl) phosphite]. In some cases, adequate analytical reference materials for substances like these are not commercially available. If reference materials are unavailable, then consider using a component with a similar structure (e.g., original antioxidant) or additional breakdown products to quantify the leachable substances. If it is not possible to identify the compound, then it is suggested to report MS spectra or other important facts, such as UV/visible spectrometry data (if available). In addition, it is possible to report the chemical substance class of an unknown signal to the extent possible through MS interpretation. A classification of extracted compounds can be proposed based on the level of knowledge collected from the analytical work; compounds can be classified as unknown, tentative, or confirmed structures.

Once leachable substances are identified and quantified, the impact of the leachable substance(s) on the drug molecule must be evaluated. This can be done on an overall basis through long-term stability studies using the PFS components as they are intended to be used. Furthermore, it is possible for leachable substances to interact with the drug molecules in such a way as to cause the drug molecules to oxidize, aggregate, or even form drug-leachable adducts, all of which have the potential to cause an immunogenic response ***(59).***

5.7 Threshold Evaluation

In 2006, the Product Quality Research Institute (PQRI) working group developed the threshold concept for safety considerations of extractables and leachable compounds for the primary packaging of orally inhaled and nasal drug products ***(60, 61).*** Subsequently, a new PQRI working group, the Parenteral and Ophthalmic Drug Products (PODP) Leachables and Extractables Working Group was formed to extrapolate these conclusions to the packaging/delivery systems of parenteral and ophthalmic drug product applications. Although the PQRI-PODP toxicology team's recommendations have not yet been finalized, the group's intermediate recommendations were published in the *PDA Journal of Pharmaceutical Science and Technology* ***(62).***

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The potential approach for the threshold evaluation of the extractables and leachable evaluation for a PFS system (**Appendix IV**) is based on the preliminary conclusions of the PQRI-PODP team ***(62).*** However, the following should not be construed as an endorsement by regulatory agencies, and each manufacturer must apply a sound scientific approach (including expertise from a toxicologist) to evaluate leachables in their products to ensure appropriate thresholds are established prior to seeking regulatory approval.

There are two different levels in which to use the threshold approach to evaluate the safety qualification of pharmaceutical containers and closures ***(61).*** First, a threshold evaluation of the analytical results of a controlled extraction study will facilitate the safety qualification of the constituent parts of the C/C system. Extractables studies on different parts of a PFS often result in long lists of impurities present in a material used in the manufacture of a packaging component. The identification status of a compound can range from confirmed identity with a high level of confidence to only tentative identification and some remaining unknowns. Applying the threshold approach to the results of a controlled extraction study also helps determine the subsequent steps to take in the leachable study (e.g., target leachables to study, additional identification efforts to make, and concentration ranges to consider when performing quantitative leachable studies).

Second, applying the threshold approach to the results of a leachable study could substantially reduce the level of effort in the leachable risk evaluation; leachable compounds present at less than their respective thresholds may not require any further individual safety assessment. Only the leachable compounds present at a concentration level that is higher than their respective threshold may need to be considered for further chemical and risk assessment.

For compounds that have been identified, the PQRI-PODP has defined in its preliminary conclusions three different classes of compounds with three different associated threshold levels outlined in **Appendix IV *(62).*** These three classes of compounds can be converted into analytical evaluation thresholds (AETs) that allow direct assessment of analytical results in relation to their respective toxicological thresholds. The AET is the threshold at or above which a particular extractable and/or leachable should be identified, quantified, and reported for further toxicological assessment ***(63).***

5.8 Conclusion

Extractable and leachable substances in the C/C system must be evaluated to provide an acceptable level of safety to the patient. This can be determined once the leachable substances have been identified and quantified, the safety of the compounds has been established, and the impact on drug product quality and safety has been evaluated (via regular stability studies). If it is determined that lower levels of leachable substances are required, then the drug company may ask its vendor for another component or work with the vendor to develop a process that has decreased levels of the undesirable leachable substance. It must be noted, however, that a change in a component or manufacturing process will necessitate additional assessments to ensure that the desired outcome (i.e., decreased quantities of leachable substances or of a particular substance) has been achieved. When materials or processes are changed, it is necessary to ensure that no additional, unintended leachable substances have been introduced because of that change.

Leachables testing should be conducted along with drug product stability testing. Periodic review of the extractables/leachables package is suggested due to changes in manufacturing processes and analytical technology. The continued process validation concept may be integrated into this approach and aligned with applicable regulatory guidance.

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The goal of this section was to provide a description of how the extractables and leachables profile can be established for a PFS, plunger stopper, needle shield, and assembled PFS syringe device. This effort is required for a drug company to demonstrate the appropriate controls and knowledge of the C/C system expected by regulatory agencies. Even after the initial evaluation of the extractable and leachable substances profile has been completed, the drug company must maintain a system of change controls with the component vendor and drug product manufacturing site to ensure that all changes to the manufacturing and/or processing steps are evaluated for their potential to affect the extractable and leachable substances.

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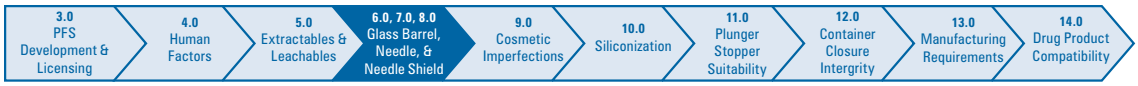
6.0 Glass Barrel Requirements



The dimensions of a 1 mL long syringe are defined in ISO 11040-4, Prefilled Syringes, but may be modified as necessary by the manufacturer or customer ***(64).*** Variations in dimensions are not uncommon since most syringes are used for "platform" purposes and originally designed for a specific use. Different areas of the syringe, including the finger flange, barrel, shoulder, and cone are defined, but can be changed due to handling requirements for use and overall customer requirements. The syringe can be used by manual application or device application (using autoinjectors, safety devices, or other devices).

In most cases, the barrel dimensions are as defined in ISO 11040-4. Changes are not common in these dimensions because most suppliers of glass tubes use this dimension of tubing only for their product. The dimension with the greatest influence is the inner diameter. This parameter affects the filling height, dose accuracy, graduation or dose markings, and dead or hold-up volumes.

ISO 11040-4 can be used as a guideline to define the dimensions; the design of the finger flange and the diameter of the bore inside the cone (related to the use of the specified needle type) should be aligned between customer and manufacturer.

6.1 Flange Geometry

Flange dimensions and tolerances should be defined in the technical portion of the product specification. Flange dimensions and tolerances can influence the main design parameters, including handling, combination with devices, and strength of the flange. ISO 11040-4 describes two types of flanges (cut and round finger flanges) that can be used as guidelines.

ISO 11040-4 provides descriptions for Dimensions and Tolerances, but these may differ according to customer requirements. Dimensions should be verified according to product specifications by inspection controls during manufacturing.

6.1.1 Finger Flange Thickness

ISO 11040-4 describes a flange thickness of 1.9 mm with a tolerance of ± 0.5 mm. Different dimensions may be used based on the definitions and application requirements of the customer and intended use of the syringe. Flange thicknesses in use range between 1.65 mm and 2.2 mm; the most common tolerance is ± 0.25 mm. These should be provided as part of the product specification.

6.2 Flange Strength

The process to define the flange strength, which is dependent on the final use of the syringe, should align with the customer and manufacturer requirements. The required flange strength depends on the mechanical load that the syringes are subjected to during the filling and packaging process. Flange breakage can result in problems, such as equipment downtime, glass particle contamination of the equipment, and additional cleaning steps during the filling and packaging process.

Additionally, it is necessary to consider the application and use requirements of the syringe (manual versus autoinjector application) when determining the appropriate design to use.

For autoinjectors, shoulder-oriented (in which finger flange strength is less important) and finger flange-oriented systems (in which finger flange strength is more important) are often used. For either-configuration, detailed tests are commonly performed to investigate the range of force required for the assembly and application phase (time of use of the syringe).

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Round (noncut) finger flanges have a high degree of planarity and symmetry of form. However, the cutting process can negatively affect the finger flange. To control for this possibility, the technical drawings must specify the tolerances of form, orientation, location, and run-out, in addition to dimensional tolerances.

The measurements and use of devices should be representative of the real impact of the insertion or assembly process ***(64)*.**

It is not uncommon to experience differences in test results due to the occurrence of flange cracks, shoulder cracks, and burst syringe bodies—possibly in combination—rather than on an individual structural basis. Therefore, only specifically trained individuals should conduct an evaluation of the breakage (e.g., to determine its origin).

6.3 Barrel Geometry

ISO 11040-4 defines the glass barrel geometry and describes the inner diameter, outer diameter, wall thickness, inner length, and total length with dimensions and tolerances. Glass with tighter tolerances is available for all dimensions and may be used if required for dose accuracy or small injection volumes.

Both dimensions (cylindrical length and total length to the top of the barrel-glass) must be verified during manufacturing because both influence other parameters and the functionality of the syringe when implemented into a device. The customer and manufacturer agree on precise dimensions based upon capabilities and intended uses.

6.4 Syringe Shoulder Design: Impact on Hold-up Volume

Shoulder design is not described in ISO 11040-4 and is generally based on manufacturers' procedures. Shoulder and cone formation often involve different steps. The shoulder forming operation is only an outside forming step; the inside shape of the shoulder area is not established by a forming tool. Therefore, variations in tolerances and shapes are accounted for in production processes at the converting sites. The effect on hold-up volume should be evaluated and measured in accordance with the existing guidelines and international standards, such as Chapter 2.9.17 of European Pharmacopeia (EP) ***(65)*.**

An adequate overfill can compensate for the hold-up volume of the syringe system. However, product viscosity and the needle itself affect this volume, and the hold-up volume should be verified during design qualification. ISO documents do not provide cone dimensions; however, all manufacturers use the same base for the cone design. Some variation in certain dimensions is possible.

The cone bore design depends on the type of needle used and the specific manufacturing and needle assembly process. The outside dimensions are the same in all cone bores ***(64)*.** Dimensions and tolerances for the shoulder can vary by customer need (**Figure 6.4-1**).

Figure 6.4-1 Glass Syringe Cone with Needle Inserted

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6.5 Additional Considerations

Depending on the product's presentation and fill/finish process, other attributes to consider include barrel compression strength related to the labeling processes, cone tip strength during manual use, and flange compression testing for specific add-on assembly processes. These considerations must align between the producer and the customer based on the application requirements (e.g., manual versus autoinjector application).

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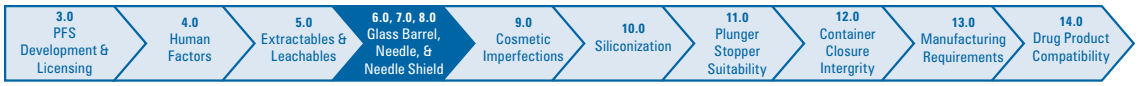
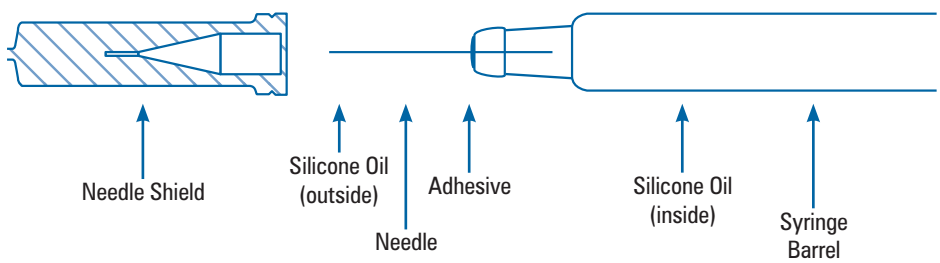
7.0 Needle Requirements



The staked needle used in a PFS for a biotechnology product is made from AISI 304 or AISI 304L stainless steel and covered with a needle shield (**Figure 7.0-1**). The needle is manufactured from an elongated stainless steel tube consistent with applicable requirements in ISO 7864, Sterile Hypodermic Needles for Single Use, ISO 11608-2, Needle-Based Injection Systems for Medical Use—Requirements and Test Methods, and ISO 9626, Stainless Steel Needle Tubing for the Manufacture of Medical Devices ***(66–68)***. The tip is made by grinding the tube, and the bevels are made by changing the angles of the needle to the equipment. The bevel is the slanted surface of the needle tip and can have varying numbers and angularities of tip facets (**Figure 7. 1.1-1**). The needle tip is beveled to create a sharp, pointed tip, which facilitates skin penetration.

The bonding of the needles in the finished syringe barrels is performed in suitable assembling automats. The needles are affixed by a UV-cured adhesive.

Figure 7.0-1 Syringe component diagram



Factors to consider when selecting a needle for a 1 mL long PFS include:

- minimizing injection site experiences or reactions (ISEs)
- needle material compatibility with drug product

These factors are the topic of discussion for this section of the report.

7.1 Minimizing Injection Site Experiences

Injection site experiences (ISEs) are characterized by pain, redness, or inflammation, and can affect the patient's adherence to a treatment regimen. As a result, many studies have focused on evaluating needle designs that could decrease ISEs.

Several parameters may play a role in ISEs, including:

- needle geometry
- needle lubrication
- needle surface smoothness
- needle defects

Sections 7.1.1, **7.1.2**, and **7.1.3** provide a discussion of these parameters.

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7.1.1 Needle Geometry

Injection needles are identified by their bevel, outer diameter or gauge, and length. Needles using thin-walled technology where the inner diameter of the needles is larger than that of standard needles are also available. This technology is included in ISO 11608-2 and allows patients to use smaller-diameter needles without a negative impact on flow rate, speed of injection, or penetration force ***(67, 69)***.

Multiple studies in models and patients have shown a positive relationship between ISE, needle gauge and bevel, as well as a positive relationship between penetration force and pain ***(70–73)***.

Needles for 1 mL long PFSs typically have a three- or five-bevel tip (**Figure 7.1.1-1**). Various studies have compared the designs of three- and five-bevel needles.

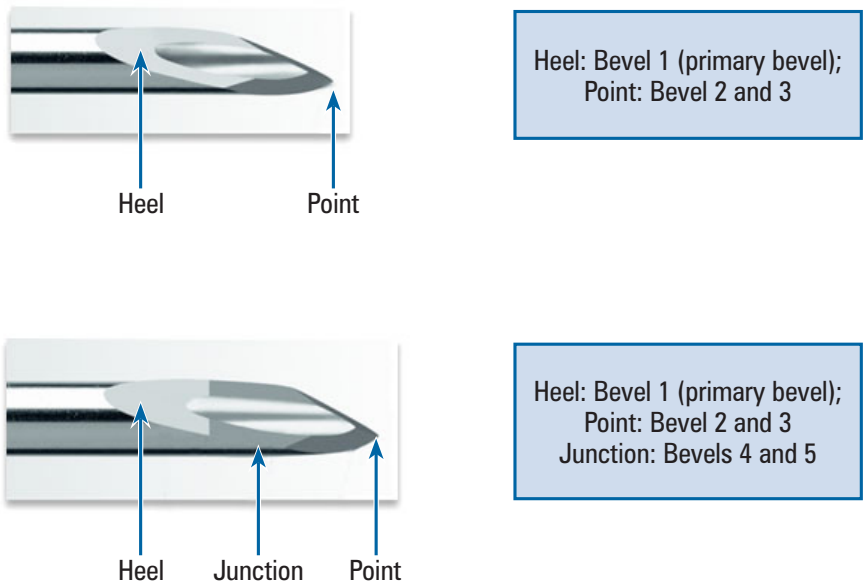

Figure 7.1.1-1 Needles with Three- or Five-Bevel Tips

A number of factors determine needle diameter and length, including desired penetration length (route of administration), drug product formulation characteristics (e.g., pH and viscosity), and patient population. ISO 7864 and ISO 9626 cover needle dimensions and materials ***(66, 68)***. The dimensions of most needles used for 1 mL long PFSs are 27G 1/2 inch and 29G 1/2 inch.

The abbreviations "27G" and "29G" describe the outer diameter of the needle. For example, a 27G needle has a nominal outside diameter of 0.40 mm, whereas a 29G needle has a nominal outside diameter of 0.33 mm. The outer diameter does not indicate the inside diameter of the needle ***(66, 68)***.

In cases where the manufacturer selects a needle with dimensions that deviate from ISO standards to improve product performance, the manufacturer should demonstrate the needle's fitness for its intended use through design control activities ***(74)***. The manufacturer should also identify designs that deviate from ISO standards in product designation and customer technical documentation.

The 1/2 inch in the needle descriptor refers to the exposed needle length or the part of the needle outside the needle channel (**Figure 7.1.1-2**). Half-inch-long needles are commonly used for subcutaneous application. However, needles of other lengths may also be available.

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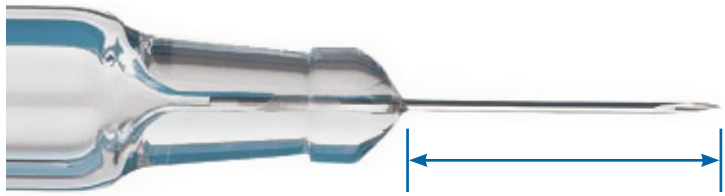

Figure 7.1.1-2 Exposed Portion of Needle

Needle diameter and length may affect the injectability (ability to inject the product) of the syringe; injectability is mainly characterized by the time, pressure, and force required for completion of an injection at a constant flow ***(75)***.

A needle with a smaller outside diameter may require lower penetration force. Such a needle, however, may also result in longer injection times and higher extrusion forces. Therefore, the inner diameter of a needle or its wall thickness is a factor for consideration. Currently, some 27G needles have a larger inside diameter. A 27G normal-walled needle has a nominal inside diameter of 0.20 mm; a 27G thin-walled needle has a nominal inside diameter of 0.26 mm. Therefore, the inside diameter of a 27G thin-walled needle is 30% larger than that of a 27G normal-walled needle. In a study that compared pen needles of similar outer diameters but different inner diameters, a 49% reduction in flow resistance was observed with the needles that had larger inner diameters ***(76)***.

The drug product formulation, including volume and viscosity, also becomes critical in selecting needle diameter and length to minimize ISE risk (**Figure 7.1.1-3**). For the rapid delivery of large volumes or high-viscosity products, larger needles may be required or the injection time may increase if a small-diameter needle is chosen. If the needle's outer diameter is increased to decrease injection time and force, then ISE risk could increase.

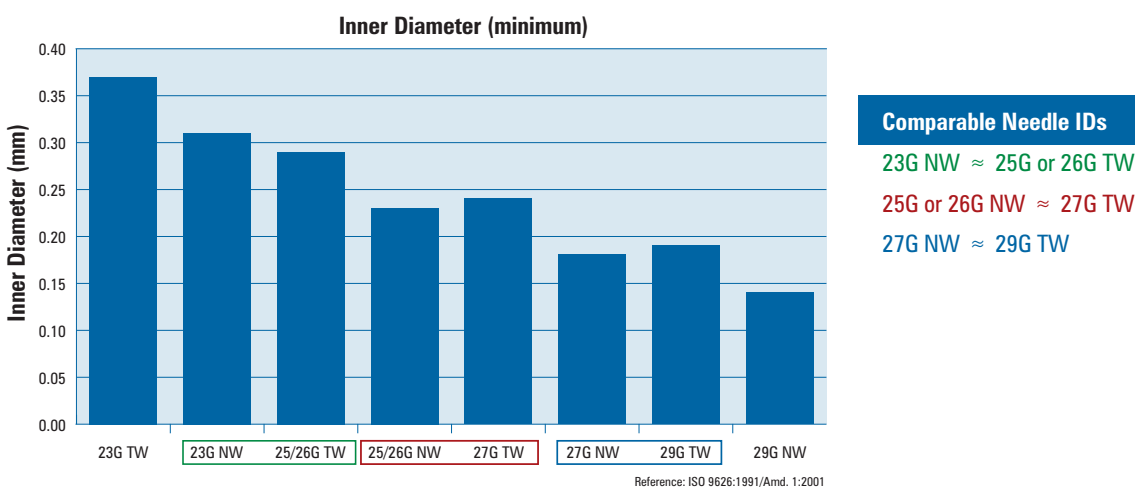

Figure 7.1.1-3 Inner diameters of various needle gauges

Using first principles of engineering, a model has been developed to predict the potential impact of critical parameters, such as needle length and inner diameter, on extrusion force or injection time ***(76)***. Based on the Hagen-Poiseuille equation, the authors of this model predicted that the injection forces would decrease as the inside radius of the needle increased. The model has been evaluated using different syringe types and vendors by measuring the pressure drop for a liquid flowing across the syringe barrel and the needle at a constant rate to determine the impact of several parameters on

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injectability. The researchers found that the inner diameters of the needles can greatly affect injection times and forces.

Thin-walled technology may be used to maintain the functionality of the needle with respect to flow rate, speed of injection, and the force applied on the syringe and drug solution ***(69).*** When designing a needle, a balance must be achieved between the drug product formulation, desired penetration length, injection time, and needle dimensions to minimize ISE risk and increase patient physical comfort and adherence to the treatment.

7.1.2 Needle Lubrication

Staked needles are treated to reduce penetration force during insertion. This treatment takes place after washing and immediately prior to mounting of the needle shield. Penetration force may be assessed following the requirements of Deutsches Institut für Normung (DIN) 13097-4 ***(66, 68)***.

The typical lubricant is medical-grade silicone oil with a viscosity of 12,500 cSt. Applying the silicone occurs by wiping the needle with a sponge carrying silicone oil before assembling the rubber needle shield. The needle tip may pierce the needle shield during the process resulting in removal of silicone oil from the needle tip or compromising CCI. Removal of the silicone from the needle tip can be mitigated with the use of alternative silicone formulations and application methods (i.e., humidity-cured and heat-cured formulations). Proper design of the components is required to avoid breaches in CCI. Intelligent (advanced) needle-shield systems in which the needle tip is free inside the needle shield are available.

For bulk syringes, the pharmaceutical company or the contract filler applies the silicone to the needles (**Section 10.0**)**.** Needles are attached to the barrel prior to siliconization and checked for plugs or clogs. For presterilized syringes, the syringe manufacturer applies the silicone to the needles after the assembled syringes are washed and before the needle shield is mounted onto the syringe. An evaluation to ensure that lubrication is present and effective is necessary. Syringes are then packed into nests and tubs, and subjected to sterilization.

7.1.3 Needle Defects

Typical needle defects are axial deviations (from the longitudinal axis of the syringe), damage or dulling at the tip, and clogs. If the needle has an axial deviation and is therefore positioned at an angle in the needle shield, then there is a risk of formation of rubber particles from the needle shield (fragmentation). In the worst case, the needle penetrates the needle shield sideways resulting in a loss of sterility of the entire system. Damage to needle tips (so-called "hooks"), dull needle tips, and burs can result in painful injections (**Figure 7.1.3-1**)**.**

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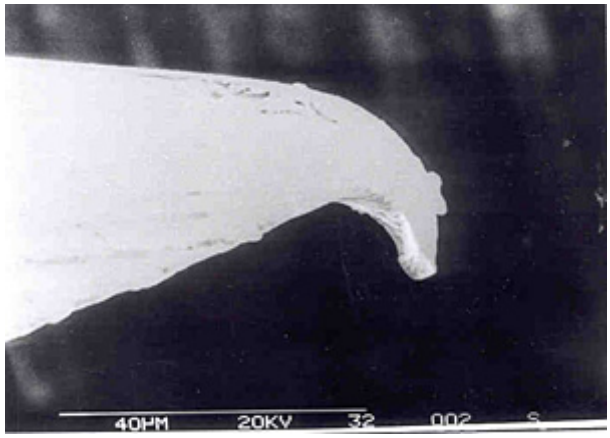

Figure 7.1.3-1 Scanning electron micrograph of a needle hook

Needle polishing is an important step in needle manufacturing because it removes burs generated during cutting and grinding.

The needle-shield material may be important in the protection of the needle by decreasing the likelihood of needle bending ***(69).*** The creation of a "hook" due to needle bending is a known defect and can increase ISE risk.

Section 14.3 provides a discussion on clogs—another defect caused by several factors. This type of defect affects the injectability of the drug product.

The construction material of the needle shield—whether rigid or nonrigid—should be evaluated during needle design. Additional factors to consider should include the smoothness of the needle and any possible blockage of the needle. Needle, needle shield, and syringe manufacturers must ensure that processes are in place to control the potential for defects (**Section 9.2**)**.**

7.2 Material Compatibility with Drug Products

There are several needle-related compounds that can affect the purity, efficacy, and safety of the drug product as discussed in **Section 5.0.** These compounds include metal ions, silicone oil, the adhesive used to connect the staked needle to the barrel, and tungsten (W salts residues). The compatibility between these compounds and the drug product is assessed during long-term stability studies.

7.3 Conclusion

There are multiple factors to consider when selecting a needle for a 1 mL long PFS, including minimizing ISE risk to increase patient comfort based on the desired penetration length, injection time, injection force required, and compatibility with the drug product. A careful balance between all of these factors should be evaluated for each drug application.

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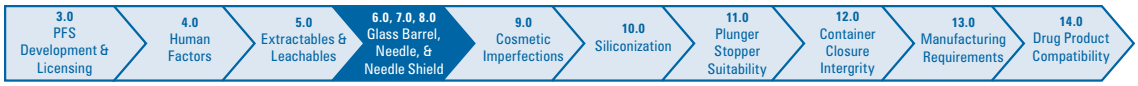
8.0 Needle-Shield Requirements



This section describes the functions, materials, and formulations used by manufacturers and suppliers to develop the needle shield.

8.1 Functions of the Needle Shield

The function of the needle shield is to cover the needle, protect the needle mechanically, minimize risk of needle contamination from the external environment, and keep the sharpness of the needle from being compromised. Because the sharp end of the needle is embedded in the elastomeric needle shield, the needle shield serves as a seal that prevents leakage or microbial ingress through the needle opening. In addition, the interference fit between the elastomeric needle shield and the glass cone provides a seal against microbial ingress into the external surface of the needle. This interference fit also provides resistance to separation of the needle shield from the syringe tip during normal handling and distribution. Furthermore, the seal isolates the needle from external contaminants.

In the market, two versions of the needle shield are in use: the soft or flexible needle shield, which consists only of the elastomer, and the rigid needle shield (RNS), which has a plastic cover over the elastomer. The RNS has higher impact resistance, and pulling it off is more intuitive for the user. Some RNS designs offer additional mechanical resistance to needle shield separation from the needle tip or to provide tamper evidence. **Figure 8.1-1** provides schematics of the two types of needle shield for reference.

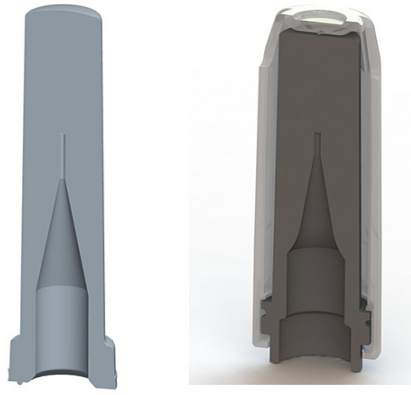


Figure 8.1-1 Nonrigid (soft) Needle Shield (L) and Rigid Needle Shield (RNS) (R)

In many cases, the syringe is provided ready-for-filling. The nonrigid needle shield or RNS material and design must be compatible with the EO sterilization process typically used for sterilization. In situations where empty syringes are supplied in bulk, the pharmaceutical company washes, assembles (siliconized) with a needle shield, and sterilizes the syringes. The needle-shield materials must be compatible with the sterilization process, which typically consists of autoclaving by steam.

The elastomeric part of the needle shield can make contact with the product after filling. Therefore, the manufacturer must demonstrate that any needle shield elastomer material is compliant with safety regulations and compatible with the drug product. The design and installation of the elastomer needle shield must be confirmed to ensure satisfactory CCI of the needle end of the filled assembled product throughout the life cycle of the drug product. For details, refer to **Section 14.3** of this report for information on compatibility requirements, and **Section 12.0** for details on CCI requirements.

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Often, a PFS product is attached to an add-on device, such as a safety device or autoinjector. The dimensions of nonrigid needle shields and RNSs should be compatible with the devices and machines that assemble them. These details are not within the scope of this technical report.

8.2 Material Property Requirements

Needle shields and the elastomeric parts of RNSs are typically made from thermoset rubber or thermoplastic rubber formulations. A thermoset rubber formulation is a polymer system in which the basic elastomeric properties are derived from covalent bonds between adjacent polymer chains. These chemical bonds are referred to as "cross-links" and are typically created through a high-temperature molding process. In addition, the elastic properties are influenced by interfaces among filler and polymer materials and the incorporation of wax and plasticizer ingredients, when required for product functionality or improved processability. Chemical cross-links are considered to be irreversible following the molding step.

The industry uses two families of thermoplastic elastomer formulations. These formulations include materials from the styrene block copolymer and thermoplastic vulcanizate (TPV) families. Thermoplastic elastomers based on the styrene block copolymer family are traditionally composed of a thermoplastic elastomer, such as styrene-isobutylene-styrene, styrene-butadiene-styrene, or styrene-ethylene-butylene-styrene; a thermoplastic polymer, such as polypropylene or polyethylene; paraffin or mineral oil; fillers, colorants, and antidegradants.

Thermoplastic rubbers (TPRs) based on the styrene block copolymer family derive their elastic properties from hard polystyrene domains that act as physical cross-links. Following the molding step, this kind of TPR can be reprocessed if necessary. TPRs from the TPV family are made of a thermoset rubber/thermoplastic polymer mixture, such as ethylene propylene diene monomer (EPDM) and polypropylene. Within TPV polymer systems, the rubber phase is highly cross-linked by means of traditional thermoset-curing systems, such as peroxide or sulfur. TPVs obtain their elastic properties from the cured EPDM segment and the interaction between the thermoset rubber and the thermoplastic polymer segment. Similar to thermoset rubber formulations and TPRs from the styrene block copolymer family, TPVs traditionally contain various other materials, including cross-linked EPDM, polypropylene, paraffin oil, antidegradants, and colorants.

Elastomer formulations are unique to the component and manufacturer. The formulations are trade secrets of the manufacturer. Formulations of different manufacturers are not chemically equivalent, and switching from one formulation to another typically requires requalification efforts, including drug product stability and regulatory filing.

Some needle shield materials have been known to be susceptible to embrittlement or cracking (e.g., through ozone attack). The selection of needle shield material should account for such aging properties. Additionally, consideration of needle shield and needle interactions that may result in corrosion should also be included when selecting the needle shield materials.

For a list of typical rubber ingredients other than the polymer and of the properties and requirements of elastomeric formulations, see **Section 11.1.1.** For sterilization purposes, needle shields typically require a sufficiently high level of gas and water vapor permeability.

The material of the elastomer needle shield must be compatible with the EO or steam sterilization process of the subassembly that consists of the barrel, needle, and the needle shield (nonrigid or RNS). Ethylene oxide or steam must be able to permeate the wall of the elastomer needle shield in order to sterilize the surface of the needle and the interface between the needle shield and glass cone.

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Moreover, residual EO must be able to move out of the elastomer needle shield at the end of the sterilizing cycle. Therefore, the elastomer needle shield material must have sufficiently high permeability for steam and gas. Needle shield elastomer formulations suitable for this process are dedicated polyisoprene and styrene-butadiene formulations, as well as certain thermoplastic elastomer formulations. Meeting this high permeability requirement is difficult for elastomer formulations used for vial closures and syringe plunger stoppers.

The requirements for compatibility with the EO or steam sterilization process are the same for the elastomeric part of a nonrigid needle shield or RNS. For the RNS, the plastic part must be designed in such a way as not to interfere with the accessibility of EO or steam to its elastomeric part ***(55)***.

8.2.1 Needle-Shield Extractables Profile

Elastomeric needle shield formulations are chosen to comply, at a minimum, with the requirements of USP <381> Elastomeric Closures for Injections, and European Pharmacopeia 3.2.9, Rubber Closures for Containers for Aqueous Parenteral Preparations for Powders and for Freeze-Dried Powders ***(21, 77)***. USP <381> requires conformity of the closures as they are shipped by the closure supplier and in their final ready-to-use state. Materials used for the "rigid" portion of the needle shield should be compliant with USP <381> and ISO 10993, Biological Evaluation of Medical Devices ***(54)***.

To minimize the risk of a negative impact of leachables from the needle shield elastomer on the quality of the drug product, the customer (e.g., pharmaceutical company) and the syringe supplier should collaborate to obtain the potential extractables profile for the elastomer. The customer should then examine the list with the toxicology group and complete the necessary studies. Refer to **Sections 5.3** and **14.0** for details on compatibility requirements.

8.2.2 Needle-Shield Leachables Profile

While the contact area between the elastomeric part of the nonrigid needle shield or RNS is small, this does not preclude migration of extractables from the elastomer needle shield into the pharmaceutical product once the syringe is filled. A number of such migration cases have been reported as leachable cases in PFS drug formulations. This possibility must be taken into account during drug product development. Refer to **Sections 5.3** and **14.3.1** for additional details.

8.2.3 Needle-Shield E-beam Compatibility

When e-beam surface sterilization is employed at the form/finish line, the manufacturer should select needle shield (nonrigid or RNS) materials that have been confirmed to be compatible with the equipment and process. Some aging evaluation may be warranted because the impact of radiation on elastomers could be delayed. Similar considerations should be taken into account if another "surface sterilization" approach is used.

8.2.4 Needle-Shield Elastomer Cleanliness

The elastomer needle shield should not contaminate the needle. Prior to assembly of the barrel, the elastomer should be appropriately cleaned so that fibers or other contaminants are not introduced.

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8.3 Mechanical Property Requirements

8.3.1 Needle-Shield Pull-off Force

Before injection, the needle shield (nonrigid or RNS) must be removed manually or in a semiautomated or automated fashion (i.e., using an autoinjector). The force to remove the needle shield (nonrigid or RNS) from the syringe tip is known as the "needle-shield pull-off force."

Depending on the type of elastomer formulation, sterilization treatment, design and other characteristics, the pull-off forces can vary from type-to-type. Therefore, it is recommended that the system supplier and customer (usually the pharmaceutical company) agree on the minimum and maximum pull-off forces for the needle shield (nonrigid or RNS). The customer can use incoming inspection of the needle shield pull-off force in empty syringes at appropriate frequency to check the robustness of the supplier's process For further details, refer to ISO 11040-4 ***(64)*.**

A specification for final product performance must consider the end user's therapeutic area and any requirements associated with combination product devices, if applicable. The specified pull-off force range or value for the empty syringe should guarantee the tightness on the syringe during the filling process, downstream processes, and storage throughout the product's shelf life.

It should be cautioned that sometimes test results from empty syringes do not necessarily correlate with the results for the final filled syringe system. It is also important to establish agreement among the suppliers and pharmaceutical companies on a control strategy to minimize risk.

For RNSs, the separation force of the plastic cap from the elastomer shield must be higher than the pull-off force of the complete RNS assembly from the syringe barrel to ensure proper functionality and avoid detachment of the plastic cap from the elastomeric needle shield during preparation for injection.

8.3.2 Influence of Needle Shield on Needle Sharpness

The sharpness of the needle tip is not likely to be reduced by assembly of the needle shield (nonrigid or RNS), although the sharp end of the needle is embedded into the elastomeric needle shield. Testing the sharpness of the needle should be in accordance with ISO Method 11040-4 and confirmed by the syringe supplier or customer/filler during incoming inspection. The quality agreement between the syringe supplier and the customer (e.g., a pharmaceutical company) should include additional controls, if desired.

8.4 Container Closure Integrity Requirements

8.4.1 Empty Syringe

During validation of the syringe manufacturing process by the system supplier, an appropriate statistical sampling plan should be used to assess the seal of the needle shield (nonrigid or RNS) and maintenance of sterility during transportation and storage of the sterilized syringe. A statistical sampling plan may be used during both the supplier's outgoing inspection and the customer's incoming inspection to evaluate empty assembled syringes for critical defects, such as tears or cracks of the needle shield (nonrigid or RNS), needle protrusion through the needle shield (nonrigid or RNS), and dimensional nonconformities that could jeopardize CCI. The quality agreement should address the sharing of steps implemented to ensure integrity of the CCI.

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8.4.2 Filled Syringe throughout the Life cycle

The needle shield (nonrigid or RNS) must not exhibit surface cracks or other surface imperfections visible to the naked eye. Application of suitable methods should ensure the CCI of the PFS throughout the product's life cycle. Refer to **Sections 12.4** for requirements related to filled and finished syringes, and common methods for CCI testing and assurance.

8.5 Dimensional Requirements

A PFS can be used without additional attachments, such as a safety device or an autoinjector. However, a PFS is often further assembled with add-on devices to meet various customer needs. Compatibility among the dimensions of the assembled syringe, including the needle shield (nonrigid or RNS) and the add-on devices is, therefore, a critical requirement for successful assembly and proper functioning of the final combination product. This subject is outside the scope of this technical report, but it should be noted that such alignment of the various dimensions should occur during the development program before the design of the add-on device or the needle shield (nonrigid or RNS) is finalized.

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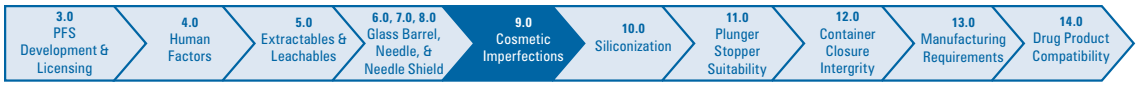
9.0 Cosmetic Imperfections



The focus of this section is on visual imperfections in unfilled glass syringes with staked needles and other components (i.e., needle shields, plunger stoppers, and plunger rods). Dimensional nonconformities, printing imperfections, and defects in packaging are out of scope for this section. Visual inspection to detect imperfections on filled glass syringes is described in **Section 13.4**.

Cosmetic imperfections in PFS components may increase the defect rate detected in the final visual inspection of the filled containers, so each component contributes to the overall cosmetic result of the filled syringe at the end. Imperfections not detected during the final inspection (i.e., imperfections with a direct impact on container integrity or drug integrity and delivery) may lead to severe issues in the market, including risk to the patient during use.

Each part of the process outlined in **Figure 9.0-1** can create specific defects, but all processes can have a common list of identified and classified defects.

Figure 9.0-1 PFS Assembly Steps

The PFS system can be divided into two subassembly parts:

- **Primary components** in contact with the drug (glass barrel, plunger stopper, and tip closure)
- **Secondary components** (plunger rod, backstop, and safety system in the combination product) interacting with the primary components and providing complete functionality to the healthcare provider

The defect evaluation list should be available and comprehensible to all stakeholders (manufacturers and suppliers). Different approaches to classify various imperfections exist, and companies should have the capability and applicable methods in place for inspection. **Table 9.0-1** presents a cosmetic defect tracking model for classification and identification through applicable standards or guidelines. An overview of visual inspection and evaluation is presented in **Section 9.3**.

Table 9.0-1 Available Guidance and standards for PFS components

			Visual inspection			
Material	Guidance/ Reference	Comments	Supplier	Incoming Inspection	In-Process Control	Final Control
Glass tube	Not Available	Defect evaluation list agreed on by glass tube producers and glass converters	•	•	•	•
Bulk Syringes	PDA *Technical Report No. 43*	Defect evaluation list agreed on by glass converters and pharmaceutical company	•	•	•	•
Ready-to-use Syringes	PDA *Technical Report No. 43*	Defect evaluation list agreed on by glass converters and pharmaceutical company	•	•	•	•
Rubber component	Not Available	Adaptation required due to the specific criticality of pharmaceutical use	•	•		•
Plastic component	Not Available	Applicability of industrial standard	•	•		

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A final visual inspection should be included in the process to ensure adequate quality output at the completion of syringe manufacturing. Any defective syringes found will result in a nonconformance requiring root-cause investigations of the unexpected event. The results of the root-cause investigation can be ultimately used to allow use of a given syringe showing an individual defect for delivery of end-user medication.

As noted above, imperfections not detected during the final inspection may lead to severe issues in the market linked to risk to the patient during use. To minimize this risk, pharmaceutical companies and producers of PFSs should implement agreed-on processes and methods to minimize cosmetic imperfections.

9.1 Defect Identification

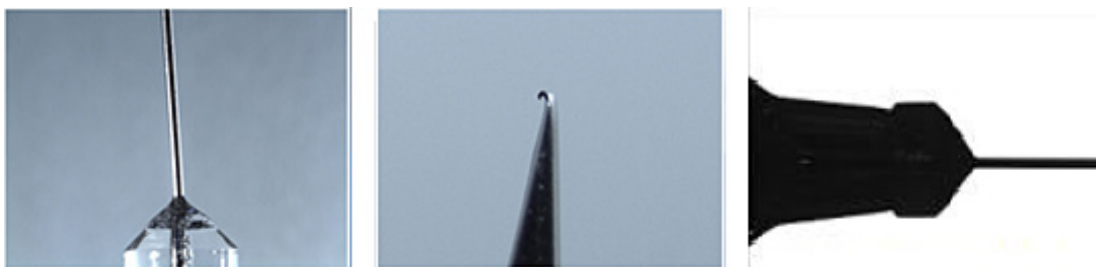
To ensure defective syringes with cosmetic imperfections relevant to the quality of the drug product are identified and addressed, imperfections need to be classified and robust processes implemented to ensure that such syringes are not used for patient medication. To accomplish this, PDA *Technical Report No. 43: Identification and Classification of Nonconformities in Molded and Tubular Glass Containers for Pharmaceutical Manufacturing* provides basic guidelines on the definition of the imperfections ***(33).*** Currently, there is no pharmaceutical guidance concerning elastomeric parts defects or plastic components. However, some basics can be found in rubber technology standards ***(21).***

The supplier of the PFS components and the pharmaceutical company should determine specifications that reflect agreement on the acceptance levels for individual cosmetic imperfections. To reach agreement on acceptance levels, the PFS supplier and the pharmaceutical company should define and agree on the limit samples that may be required.

Pharmaceutical companies and suppliers of PFSs should align their methods for identification of cosmetic imperfections. Visual inspections can be performed manually or by automatic camera systems.

9.2 Defect Classification

Currently, the defect evaluation list may not be supported by pharmaceutical standards or requirements. However, the industrial production references can be a source of defect definition and help define a common language for suppliers and users. **Figure 9.2-1** illustrates some defects that may occur during syringe production.



Bent needle **Hook needle** **Glue deposition profile**

Figure 9.2-1 Sample needle defects

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Based on PDA *Technical Report No. 43*, each nonconformance is judged using the criteria of critical, major A, major B, minor, and not applicable (N/A). The following list provides the definition for each category of nonconformance:

- **Critical:** Likely to result in personal injury or potential hazard to the patient; assigned to any nonconformity that compromises the integrity of the container and could lead to microbiological contamination of a sterile product
- **Major A:** Leads to serious impairments, such as a malfunction that makes the packaging unusable
- **Major B:** Leads to impairments of a lesser degree, such as reduced efficiency in production
- **Minor:** Does not affect product quality or process capability
- **N/A:** Of minimum size, and therefore acceptable
- **Limit sample:** Actual physical unit that shows the nonconformance and represents the maximum degree of acceptability

N/A could be applied to nonconformities that may arise out of a glass manufacturer's process-related or a non-process-related error that is not sampled or estimated statistically, and therefore should not be assigned an AQL.

Nonconformities could be reviewed and classified (with a rationale) using an ICH Q9 quality risk management approach for risk assessment. Verification of nonconformity classifications may warrant additional testing to establish their rationale.

9.3 Visual Inspection of Unfilled Syringes

9.3.1 Manual Visual Inspection

Two methods for performing a manual visual inspection are inspection by means of the naked eye or through use of magnifying optical systems**.** In both cases, the ability to identify the defect correctly is influenced by the qualifications and experience of the personnel performing the manual inspection. Personnel must be periodically trained to meet the constantly increasing demands of productivity and product quality.

Additionally, a suitable support capable of rotating the syringes in order to view all sides provides a suitable method to inspect the syringes without creating additional defects. This can be managed by avoiding direct contact with the sample by means of manipulation using a suitable device or by direct use of the support in which the sample is contained (e.g., rondo tray or nest).

The apparatus usually includes a suitable white light source, and the illuminated field permits inspection of the entire surface of the syringe. The use of a dark panel and the ability to vary the intensity of light enable detection of some defects that would be difficult to identify if illuminated by white light only. Other detection devices include magnifying systems, such as lenses or magnifying opticalmicroscopes. Operators should promptly classify and record the fault when identified.

9.3.2 Automated Visual Inspection

The rapid development of modern manufacturing processes has led to an increase in automation and production rates, and fostered the development and adoption of increasingly sophisticated detection systems. These systems reduce human error during visual inspection. Automated inspection systems are able to identify and recognize syringe defects normally identified through manual inspection by an operator. A new detection system should undergo a thorough analysis first to determine under what situations the system fails to detect the various types of defects.

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The most commonly used automated detection system is equipped with high-intensity focused lights designed to highlight a specific area of the container and is supported by software. The software uses image algorithms to identify defects without contacting the samples, which reduces the inspection work required of operators. Multiple areas of inspection (lateral surface, bottom, and front surface) allow the acquisition of different images in different areas of a syringe, permitting easier identification of specific types of defects (**Figure 9.3.2-1**).

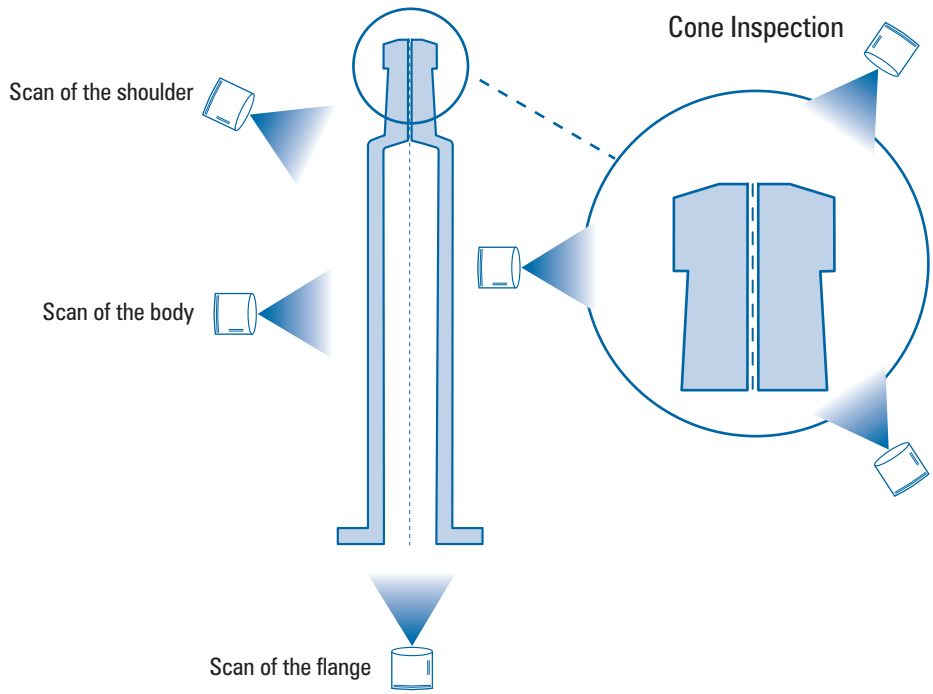

Figure 9.3.2-1 Example of automated visual inspection of an empty glass barrel

Configurable detection criteria levels for each inspection area permit optimal identification of defects on the barrel and the needle in relation to the requirements agreed by the supplier and pharmaceutical company. The automatic camera system should record all statistical parameters for later use in specific reports, certificates for internal use, or to share with the customer.

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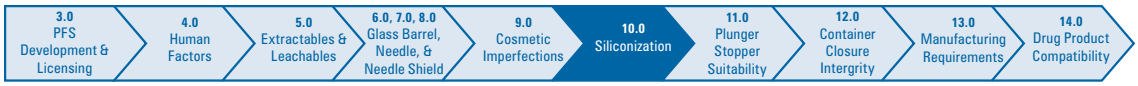
10.0 Siliconization



Major suppliers of glass syringes have advanced capabilities now to provide a wide variety of siliconization solutions for lubrication purposes. The reader is advised to work closely with the supplier to identify the appropriate solution. This section provides a collection of best practices for siliconization from industry leaders. Alternative methods for achieving lubrication and silicone oil-free systems are not discussed here.

Current standard glass syringe systems rely on lubrication for syringe functionality. Lubrication enables the plunger stopper to move in the glass barrel when the plunger is depressed. The standard method of lubrication is to apply (liquid) silicone oil to the plunger stopper and barrel. In some functional cases, the silicone oil layer acts as a barrier to prevent interactions or side reactions with the glass. In addition to the plunger stopper and barrel, the supplier lubricates the needle to facilitate needle insertion and retraction.

Overall, the goal of lubricating glass syringe systems is to apply the optimal quantity of oil without exposing the patient to unnecessary processing aids, and yet apply enough oil to guarantee process functionality throughout the product's shelf life. Additionally, potential interactions of silicone oil with rubber compounds and the drug product must be assessed as part of the stability and functional studies.

10.1 Type of Silicone and Areas of Application

The silicone oil (polydimethylsiloxane) used for lubrication must be of medical grade and appropriate viscosity. Application of the silicone oil can occur in various viscosity ranges (typically 1,000 and 12,500 cSt) in compliance with monographs from the USP, European Pharmacopeia, Japanese Pharmacopeia, and the regulations of other countries as required ***(78,79,80).***

10.1.1 Empty Staked needle Syringe Barrel

Processing to date involves spraying silicone oil, either with a static spraying nozzle (spraying into the barrel from the finger flange), or by means of a more advanced, dynamic-diving nozzle that enters the syringe barrel and distributes silicone oil more evenly. Spraying with dynamic nozzles results in a very even distribution of the silicone oil, an overall reduction in the quantity of the silicone oil used, precision dosing, and maintains low and consistent break-loose and gliding forces. Diving nozzle systems improve the control of siliconization, which can be tailored to improve not only the syringe's functional performance, but also its suitability for filling and finishing processes.

To optimize the filling process, it is possible to program the silicone oil application system to apply the silicone oil in specific patterns. For example, vacuum plunger stopper placement may require more silicone near the flange than in the product area. The specific profile for this requirement can be programmed with a diving nozzle system. Similarly, for mechanical placement (e.g., vent tube placement), the profile can be optimized to deliver more silicone oil where the plunger stopper is placed.

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10.1.2 Effects of Fixed-nozzle Systems

Figure 10.1.2-1 depicts a siliconization profile based on a fixed-nozzle system. The syringe is oriented as shown below. The different sections (based on the setup of the ZebraSci® system) show the uneven silicone profile inside the syringe based on the technology used. Sections 1, 2 and 3 have the most silicone as evidenced by the "Schlieren" patterns (optical interference due to local variations of silicone oil thicknesses, which becomes observable when combined with a specific background pattern). Section 4 is the first section covered with only a small amount of silicone oil. Single droplets are detectable in this area. The single droplets in sections 5 and 6 cannot spread all over and build up a homogeneous silicone layer inside. Section 7 demonstrates the major disadvantage of a fixed-nozzle system because only single droplets are located in this area.

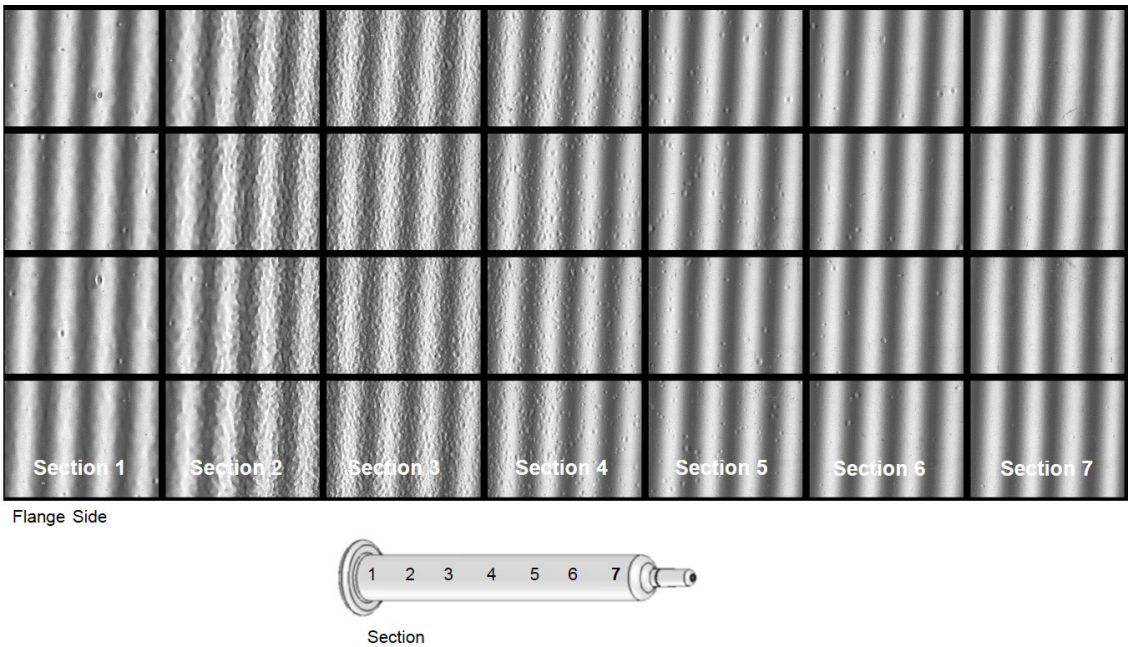

Figure 10.1.2-1 Fixed-Nozzle Siliconization (injected mass 0.8 mg, image based on ZebraSci® system)

Figure 10.1.2-2 reflects a typical break-loose and gliding force profile of a syringe siliconized with a fixed-nozzle system. The graph shows the force results of syringes with identical siliconization profiles to that of the ZebraSci® system in **Figure 10.1.2-1**.

Break-loose force and the first millimeter of the path have good gliding force results. Section 3 of **Figure 10.1.2-1**, which represents approximately the area of a 25 mm path, shows the increase in gliding force due to the reduced silicone overlay in this area and worsens through the end of **Section 7.**

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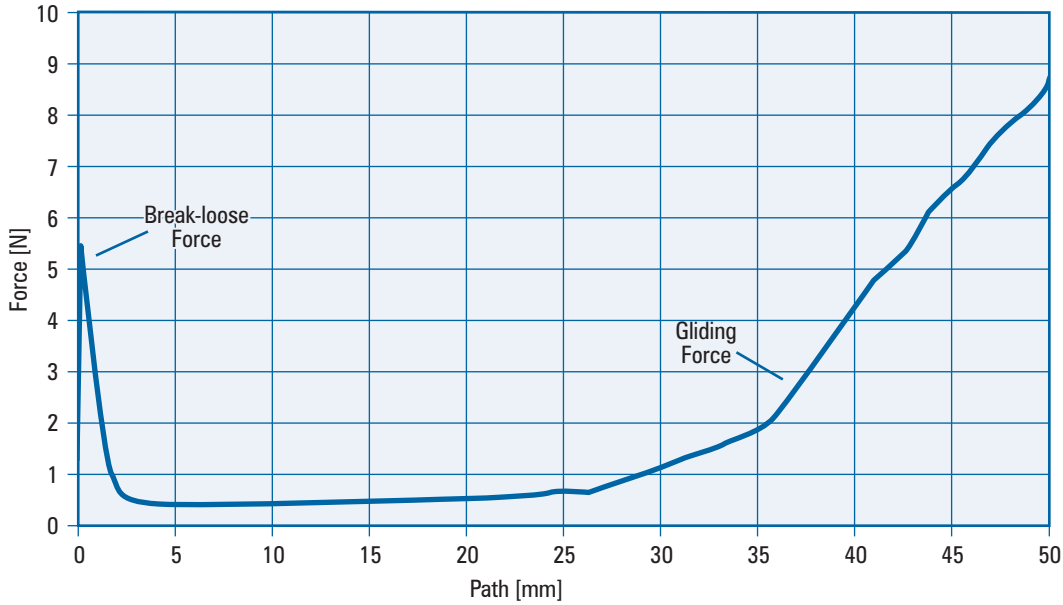

Figure 10.1.2-2 Fixed-Nozzle Siliconization – Break-Loose and Gliding Force Profile

10.1.3 Effects of Diving Nozzle Systems

Figure 10.1.3-1 shows a siliconization profile based on a diving/movable nozzle system. The syringe is oriented as shown. The different sections (based on the setup of the ZebraSci® system) show the even silicone profile inside the syringe based on the technology used. The large droplets in Section 1 (finger-flange side) are a result of the laboratory-based siliconization system and are not typical. The syringe was washed and transferred to the laboratory. All sections show the same profile of silicone distribution (e.g., spraying parameters and velocity) and diving parameters based on a special design of parameters for this type of syringe.

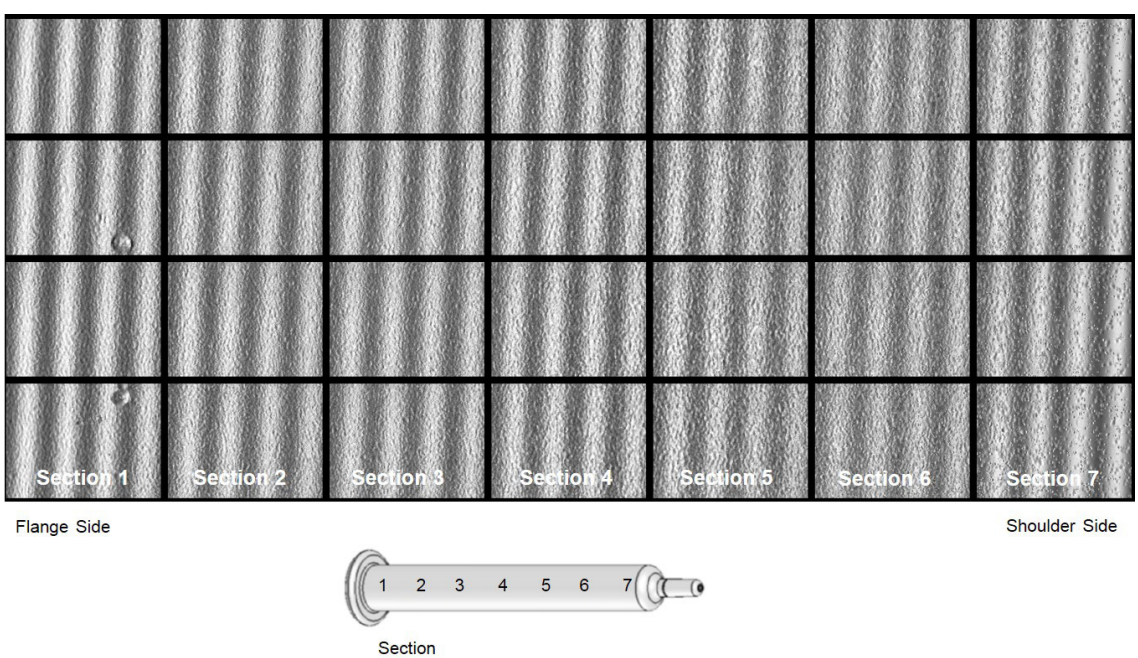

Figure 10.1.3-1 Diving Nozzle Siliconization (image based on ZebraSci® system)

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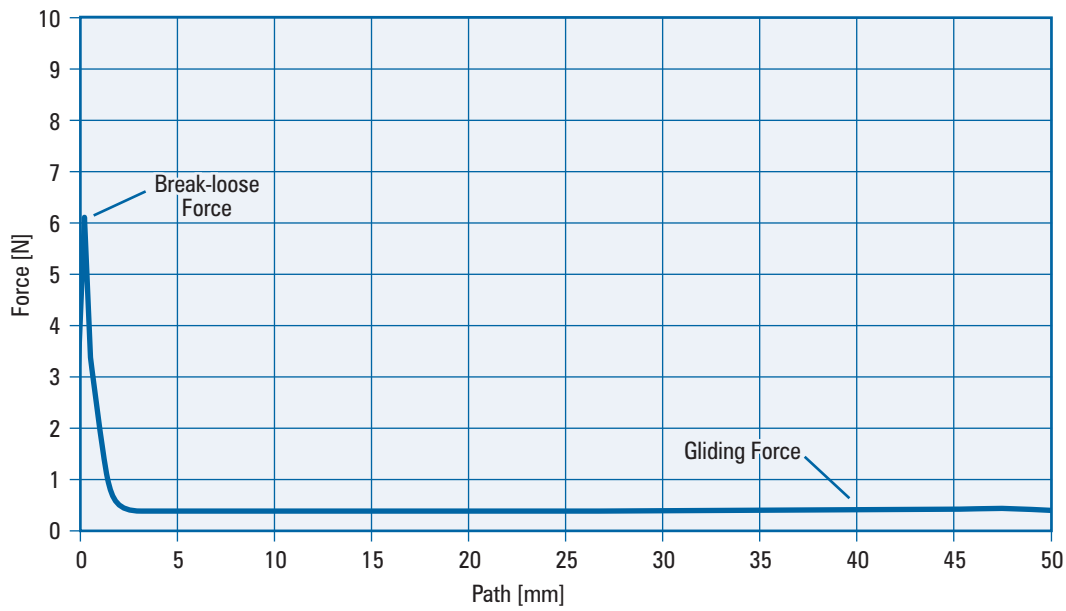

Figure 10.1.3-2 Diving Nozzle Siliconization: Optimal Break-loose and Gliding Force Profile

Figure 10.1.3-2 shows the optimal break-loose and gliding force profile of a syringe siliconized with a diving/movable nozzle system. The graph demonstrates the force results of syringes with identical siliconization profiles shown by the image based on the ZebraSci® system (**Figure 10.1.3-1).** There is a more homogenous distribution of silicone within this syringe than the one depicted in **Figure 10.1.2-1** resulting in a constant gliding force.

10.2 PFS Barrel and Plunger Stopper Combination

The combination of a well-lubricated barrel and plunger stopper as defined in the user requirement ensures consistent force for delivering the drug product during injection and easy assembly of the plunger into the syringe barrel.

Typically, plunger stoppers are lubricated with silicone oil. The lubrication serves multiple purposes by ensuring machinability (in vibrator bowls and tracks), lubrication of insertion tubes, and plunger stopper functionality in the syringe.

The oil for the plunger stoppers is commonly added during the final washing steps of component preparation. Studies are necessary to establish suitable siliconization levels that balance machinability and break-loose/extrusion force. Silicone oil can also contribute to the level of subvisible particles in the drug product solution and may affect subvisible particle test results. Subvisible particles are routinely monitored during drug product release and stability studies for PFS products. Additional characterization techniques that differentiate silicone particles from other particles may enhance product characterization. The reader is encouraged to start with the highest level of silicone oil and slowly reduce the amount applied to find the optimum level that achieves this required balance.

An alternative solution is precoated plunger stoppers that do not require the addition of silicone oil. The coating applied is cross-linked silicone oil with or without polytetrafluoroethylene coating.

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10.3 PFS Barrel and Needle-Shield Combination

The combination of a barrel and needle shield ensures an appropriate pull-off force for the patient or healthcare provider to remove the needle shield and allow the injection. Care must be taken not to overlubricate the plunger stopper, which could compromise CCI during shipping and throughout shelf life.

For an RNS, additional balancing of siliconization and the force required for plastic overcap removal must be considered. If the deshielding or separation force is lower than the pull-off force, the rigid plastic cover may separate from the rubber needle shield. Typically, the syringe suppliers have already optimized this careful balance of siliconization with other requirements.

10.4 Siliconization of the Injection Needle

The injection needle is often siliconized to manage the friction during needle insertion and removal during drug administration. Typically, 12,500 cSt (high viscosity) silicone oil is applied using a wiping process. However, the wiping process requires frequent replenishing of silicone oil.

10.5 Drug Product Requirements

The PFS with all its components (including siliconization) is—like any primary packaging—an integral part of the drug product. Therefore, development (including stability testing of the drug product) should be performed in packaging systems representative of the final commercial configuration. This should include assessing (e.g., during stability studies using relevant quality attributes of the drug product, including particle analysis; see also **Section 10.8.1**) the compatibility of primary container components and associated process-related impurities with the active ingredient in its formulation.

Interaction may also affect the functionality of the syringe system. Therefore, it is necessary to assess functionality during development (extended characterization) and in the final drug product configuration to determine the effects of such factors as manufacturing operations, storage, and transportation. Appropriate testing for functionality can include break-loose forces, gliding forces, and injection time (e.g., for autoinjectors).

10.6 Shelf-life Effects

Filled syringe stability programs should consider all endpoints routinely assessed for parenteral products, such as CCI, visible and subvisible particles, physicochemical stability, and microbiological quality (e.g., sterility and endotoxins). Control of the orientation of empty syringes during storage and shipment is a relevant factor. Empty syringes are typically stored and shipped with the needle pointing up to prevent the silicone from accumulating in the needle opening.

10.7 Methods Toolbox

This section provides a discussion on available analytical methods suited for characterizing/assessing the silicone content of an empty syringe.

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10.7.1 Fourier Transform Infrared Spectroscopy

Fourier Transform Infrared Spectroscopy is one method used to identify the silicone oil content of an empty syringe. In this method, silicone oil identification occurs through organic extraction from the glass and elastomer surfaces of the syringe followed by solvent evaporation under vacuum. In some applications, it is also possible to measure the silicone oil directly on the rubber surface with an infrared probe.

It is possible to measure break-loose and gliding forces in empty syringe barrels indirectly using mechanical force systems. This method is advantageous because it is quick and also suitable for in-process controls.

10.7.2 Powder Tests

Other methods described add glass powder or flour to the siliconized syringe, and the dry powder's adherence to siliconized surfaces provides an approximate visualization of the silicone oil distribution ***(81)***. The images below show the powder distribution within syringes sprayed with silicone oil via fixed nozzles (**Figure 10.7.2-1**) and diving (dynamic) nozzles (**Figure 10.7.2-2**).

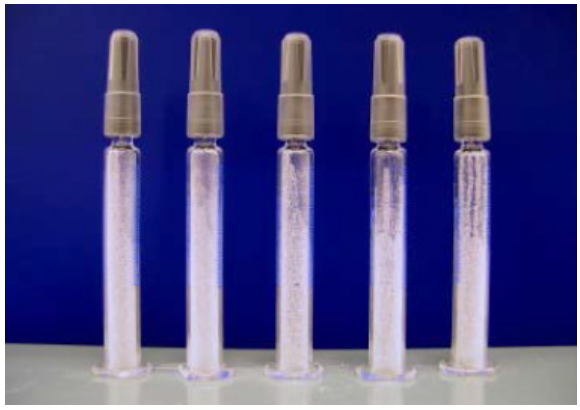

Figure 10.7.2-1 Siliconization with fixed nozzles

Figure 10.7.2-2 Siliconization with diving nozzles

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10.7.3 Reflectometry Measurements

Reflectometry is based upon the different refractive indices of the syringe glass and the silicone.Reflectometry measurements can determine the distribution and thickness of the silicone oil layer.

Figure 10.7.3-1 shows the radial silicone oil distribution on the barrel of the empty syringe. The syringe is stored horizontally and the silicone oil is redistributed. On the lower part of the barrel, more than 10 times the amount of silicone is visible. The uneven distribution is due to the redistribution of the oil during horizontal storage. Because some areas have less silicone oil, the impact on break-loose extrusion force should be assessed. This method is used primarily for silicone layer characterization during research and development, but not for batch release or incoming goods control during commercial supply of syringe barrels.

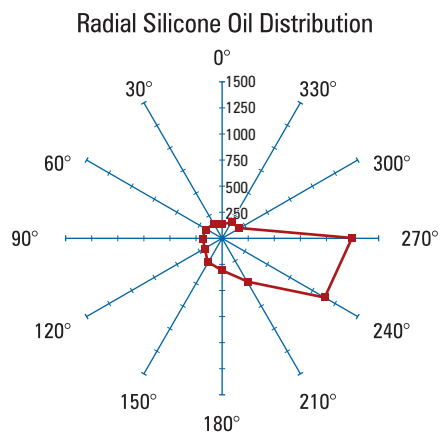

Figure 10.7.3-1 Radial silicone oil distribution measurement

10.7.4 Schlieren Effect Analysis

This method uses high-resolution cameras, special lighting, and software (to calculate coverage) to visualize oil droplets that form at the interface of glass and product solution (**Figure 10.7.4-1**). This method is qualitative, fast, and mainly used for silicone layer characterization during research and development, but not for batch release or incoming goods control during commercial supply of syringe barrels. Key considerations for this method include plunger stopper selection (siliconization or no siliconization), age, sterilization or no sterilization, plunger stopper placement, position, timing after plunger stopper placement, and velocity.

Measuring break-loose force and gliding force in an empty syringe requires the use of different test procedures. It is possible to measure gliding force results directly after siliconization of the barrel, whereas break-loose force measurement requires preparation time. The plunger stopper must be placed in the correct/defined position using vent tube equipment, and the syringe must be stored for at least 24 hours up to 72 hours in real-time conditions, or undergo accelerated aging in a climate chamber.

A common setup for the measuring devices is 100 mm/min up to 150 mm/min speed. The measuring path depends on the barrel length and plunger stopper position.

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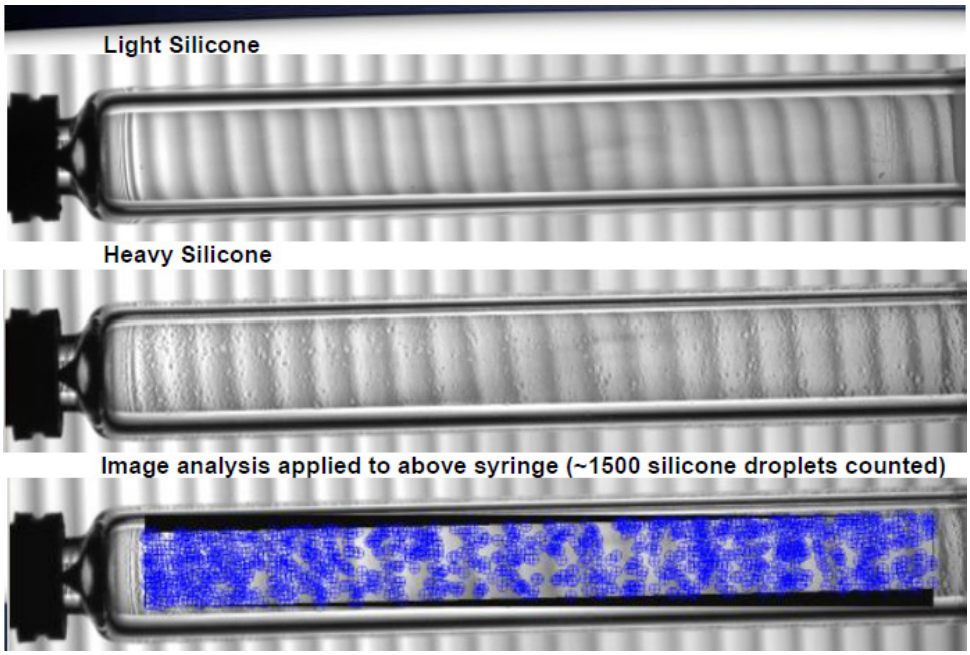

Figure 10.7.4-1 Silicone Oil Droplet Visualization

10.7.5 Break-loose and Extrusion Force

It is possible to measure break-loose and gliding forces in empty syringe barrels indirectly using mechanical force measurement systems. Advantages to this method are that it is fast and suitable for in-process controls. Major disadvantages are that the test is performed into air (not into tissue or any material more closely representing tissue), and test conditions are artificial and often do not reflect actual product use conditions (e.g., injection force, speed, and time).

10.8 Measuring Silicone Oil in Solutions

Silicone oil can mechanically detach from the wall during filling operations, storage through vibration, or as a result of plunger movement during injection. This last scenario (plunger movement during injection) occurs because the plunger stopper not only glides upon a silicone layer, but also pushes silicone oil toward the needle upon injection. In water-based drug product formulations, which make up the majority of drug products presented in PFSs, the partitioned silicone oil may enter the solution as dispersed microscopic silicone oil droplets. The droplets may be stabilized in the aqueous dispersion aided by surface-active drug product formulation ingredients, such as proteins and surfactants. In many parenteral drug product formulations, especially in therapeutic protein drug products, surfactants are used as stabilizers. This silicone oil partitioning effect, observed in systems lubricated by free silicone oil and to a lesser extent, in systems lubricated by baked on silicone oil, must be evaluated carefully for product interactions.

One method for determining the amount of silicone oil dispersed in the drug product solution (as presented in **Section 5.5**) is inductively coupled plasma optical emission spectrometry (ICP-OES). The amount of silicone oil in the drug solution may be of interest from a patient safety perspective because one must assume that the dispersed silicone oil—together with the drug solution—will be injected into the patient. However, the potential amount of silicone oil that could be injected using siliconized syringe presentations—even with repeated applications—is typically much lower than any toxicologically relevant doses.

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Another area of interest in which the amount of silicone oil in the drug product solution may be relevant is the potential interaction of silicone oil with the active ingredient or stabilizing formulation excipients. Drug/silicone oil or excipient/silicone oil interactions are very product specific and the drug product development program of any PFS presentation should address this (see **Section 14.0).**

10.8.1 Subvisible Particle Analysis

Subvisible particle analysis of particles larger than 10 μm and 25 μm by light obscuration is a mandatory compendial test method for any injectable drug ***(82–85).*** In recent years, testing of particles smaller than 10 μm has also gained interest from regulatory bodies. Silicone oil droplets dispersed in solution typically contribute to these subvisible particle counts because the droplets are often in the relevant size ranges, and the light-obscuration technique cannot differentiate particle counts by particle composition. As a result, subvisible particle counts by compendial methods for PFS products are often higher and more variable than counts in corresponding presentations using less or no silicone oil (e.g., vial presentations).

To gain insight into particle type, particle number, size distribution, and particle variability, it may be useful to apply extended characterization techniques that help differentiate silicone oil droplets from other particles that might be present in the product solution. For particles and silicone oil droplets in the micrometer size range, flow-microscopy techniques are useful and commercially available from several vendors. The differentiation of particle types in flow-imaging techniques is based on morphological differences between spherical silicone oil droplets and oddly shaped or amorphous other particles made visible by the imaging feature of the instruments. A visual assessment of images provides an estimate of particle type, as well as distribution of particle sizes in the micron range in the product. This technique is, however, time consuming and not suitable for routine operations, and the equipment is not suitable for quality control use. A higher degree of automation may be achievable with software filters for automated counting of different particle types. Several basic options are available in the commercial software, and more advanced options have been proposed in the literature. Applicability must be tested product-by-product.

Other exploratory techniques (such as resonant mass measurement) for studying particles in submicron size ranges are under development. However, the various emerging methods are currently in the research and feasibility stage and are far from routine use for characterization purposes, let alone for quality control.

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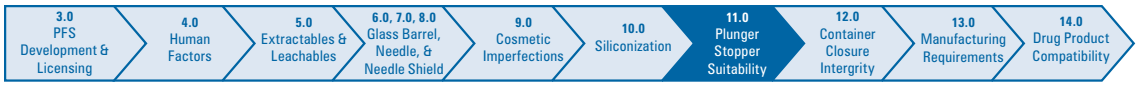
11.0 Plunger Stopper Suitability



After the syringe body is filled with the drug substance, the plunger stopper is inserted to complete the seal and provide the mechanism for dispensing the drug product. This chapter focuses on the design and selection of the plunger rod and plunger stopper combination.

11.1 Materials, Coatings, and Laminates

An elastomeric plunger is a critical element of the syringe system because it serves as the primary seal for CCI, maintains purity of drugs throughout shelf life, and delivers the contents of the barrel to the patient, either by means of a manual injection or by means of a secondary device, such as an autoinjector *(57)*.

11.1.1 Formulations

Typically, manufacturers/suppliers use thermoset halobutyl to make plunger stoppers. A thermoset elastomer is a polymer system in which the elastomeric properties are derived from chemical cross-links that are irreversibly created when the polymer is heated. Elastomer formulation technology is unique, and the components are manufacturer specific.

The formulations are trade secrets, and the formulations of different manufacturers are not chemically equivalent. Changes to these formulations typically require requalification efforts, repeated testing of drug product stability, and new regulatory filings.

Each formulation has a common list of raw materials. A typical elastomer formula with associated ingredient functions contains components outlined in **Table 11.1.1-1** with consideration toward the application's targeted elastomer properties.

Table 11.1.1-1 Elastomer Components and Target Properties

Elastomer Component	Component Properties	Overall Elastomer Target Properties
Polymer	Elastic properties	• Low gas permeability and moisture vapor permeation • Chemical and biological inertness • Low levels of extractables • Compliance with global compendial requirements • Resistance to heat, ozone, and UV light • Durometer hardness to maintain seal integrity • Sterilization compatibility • Stress-crack resistance
Curative	Cross-linking for elasticity and strength	
Filler	Hardness, modulus, strength, and reinforcement	
Processing aides	Enhance flexibility, fatigue resistance, and mold release	
Antioxidants/ Antidegradants	Stabilization for UV light, oxygen, and ozone	
Plasticizer	Processing aide and promotes flexibility	

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11.1.2 Laminates and Coatings

Most elastomeric plunger stopper components contain either a coating or a combination of coating and laminate film. These components ensure that proper mechanical feeding on fill-finish equipment adequately expels the drug from the syringe and reduces the level of extractables that migrate from the elastomer formulation. Manufacturers use direct spraying, mechanical tumbling, or addition during washing of components to apply coatings onto plunger stoppers surfaces. Laminate films, which are applied during heat forming (molding) of the plunger stopper, are bound physically and chemically to the component surface. Plunger stoppers typically fall into four main coating and laminate film categories with different associated benefits, as summarized in **Table 11.1.2-1.**

Table 11.1.2-1 Typical 1 mL long syringe plunger stopper coatings and laminate films

Category	Material Class	Benefits
Coating	Non-cross-linked surface silicone oil (multiple viscosities, e.g., 1,000 or 10,000 cSt)	Machine performance Syringe functionality Particle reduction
Coating	Cross-linked silicone oil	Machine performance Syringe functionality Particle reduction
Coating	Fluoropolymer	Machine performance Syringe functionality Particle reduction Barrier properties
Laminate Film	Fluoropolymer	Machine performance Syringe functionality Particle reduction Barrier properties

Silicone oil-based coatings, fluoropolymer coatings, and fluoropolymer laminate films increase lubricity during machining of plunger stopper feeder bowls, tracks, and placement tubes. They also improve syringe functionality (e.g., break-loose and extrusion forces), and serve as barriers between the drug and the elastomer, reducing the potential for movement of extractables inherent in all elastomer formulations.

Polydimethylsiloxane coatings do not have barrier characteristics. Silicone oil coatings applied to plunger stoppers can be classified as non-cross-linked or cross-linked. Non-cross-linked coatings typically have a single viscosity (e.g., 1,000 cSt); they are added in a pharmaceutical wash process and considered loose on the plunger stopper surface. Cross-linked coatings are blends of several silicone oil viscosities mixed with a chemical bonding agent. These coatings are sprayed on plunger stopper areas at a controlled spray level and bonded with UV light. Development of new and innovative coatings and films for plunger stopper components is ongoing. Therefore, it is important to discuss all plunger stopper product specifications and associated attributes with manufacturers.

Table 11.1.2-2 reflects possible plunger stopper coating options for use with 1 mL long glass barrel applications.

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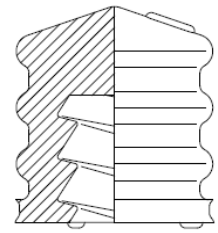
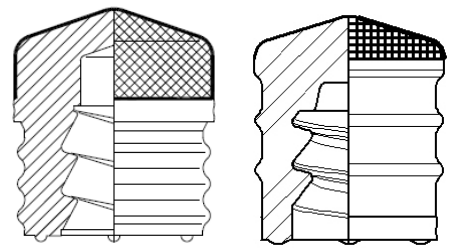
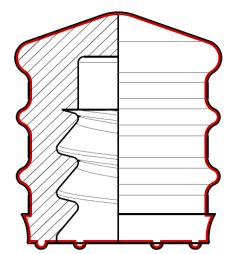
Table 11.1.2-2 Plunger stopper coating options

Plunger Stopper Coating	Plunger Stopper Schematic
Non-cross-linked or Cross-linked Silicone Oil	
Fluoropolymer Laminate Film and Cross-linked Silicone Oil (**Note:** Dark outline designates fluoropolymer laminate film)	
Fluoropolymer (**Note:** Front red outline designates fluoropolymer coating)	

11.2 Function of Processing and Sterilization

The choice of plunger stopper formats is dependent upon the planned manufacturing process for bulk, ready-to-sterilize, and presterilized applications. Presterilized glass syringes and presterilized plunger stoppers, which are the standard benchmark, are typical choices for current and new market applications.

11.2.1 Bulk

In some applications, the drug manufacturer or a contract manufacturing organization manages pharmaceutical washing, post-treatment, siliconization, sterilization, and final sterile component packaging control. In these cases, after molding and trimming of plunger stoppers, the plunger stopper manufacturer applies a general wash process to remove processing aids. The plunger stoppers are then packaged using standard methods. Plunger stopper manufactures may also add silicone oil to non-fluoropolymer-coated or laminated plunger stoppers.

11.2.2 Presterilized Plunger Stoppers

Prior to delivery to the drug manufacturer, presterilized plunger stoppers are washed and sterilized in conjunction with validated processes aligned with specifications for particulates and endotoxin levels.

After molding, trimming, and in-process treatments, presterilized plunger stoppers are washed using a pharmaceutical-quality washing process. This process reduces the levels of loose particulate matter

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and endotoxins/bioburden by using multiple rinse and agitation cycles of WFI for the rinse portions. In addition, washing and sterilization of components may have an impact on the potential extractables profile of elastomeric plunger stoppers. This impact must be understood and considered for drug compatibility and long-term stability.

11.2.3 Sterilization

In most applications—and the industry standard—manufacturers supply plunger stoppers in a pre-sterilized format (gamma or steam) for direct entry to a fill-finish line. Plunger stoppers may also be supplied in bulk or ready-to-sterilize formats that require pharmaceutical companies to complete the additional post-processing steps (e.g., pharmaceutical wash, siliconization, and sterilization. Refer to **Figure 11.2.3-1**) on plunger stoppers before drug filling. Pharmaceutical companies commonly consult with the manufacturers to determine proper post-processing and sterilization conditions to apply to their plunger stopper elastomer formulation or product so that components are not over-processed (e.g., excessive temperature and time). Over-processing of components could have a negative impact on elastomer properties.

Manufacturers typically develop plunger stoppers and elastomer formulations to withstand validated radiation sterilization ranges (typically gamma) and validated steam autoclave sterilization cycles. Certain formulations and coatings are known to be sensitive to the effects of irradiation. Therefore, applying radiation or steam can minimize the impact on chemical and functional stability. ISO 11137, Sterilization of Health Care Products – Radiation, is used to validate radiation-sterilized plunger stoppers ***(86)***. This standard provides guidance for multiple methods to validate sterilization using irradiation. Autoclave steam sterilization cycles are developed and validated to comply with a sterility assurance level (SAL) of 10^{-6}. Component and syringe manufacturers can provide guidelines, validation summaries, general technical information, and studies in support of component claims.

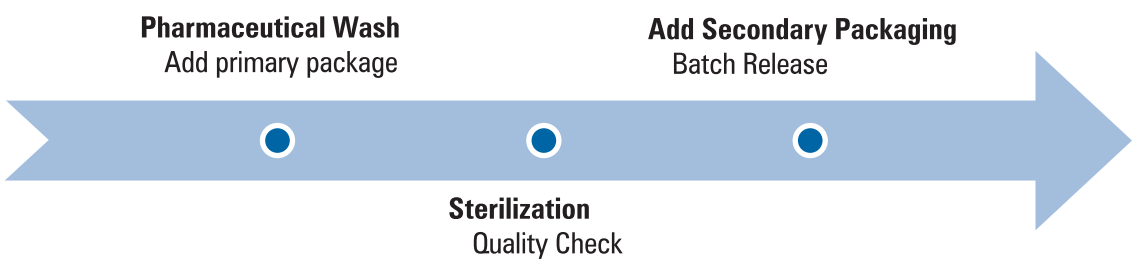

Figure 11.2.3-1 Generalized Bulk sterilization process

11.3 Functional Properties

11.3.1 Seal Integrity

The plunger stopper provides a seal on the interior of the barrel between filling and drug administration. This seal—along with the needle cover—reduces the likelihood that microbes will enter the syringe, thereby preserving the sterility of the syringe contents. At the same time, the seal prevents the syringe contents from leaking.

The plunger stopper may have more than one sealing rib. The function of the sealing rib(s) is to preserve sterility. To enter the syringe, microbes must pass all barriers formed by the sealing ribs, starting with the rib that is farthest from the drug product and ending with the rib that is closest to the drug product. Each rib acts independently as a seal, and the redundancy of ribs increases the robustness of the system.

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Care must be taken with respect to transitional movement of the plunger resulting from pressure fluctuations such as those that can occur during air transport. In this case, the plunger first moves in the direction of the lower pressure (the cargo space), and then in the opposite direction when pressure is equalized again. Movement must be controlled to protect the syringe from microbial contamination.

To prevent any loss of content through leakage, the order of ribs that the drug product must pass is reversed. Drug product must first pass the innermost sealing-rib to enter the space between the first and second sealing-ribs. The drug product must then pass the second sealing-rib to enter the space between the second and third ribs, and so on. Redundancy of ribs, in this respect, also increases the robustness of the system. Liquid trapped between the ribs closest and second closest to the syringe contents, but not progressed to the next rib may be considered a cosmetic but not functional defect. This defect can result in customer complaints.

To preserve sterility and syringe content, the plunger stopper and barrel must have an appropriate interference fit. This implies that the sealing-rib diameters [of the plunger stopper] before assembly into the barrel are sufficiently large compared to the internal diameter of the barrel. At placement of the plunger stopper into the barrel, the sealing ribs are compressed and forced to adopt the dimension of the inside of the glass barrel, thereby forming sealing "rings" on the barrel. A higher number of sealing ribs and a higher interference fit not only leads to a greater system robustness in terms of seal integrity, but also entails less favorable gliding behavior. For uncoated plunger stoppers, ISO 11040-4, Prefilled Syringes—Part 4: Glass Barrels for Injectables and Ready-to-Use Prefillable Syringes, and ISO 11040-5, Prefilled Syringes—Part 5: Plungers for Injectables, list barrel and plunger stopper dimensions that lead to an acceptable interference fit ***(64, 87).*** Fluoropolymer-laminated/coated plunger stoppers are currently outside the scope of ISO 11040-5. Fluoropolymer-laminated/coated plunger stoppers are dimensionally designed to fit to glass according to ISO 11040-4. The shape and rib diameters, however, do not follow ISO 11040-5, and the rib closest to the liquid may have a higher interference fit.

According to ISO 11040-5, uncoated plunger stoppers for 1 mL long syringes have either two or three sealing-ribs plus a trim edge. The diameter of the sealing ribs is largely determined by the corresponding dimensions in the mold the manufacturer uses for the product.

The trim edges of non-fluoropolymer-coated and fluoropolymer-coated plunger stoppers are not considered to be sealing ribs, even if there is contact with the wall of the barrel. For fluoropolymer-laminated plunger stoppers, the trim edge may provide a sealing function.

Seal integrity between the plunger stopper sealing-ribs and the walls of the syringe must be maintained during storage. Between filling and drug administration, the seal of the plunger stopper on the barrel is essentially a static seal. Under the influence of changes in pressure (e.g., during air transport), the plunger can move. If this occurs, then the plunger stopper acts as a dynamic seal. Typically, movement during air transport is minimal (depending on the air space within the syringe). Subsequent transport studies (real or simulated) to measure plunger movement can be used to evaluate this type of potential sterile boundary compromise.

The plunger stopper provides a dynamic seal when the syringe is activated. It prevents loss of any syringe contents due to leakage through the ribs of the plunger stopper, which is in dynamic contact with the interior of the barrel. The rib-barrel interfaces must be protected from defects in materials or coatings. Refer to **Section 12.0** of this report for more information on CCI requirements.

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11.3.2 Extrusion Behavior

The plunger must move at the time of activation. The force required to start the plunger moving is—in the majority of cases—greater than the force required to sustain the movement of the plunger (**Figure 11.3.2-1**). The starting force is usually referred to as the "break-loose force" or "activation force."

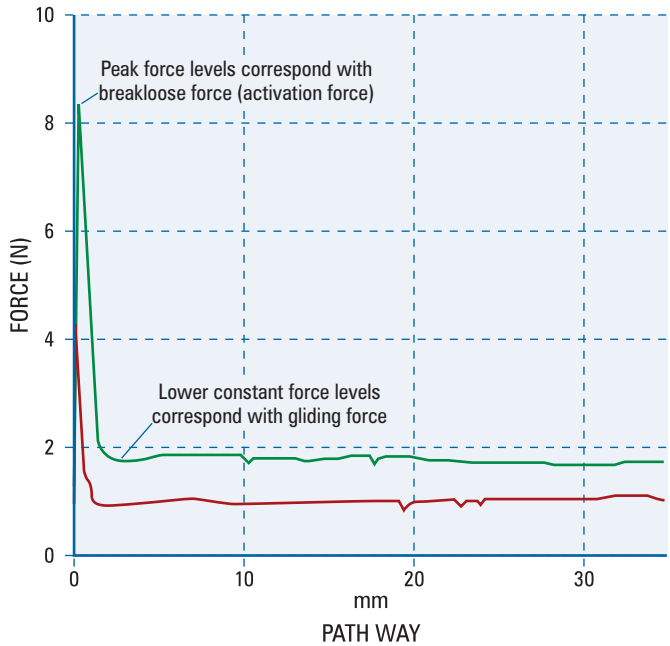

Figure 11.3.2-1 Gliding Curves of Two Plungers in the Same Type of Syringe Barrel

Break-loose and extrusion force is a property that is a complex function of many factors, including the plunger stopper, barrel, drug solution, needle and injection device (if one is used), and test conditions—especially the propagation speed imposed on the plunger. Extrusion behavior assessed in empty syringes can be substantially different from gliding behavior assessed in filled syringes. Measurements of extrusion behavior in empty syringes are valuable for assessing the surface state of the syringe barrel—especially its siliconization, whereas measurement in the filled syringe is relevant to the intended use of the syringe. The material properties of the plunger stopper (such as hardness and permanent deformation), its dimensions (especially its rib diameters), and its surface state (siliconization or presence of a polymer coating), are the main parameters to control. See **Section 10.0** for a more in-depth discussion of siliconization.

11.4 Plunger Rod/Plunger Stopper Interactions

To ensure proper force transmission from the plunger rod to the plunger stopper at the time of administration, the plunger rod should be designed so that the plunger rod matches the threads of the plunger stopper and meets the length and diameter requirements dictated by the glass barrel. Plunger rods are available in different polymeric materials, colors, and thumb-pad designs. These characteristics should be included in the evaluation of the assembled syringe (see example in **Figure 11.4-1**). The plunger rod does not have any specific processing or sterilization requirements. Thumb-pad design may vary according to specific human factor needs.

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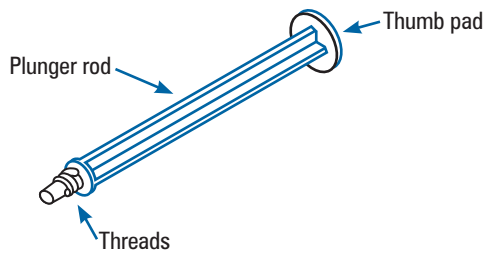

Figure 11.4-1 Plunger Rod

11.5 Plunger Stopper Suitability: Hold-up Volume

The geometric design of the inner syringe shoulder area mainly defines the hold-up volume (or dead volume). However, it is possible for an optimized plunger-stopper design (light taper/cone instead of a flat top of the plunger stopper, as shown in **Figure 11.5-1**) to affect accurate delivery of the syringe contents by minimizing the hold-up volume.

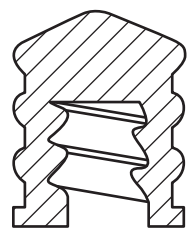

Figure 11.5-1 Plunger Stopper with Lightly Tapered Tip

11.6 Dimensional Control

11.6.1 Defining Critical Plunger Stopper Dimensions

Since the dimensions of the plunger stopper affect seal integrity, functional performance, and manufacturability/machinability of the PFS, the pharmaceutical company should collaborate with the component manufacturer to determine the critical plunger stopper dimensions. Several methodologies, such as quality function deployment, quality by design, and stacked tolerance analysis, serve as useful tools, and may be employed to identify and define the critical plunger stopper dimensions. These methodologies may be used to correlate the system-level C/C attributes (e.g., functionality, CCI, drug-product formulation interface, and manufacturability) with the component-level attributes (e.g., material properties, dimensions, and cosmetics). Such tools aid in determining the impact of out-of-specification components and specification changes on the C/C system, and may serve as a basis for risk assessments to define commercial control strategies.

11.6.2 Plunger Stopper Dimensional Control at Component Manufacturer Site

The end user should establish specifications with the manufacturer of the plunger stopper component to ensure critical dimensions, tolerances, and any other features that are critical to the function of the plunger stopper are adequately controlled. The component manufacturer must demonstrate that the appropriate in-process controls are in place to monitor and control the plunger stopper dimensions, qualified or validated methods are used, and appropriate mold monitoring and maintenance programs are implemented. To understand and evaluate the dimensional control capability of the plunger stopper, the component manufacturer should provide and discuss historical data for the critical plunger stopper dimensions, and if applicable, discuss dimensions measured as part of a mold wear and control-monitoring program with the end user.

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11.6.3 Sampling Plans and Acceptance Sampling System for End User Incoming Inspection Site

The component manufacturer and the end user should discuss and agree on the sampling plan and the acceptance sampling system (e.g., AQL or process capability index) criteria. Commonly used standards for sampling procedures for inspection by variables or attributes and corresponding quality levels or limits include ANSI/ASQ Z1.4, ISO 2859-1, and ISO 3951-1 ***(88–90).***

11.6.4 Dimensional Measurement Methods and Harmonization

In defining user geometric requirements, special attention must be given to the nonrigid behavior of the material. Due to the inherent compressive nature of the elastomer, the use of noncontact measurement tools, such as a coordinate measuring machine with vision system capability, is highly recommended as an alternative to the use of contact measurement tools. Contact measurement tools (such as calipers) used to measure plunger stopper dimensions should be employed with extreme care and avoided if possible to minimize elastomer deformation resulting from the contact measurement tool. The selected measurement system must be properly qualified.

11.7 Product Interactions

FDA industry guidance on C/C systems for human drugs and biologics requires packaging components that are compatible with a dosage form not to interact in a way that causes unacceptable changes in the quality of either the dosage form or the packaging component ***(5).***

In addition, the materials used to construct packaging components must not leach harmful or undesirable amounts of substances to the patient. The plunger stopper material must, at a minimum, meet the appropriate compendial standards to demonstrate material safety for its intended use ***(29).***

Certain buffer solutions, solution concentrations, high temperatures, and sensitivities of the drug product to the plunger stopper formulation may cause interactions between the drug product and the plunger stopper when in contact for extended periods of time. Examples of such interactions include discoloration of the drug product or the plunger stopper; pH shifts; reductions in the concentration of any of the drug product components due to adsorption, absorption, or leachable-induced degradation; and loss of potency of the active drug substance induced by a chemical entity or a degradation product leached from the plunger stopper.

Section 14.3.2 describes considerations for assessing the compatibility of the drug product formulation with the plunger stopper material. Results from such studies may be used to aid in choosing the lead C/C component and in designing the formulation stability studies to further characterize the chemical, physical, and functional interactions between the drug product and the plunger stopper.

11.8 Secondary Packaging

When the pharmaceutical/biotechnology company or the contract manufacturing organization chooses to process bulk plunger stoppers, typically these components are supplied in a basic package configuration of fibrous corrugated cartons and non-cleanroom-quality polyethylene bags intended only for prehandling and not for sterilization. Plunger stoppers must be removed and placed in suitable containers for any additional post-processing, including washing, treatment (e.g., siliconization), and sterilization.

Because of the different sterilization techniques and multiple technologies in use to fill and seal syringes (e.g., aseptic transfer, isolators, or RABS), different construction materials and packaging design formats are available. The drug manufacturer must select the appropriate sterile package to meet the filling tech-

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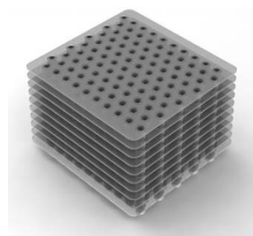
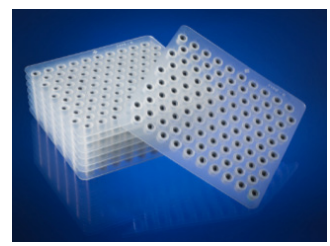
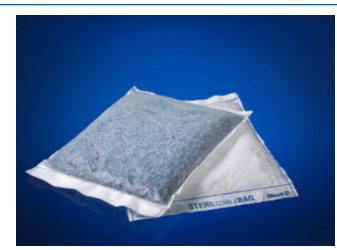
nology hardware requirements. Some examples of sterile package formats (see **Figure 11.8-1**) readily available to drug manufacturers are given below:

- 10×10 nest plunger stoppers in sealed multilayer polymer film bags (radiation sterilized)
- Bulk plunger stoppers in sealed multilayer polymer film bags (radiation sterilized)
- Bulk plunger stoppers in sealed multilayer polymer film bags with an attached rapid transfer port (radiation sterilized)
- Bulk plunger stoppers in sealed Tyvek®/multilayer polymer film bags (steam sterilized)
- Bulk plunger stoppers in sealed Tyvek®/multilayer polymer film bags with an attached rapid transfer port (steam sterilized)

Figure 11.8-1 Examples of Sterilized Plunger stoppers in Secondary Packaging

10X10 Nest		
Bulk		
Rapid transfer port (RTP)		

The packaging material used to maintain the physical and sterile barrier for presterilized plunger stoppers must also be qualified for integrity following the gamma or steam autoclave sterilization process, transportation simulation, and aging performance. The packaging material may also be evaluated for any potential extractable or leachable components (**Section 5.0**). The product is typically packaged with at least three sealed layers (a primary layer for the plunger stopper and two secondary layers for protection) to ensure product integrity during transportation and handling. This design also allows cleanroom decontamination wipe down and shedding of layers as the product is staged for aseptic filling. Care must be taken to avoid particle contamination. Validation and qualification of the packaging are typically performed according to the following standards:

- ISO 11607-1 ***(91)***
- ASTM D4169 ***(6)***
- ASTM F1980 ***(92)***
- Center for Devices and Radiological Health consensus standards (see **Section 18.0**)

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11.9 Shelf Life and Storage

Plunger stopper shelf life or storage time is often specified in conjunction with a specific product and packaging conformation. The total period, from the date of manufacture to the date by which an item must be used (expiration date), must be determined. Plunger stopper shelf life is influenced by several factors, such as exposure to temperature, moisture, and light, as well as any processing. Elastomers that are stored in a bulk format with no additional processing, such as sterilization, typically have a longer shelf life than those that have been sterilized. The type of sterilization used can also alter the shelf life of the plunger stopper. Temperature is one of the most important factors in determining shelf life. A general rule is that as component storage temperature drops, shelf life increases. The component manufacturer considers these factors when assigning the shelf-life values to their product.

The customer must consider the useful life of the plunger stopper, which includes the shelf-life date, as well as the duration of the actual use of the component. The useful lifetime is based on the ability of the component to perform its intended function, compatibility of the drug after any post-processing that the customer performs, and the duration of use. The useful life can only be determined during the component's contact with the drug product by means of accelerated or real-time stability studies.

11.10 Processing Interactions (Plunger Stopper Design vs. Equipment Requirements)

A number of plunger stopper design features can influence the processing ability of the plunger stopper. This section discusses in brief several of these features and their interactions relative to the plunger stopper.

11.10.1 Elastomer Selection

Different elastomers stretch and compress at different rates affecting insertion tube compression of any laminates/films on the plunger stopper surface, as well as the final length of the plunger stopper once it is inserted into the syringe. The length of the inserted plunger stopper is generally noticeably different from the length of the plunger stopper as molded. This changes the point where the insertion rod must push the plunger stopper within the insertion tube to obtain the proper alignment at the bottom of the insertion tube/insertion rod. Proper insertion is required to ensure proper venting of the syringe before the plunger stopper is set.

11.10.2 Surface Treatments

Surface treatments are used, in part, to prevent elastomer from sticking in the packaging and aid in the processing of the plunger stoppers through the bowls, tracks, and other processing equipment, as well as functionality in a glass syringe barrel.

The type of surface treatment can affect the consistency of plunger stopper siliconization within the plunger stopper bags. The sprayed-on treatments have a tendency to migrate and can result in some pockets of dry plunger stoppers within the bags (more prevalent with the larger packaging formats, such as containers for alpha-beta rapid transfer ports). This dry plunger stopper phenomenon does not appear to be as common with cross-linked silicone oil coatings, indicating that—in addition to amount—the type of siliconization matters. Surface treatments may also influence the type of plunger stopper placement used (insertion tube versus vacuum placement). Depending on the insertion tube quality (e.g., lack of surface imperfections, straightness, and strength), vacuum insertion is less likely to damage surface lamination.

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Finally, surface treatments may influence the shape of the plunger stopper. Smooth finishes on the product contact face of the plunger stopper may require special consideration of the surface finish of the bowl to ensure that components do not stick together.

11.10.3 Plunger Stopper Dimensions

For plunger stopper designs, the trim edges must have specifications for the maximum outer diameter. However, the height and consistency/smoothness of the trim edge can also be important for vision systems. The smoothness and relative outer diameter of the trim edge compared to the rest of the plunger stopper are important for the machining through the insertion tubes. Friction/resistance increases if the trim edge is close to the outer diameter of the plunger stopper and is not smooth or larger than the outer diameter. Rougher trim edges can also wear down the equipment bowls and tracks.

The center of gravity defined by the plunger stopper dimensions is also a key factor in the design of plunger stopper bowls to ensure proper orientation of the components before placement into the syringe barrel.

11.11 Plunger Stopper Geometry

Pharmaceutical companies should work closely with plunger stopper venders to identify/develop plunger stoppers with a geometry that is suitable for meeting the requirements of the product. This includes examination of details discussed in previous sections:

- Functional properties
- Plunger rod/device interactions
- Plunger stopper suitability based on hold-up volume
- CCI
- Dimensional control

At present, the number of reference standards related to the suitability of PFS plunger stoppers is limited. ISO 11040-5 provides the most direct reference ***(87)***. This standard provides guidance on the shapes and dimensions of elastomeric plunger stoppers that are used in combination with glass barrels and is to be used in conjunction with ISO 11040-4 ***(64)***. Neither ISO 11040-5 nor any other part of the ISO 11040 series on PFSs provides test methods or limits for assessing plunger stopper suitability. Plastic barrels are described in ISO 11040-6, but there is no separate corresponding document for plunger stoppers for plastic barrels at this time ***(93)***.

A number of indirect references to plunger stopper suitability exist, such as USP <1207> Sterile Product Packaging—Integrity Evaluation, and documents from the FDA ***(5,94)***.

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12.0 Container Closure Integrity Requirments

Container closure integrity is an important element of the overall sterility assurance program for sterile products. Various layers of controls, including design verification, stacked tolerance, qualification, manufacturing process controls, inspection and handling, process simulation (media fills), microbial challenge testing, and physical challenge testing ensure the integrity of the C/C system. CCI testing should ultimately demonstrate that the PFS can maintain the integrity of its microbial barrier, which is typically a product design criterion.

Consideration should be given to CCI testing for the PFS during C/C design, development, and qualification phases, in addition to routine product manufacturing and product shelf-life stability programs. A risk-based, phase-appropriate approach should be used to identify the suitable application of CCI testing.

This section addresses best-practice approaches and considerations for control-strategy development related to CCI and the application of CCI testing to the PFS. The scope spans the temporal continuum, including component manufacturing at suppliers, through the pharmaceutical process of receipt, inspection, primary packaging assembly, secondary packaging, and distribution. It also presents a toolbox approach with the most useful available test methods and approaches to method selection, development, and validation.

12.1 Regulatory Basis for Container Closure Integrity

A variety of global regulatory guidance documents address expectations for CCI of sterile products at the time of manufacturing and throughout a drug product's shelf life. A C/C system—such as a PFS—used in the packaging of a drug or biologic is a critical part of the submission to gain regulatory approval. A successful licensure application requires substantial testing to demonstrate the suitability of the selected C/C system, which can be qualified for its intended use based on the quality target product profile.

Suitability has four aspects: protection, safety, compatibility, and performance ***(12)***. In particular, protection from microbial contamination is critical for a parenteral product that purports to be sterile as required by regulations to ensure product safety ***(38)***.

Demonstrating absolute sterility of a lot requires testing every filled finished article in the entire lot, which is not practical ***(95,96)***. Therefore, a holistic approach provides the best means to accomplish sterility assurance. First, pharmaceutical companies must adequately validate products manufactured by terminal sterilization and/or aseptic processing under cGMP. Second, manufacturers/suppliers must demonstrate that the CCI for the primary packaging configuration (C/C system) can provide assurance and maintain microbiological quality throughout the raw material and product shelf life ***(5)***. Third, for each product lot purporting to be sterile, conformance to sterility requirements must be determined by a validated sterility testing method in accordance with compendial guidelines prior to release ***(97)***. This holistic approach is consistent with EU Guideline 10, Annex 1, which emphasizes that demonstration of sterility or other quality aspects of the product must not rely solely on any terminal process or finished product test ***(98)***.

In summary, protection of the drug product from microbial contamination by the C/C system throughout the product's shelf life is an essential element of the holistic approach to sterility assurance for patient safety as required by regulations. Sterility testing must be performed during product lot release unless parametric release has been approved. It is also essential to demonstrate the ability

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of the C/C system to maintain the integrity of its microbial barrier, and therefore, the sterility of a drug product throughout its shelf life.

12.2 PFS Design Issues

Component selection for the PFS is based on a variety of design requirements, including the assurance of CCI. In its role of protecting the dosage form, the syringe must prevent product contamination and leakage. The syringe components that prevent contamination and leakage are the barrel, the plunger stopper, and the needle shield and their interfaces. **Figure 12.2-1** shows the component interfaces of relevance to CCI.

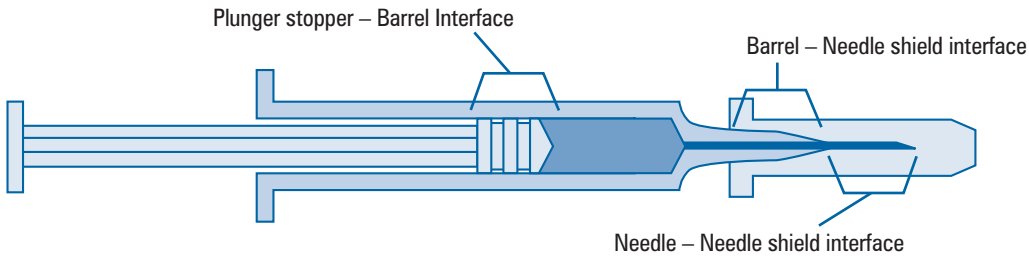

Figure 12.2-1 Syringe component interfaces involved in CCI

12.2.1 Plunger Stopper/Barrel Interface

The plunger stopper/barrel interface is the circumferential contact between the plunger stopper and the inner diameter of the syringe barrel, typically in the presence of a silicone lubricating fluid. The primary functions of this interface are to protect the dosage form from leakage and solvent loss over the shelf life of the product and to prevent contamination by microbial agents or foreign material. Plunger stopper designs vary from supplier to supplier. Plunger stopper designs typically include a leading sealing surface closest to the drug product, followed by one or more secondary sealing ribs that also provide stability during travel in the barrel. An additional consideration of the design of these sealing interfaces is functionality, including break-loose and extrusion forces.

12.2.2 Barrel/Needle-Shield Interface

The barrel/needle-shield interface is composed of the tip of the syringe and the corresponding sealing surface of the needle shield. The primary function of this interface is to protect the needle from contamination by microbial agents or foreign material. Understanding the specific needle-shield designs is necessary because different needle-shield designs respond differently to changes in environmental conditions (e.g., during sterilization cycles or CCI test method cycles). Additional considerations for this interface include performance requirements, such as permeability to sterilizing agents, removal forces, and device compatibility, if applicable.

12.2.3 Needle-shield/Needle Interface

The needle shield/needle interface protects the dosage form from contamination by microbial agents or foreign material, leakage, and solvent loss over the shelf life of the product.

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12.3 Supplier and Pharmaceutical Processes

Factors that ultimately ensure CCI for the PFS are many, spanning the manufacture of components and subassemblies through filling, closing assembly, inspection, final packaging, and transport.

12.3.1 Supplier Components, Subassembled Components, and Component Packaging and Distribution

The supplier is responsible for establishing design requirements for performance and compatibility among system elements during product development, including requirements related to ensuring CCI. The design/development process should include a risk assessment of the potential failure modes of the system. The supplier should validate the output of the design process to ensure that the resulting product meets established criteria and is suitable for its intended use. Design requirements related to CCI and sterility must include the C/C components, C/C system, and the secondary packaging system containing the PFS. In addition, suppliers must establish requirements for transport packaging and shipping processes.

The supplier is responsible for integrating design outputs into product specifications agreed to by the supplier and the pharmaceutical partner. The supplier is also responsible for establishing a quality system that maintains processes in a state of control to deliver consistently the product in conformance with the quality specifications established.

Acceptance criteria should be established as part of product design/development activities for the critical product parameters related to CCI, such as dimensional specifications, relevant ISO standards, cleanliness (e.g., freedom from visible contaminants), and integrity of components (e.g., lack of elastomer tears, glass cracks, or needle puncture through the needle shield) required for maintaining CCI over the shelf life of the PFS. Suitable supplier in-process controls should ensure CCI compliance (needle puncture, glass integrity, and geometry). The supplier and pharmaceutical partner should agree on the drawings and specifications output from design activities.

Appropriate statistical techniques should validate the supplier manufacturing and packaging processes, and demonstrate that the process is capable of delivering the product in compliance with specifications. To maintain product quality in a state of control, suppliers should establish appropriate in-process control strategies related to dimensions, cleanliness, and defects. Appropriate sampling plans should demonstrate that the product released for sale meets established criteria for fitness for use and provide the basis for releasing finished good lots.

12.3.2 Incoming Receipt and Inspection

Development of the incoming control strategy should use a risk-based approach to ensure that it is possible to assemble incoming components properly to form integral C/C systems. The incoming control strategy should include identifying component attributes and assigning a criticality classification (e.g., critical, major, or minor). For example, attributes potentially affecting the CCI would be considered critical.

Pharmaceutical companies should take into consideration the supplier's manufacturing process and control strategy, as well as nonconformance probability. When appropriate, pharmaceutical companies may rely on the supplier's certification program to ensure that the component meets specifications, and accept it based on a certificate of analysis. However, for higher-risk critical attributes, incoming testing may be required to measure the conformance of the components to the specifications for each batch.

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The incoming testing strategy should identify the quality attributes to measure, the sampling plan and the acceptance criteria. A statistical analysis should provide the basis for these, including AQL or process capability. The testing methods should be appropriately validated and harmonized with supplier test methods. Upon initial implementation, the control strategy should be closely monitored and improved as needed to ensure its continued suitability.

12.3.3 Semifinished Syringes

A semifinished syringe is one lacking a plunger rod. This term refers to the primary packaging in which the product is filled and sealed. CCI requirements are directly associated with the life cycle of a sterile product. During the development phase, which includes the process parameters, the product design and its specifications, the initial assessment of CCI should encompass worst-case conditions that may explore the limits of the final product and process. This can include the use of components that have undergone preparation process steps (e.g., washing and sterilization) with parameters exceeding the ones that have been defined. For example, a higher irradiation dose for sterilization can be used to demonstrate the absence of impact on the mechanical properties of the elastomeric compounds, and therefore, on the CCI of the product. The use of components from different batches and of different ages is also recommended to ensure that batch-to-batch variability and component degradation over time are adequately addressed. During this development phase, these attributes are commonly assessed via a formal risk assessment of how CCI may be affected and/or whether further evaluations are required. As per regulatory guidance, a robust rationale should be built when the actual product is not used to justify the use of these alternatives ***(98, 99).***

The assessment of CCI should follow the evolution of the product/process development. It should encompass any change in the critical parameters of the process or in the primary packaging components. Any change in the process or product should be assessed using a risk-based approach to identify any potential impact on the CCI of the final product and to conduct the appropriate studies.

When the product reaches the routine production stage, CCI should be monitored on a predefined basis. Depending on the validation strategy used and the risks identified in the process, this can include in-process checks of the integrity or release testing for integrity of the product. Additionally, it may be necessary to consider the aging of the packaging components until the end of the primary packaging shelf life to determine the potential impact on CCI, depending on applicable regulatory requirements. CCI assessment may also replace sterility testing on stability studies, which is in line with the recommendation from the FDA on CCI testing for sterile products ***(5).***

Finally, distribution aspects should receive consideration at any stage of product development. A risk evaluation associated with the different distribution channels of the product should provide the basis for the evaluation ***(100).*** Temperatures, vibrations, pressure differences, and the sizes of the air bubbles within finished syringes should also be considered. These variables can have an impact on excessive plunger movement that could lead to a loss of sterility of the product, and should therefore receive a careful assessment, especially for semifinished syringes in which it can be more difficult to prevent plunger movement.

12.3.4 Syringe Assembly and Secondary Packaging

The CCI of the assembled PFS (including finger-flange and plunger rod components) is predicated on the sterility assurance and CCI of the semifinished syringe introduced into the assembly process. The design/development of the assembly and secondary packaging processes should include risk assessment of the failure modes related to potential loss of drug product sterility and/or CCI of the semifinished syringe, including evaluation of the potential for plunger stopper and needle-shield

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movement, and damage to the semifinished syringe. The processes should be validated to ensure that the resulting product meets established criteria and is suitable for its intended use.

12.3.5 Shipping and Distribution of Semifinished Syringes and PFSs

The semifinished syringe and PFS may be exposed to transportation conditions that can include mechanical forces, pressure changes, and vibrations during shipping and distribution throughout the life cycle of the product. Shipping studies should provide an understanding of the capability of the product to maintain CCI following exposure to representative and/or worst-case transportation conditions.

12.4 CCI Test Method Toolbox: Method Selection, Development, and Validation

CCI testing ensures that the sterility of the syringe contents is maintained. Physicochemical CCI testing methods are typically designed to challenge the PFS. Key technology options include microbial ingress, dye immersion, pressure/vacuum decay, mass extraction, high-voltage leak detection, and helium-leak detection. **Table 12.4-1** compares and contrasts the major characteristics of the methods of CCI testing.

The CCI test method selected must be suitable for its intended use. The use of visual inspection and probabilistic methods (e.g., dye ingress) may aid in gross defect detection, whereas deterministic methods are more appropriate for definitive CCI testing, if applicable.

Nondestructive methods are generally preferred, which conserves the sample for further analyses. The use of quantitative methods should indicate if leak size information is required. Testing speed and throughput are key considerations for in-process and in-line CCI testing. In addition, consideration of package content attributes should aid in ensuring that the selected methods are applicable. For example, contents with low electrical conductivity may preclude use of high-voltage leak detection methods. Another scenario may be one in which a proteinaceous drug product may clog leak defects and prevent reliable detection by vacuum-based methods (e.g., vacuum decay).

Critical method parameters must be optimized for the specific drug product package and fully demonstrate the method's applicability. Testing a number of intact samples bracketing typical drug product variations (e.g., multiple batches of C/C components, drug products of different concentrations, and packages at different sites/lines) should ensure a low false-positive probability. The use of positive control samples of known defect sizes (prepared using micropipettes, wires, capillary columns, laser drilled defects, and/or microelectromechanical systems) can optimize and demonstrate method reliability and sensitivity. Evaluation of additional types of actual and simulated real-world defects may further verify method effectiveness.

CCI test method validation should be specific to the drug product or product type and appropriate for various drug product development phases. After selection of the C/C system, packaging process, and drug product formulation, the CCI test methods selected should support various development studies, such as engineering and stability studies. Prior to licensure application, validation of the method must follow applicable ICH guidelines and regulatory requirements. Complete method robustness studies should enable future long-term routine use in commercial QC laboratories.

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Table 12.4-1 Comparison of CCI testing methods ***(101)***

Considerations	Vacuum/Pressure Decay and Mass Extraction	High-Voltage Leak Detection	Helium-Leak Detection	Dye Ingress	Microbial Ingress
Best-Case Practical Detection Limit	2-5 μm	2-5 μm	0.1-2 μm	20-50 μm	20-50 μm
Detection	Deterministic	Deterministic	Deterministic	Probabilistic	Probabilistic
Destructiveness	Nondestructive	Nondestructive	Nondestructive	Destructive	Destructive
Quantitative	Qualitative and quantitative	Qualitative only	Qualitative and quantitative	Qualitative only	Qualitative only
Typical Applications	Container/closure (C/C) development and qualification Routine QC testing 100% in-line testing potentially feasible Sealing process validation	C/C development and qualification Routine QC testing 100% in-line testing potentially feasible Sealing process validation	C/C development and qualification Routine QC testing 100% in-line testing potentially feasible Sealing process validation	C/C development and qualification Routine QC testing Sealing process validation	C/C development and qualification Sealing process validation
Major Advantages	Potentially applicable to injection devices	Testing under normal atmospheric pressure Applicable to high-concentration proteinaceous products	Highly sensitive	Widely used for decades Industry and regulatory familiarity "Last resort" used when all else fails	Widely used for decades Industry and regulatory familiarity "Last resort" used when all else fails
Major Limitations	Possible interference with defect detection by proteinaceous products	Requirement for electrically conductive fills Not applicable to most injection devices Possibility of causing drug product degradation	Requirement for helium-containing headspace for drug product-filled syringes	Less sensitive Probabilistic detection Destructive	Less sensitive Probabilistic detection Destructive Not applicable to drug-product-filled syringes
Key Considerations	Assurance of proper fixture design to prevent plunger stopper movement and deformation Sample surface and vacuum chamber preparation	Possibility that syringe barrel area covered by needle shield is not assessable	Need to ensure that sample surfaces are free of adsorbed helium prior to testing to avoid false detection	Need to ensure that a true failure can be detected (e.g., dye in syringe, liquid in ribs)	Inability to ensure the absence of pathogens
Reference or Standard method	ASTM F2338 ***(102)***				

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13.0 Manufacturing Requirements

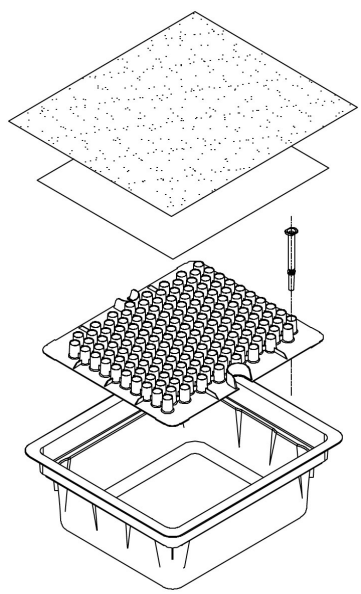
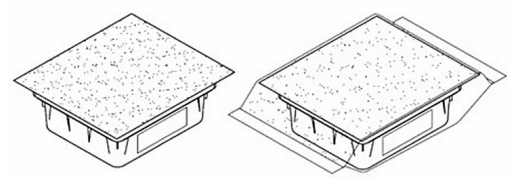
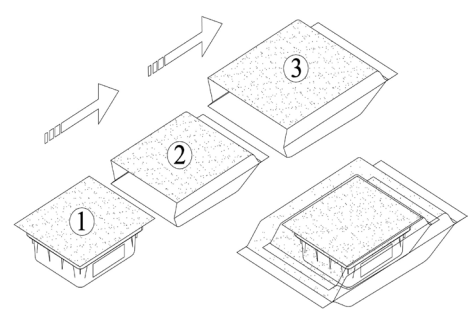
13.1 Component Packaging from the Component Manufacturer

This section discusses the following principle steps of processing of prefilled syringes: preparation of syringes, filling, and secondary packaging (labeling, blistering, cartoning) for both presterilized syringes and bulk.

13.1.1 Packaging of Presterilized Syringe Systems

To ensure the isolation of the product, the assembled syringe is placed into a nest to prevent contamination from the external environment. The nest is then placed into a tub that serves as primary packaging. The tub is sealed with a Tyvek® sheet before being placed into a single or double steribag, which is also sealed using the system shown in **Table 13.1.1-1.**

Table 13.1.1-1 Packaging methods

Primary Packaging system	Material	Steribag Packaging
	Tyvek® lid Tyvek® sheet Syringe Nest Tub	Single steribag  Double steribag 

13.1.1.1 Nest

Nest supports are made of rigid plastic material with slots that can vary in number and diameter. The design of the supports ensures the syringes remain at a proper distance from one another and reduces the risk of breakage during subsequent manipulations and transfers.

Nests with 100 or 160 syringes (i.e., slots) are common for 1 mL long syringes. These nests can be made of polystyrene, polypropylene, or another material. The design minimizes deflection or warping of the tub and nest during filling. Warping leads to variations in plunger stopper position between the middle and edges of the nests.

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13.1.1.2 Tub

The tub, which is a container of a plastic rigid material, maintains the nest in the correct position. It is usually white to permit the detection of the printing on the side of the tub by automatic camera systems. ISO 11040-7 describes common dimensions for tubs ***(103)***.

13.1.1.3 Tyvek® sheet and Tyvek® lid

A Tyvek® lid is coated and heat-sealed on top of the tub to isolate the product from the external environment before and after sterilization to obtain a sterile barrier. Single or double Tyvek® sheets consist of one (single) or two sealed foils (double) to ensure sufficient protection of the glass barrels during depackaging operations.

The Tyvek® lid is suitable for the sterilization methods described under **Section 13.2**. For further processing, the correct position of the sealing, the strength of the sealing (which might be influenced by the type of glue used to seal the lid), and any imprinting on the lid must be considered. The sterile barrier system consists of the tub sealed by the Tyvek® lid. The protective bag (single or double steribag) covers the sterile barrier system and its contents, preventing particle contamination and damage. **Section 13.1** discusses different methods of tub openings.

13.1.1.4 Steribag

Enclosing the tub into a steribag that is welded shut creates another barrier against possible environmental contamination and facilitates the movement operations of the primary packaging of assembled syringes. Again, consideration must be given to the size and position of the seals on the steribag for automated bag opening prior to filling.

The steribag is composed of a porous plastic portion (Tyvek®) for the penetration of the sterilization agent and a nonporous plastic-portion low-density polyethylene or high-density polyethylene (LD-PE or HD-PE). The configuration of the steribag, header bag, and reel bag depends on the dimension of the porous portion.

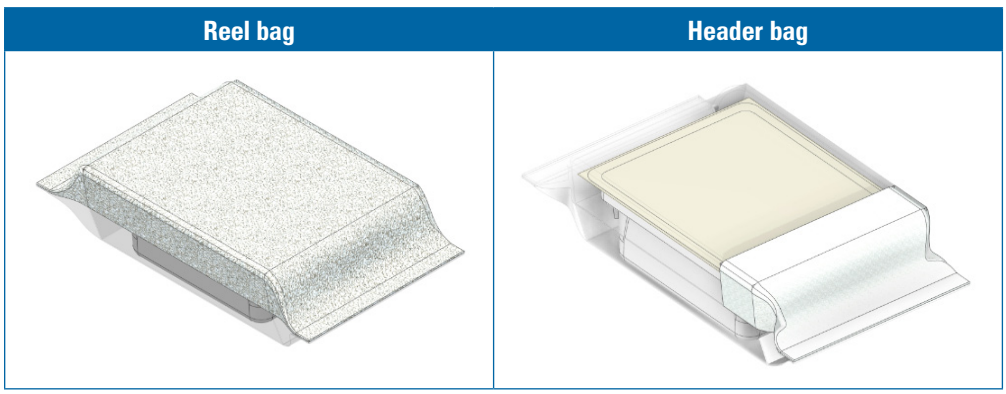

Figure 13.1.1.4-1 Steribag Components

Steribags are available in different dimensions for packaging. The selection of the steribag depends on the depackaging equipment. Depending on the depackaging process, single or double bagging might be more suitable. These options must be identified when sterilization validation occurs.

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13.1.1.5 External Box

To facilitate the movement of the primary packaging and avoid creating a barrier to the sterilization agent, the tub is ideally packaged in a plastic box that can be sterilized (see **Figure 13.1.1.5-1** for an example).

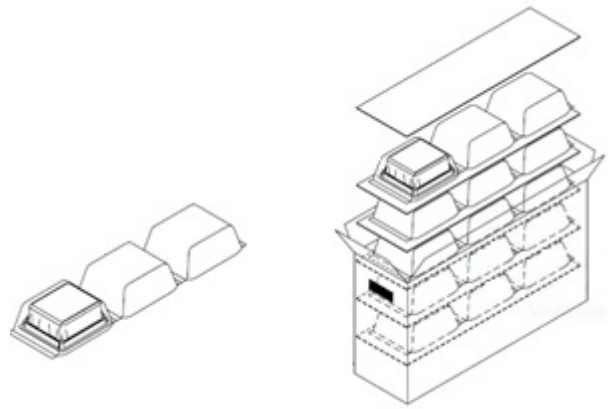

Figure 13.1.1.5-1 Packaging configuration

13.1.2 Presterilized Component Sterilization

The packed and palletized units described above undergo a sterilization process. If the sterilization process uses EO, then an aeration phase is needed to eliminate the chemical residues created during the process. Validation of the sterilization process demonstrates the quality of the process, the removal of EO residues, as well as the sterility and endotoxin levels ***(55).*** A qualified subcontractor versed in using EO can develop these processes.

If validated in accordance with ISO 17665-1, Sterilization of Health Care Products – Moist Heat – Part 1: Requirements for the Development, Validation and Routine Control of a Sterilization Process for Medical Devices, ISO 11137, Sterilization of Health Care Products – Radiation – Part 2: Establishing the Sterilization Dose, and ISO 14937, Sterilization of Health Care Products – General Requirements for Characterization of a Sterilizing Agent and the Development, Validation and Routine Control of a Sterilization Process for Medical Devices, then other sterilization methods can also be considered ***(86,104,105).*** These methods can include gamma sterilization (not suitable for glass syringes due to discoloration [turns clear glass amber]), steam sterilization, and electron beam sterilization.

13.1.3 Bulk Syringes

A supplier can deliver prefillable syringes to the pharmaceutical company in the "not washed and not sterilized" form (i.e., as bulk syringes). The pharmaceutical company performs the processing steps, such as washing, siliconization, assembly of the needle shield, and sterilization. After the glass barrel formation and needle assembly steps, the bulk syringes are placed into a plastic rondo tray, then into a plastic or cardboard box.

13.1.4 Rondo Trays: Materials, Dimensions, and Designs

Bulk syringes are packed in plastic (polystyrene and polyethylene terephthalate) rondo trays. The pharmaceutical company and the supplier agree to the material, dimensions, configuration, and number of syringes per rondo tray. A certain number of rondo trays are stacked in a specific orientation (**Figure 13.1.4-1**) and wrapped with a plastic bag for protection.

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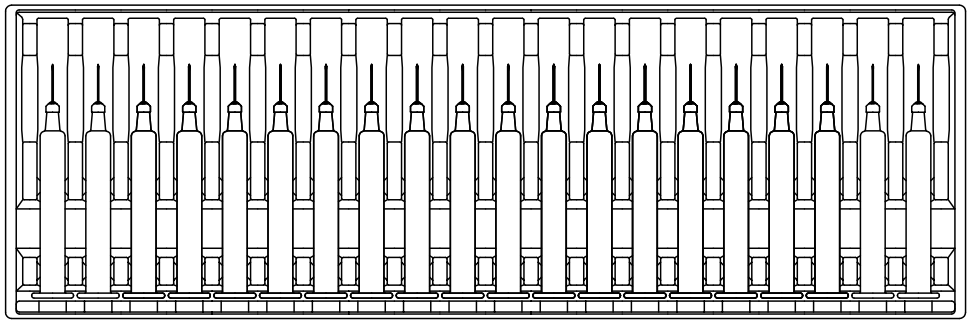

Figure 13.1.4-1 Rondo Tray

13.1.4.1 Secondary Packing of Rondo Trays

The filled rondo trays are packed in plastic (corrugated polypropylene boxes) or cardboard boxes to avoid damage during transport. For the biotechnology applications that are the subject of this technical report, corrugated polypropylene boxes are considered state-of-the-art. The supplier and pharmaceutical company agree to the material, dimensions, and configuration (see **Figure 13.1.4.1-1** for an example).

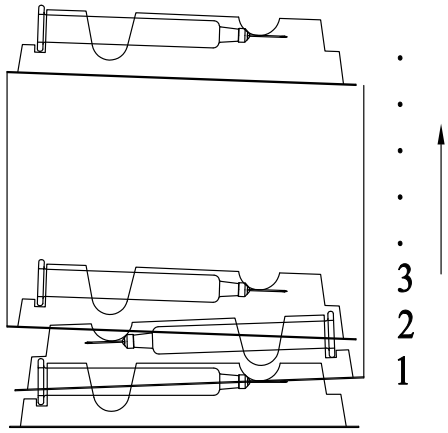

Figure 13.1.4.1-1 Packaging orientation illustration

13.2 Machinability

13.2.1 Filling of Presterilized Syringes (Tub-Nest Configuration)

Introduction of syringe components (syringes and plunger stoppers) into the filling line can be accomplished with presterilized (ready to be filled) syringes. There are different ways to introduce syringes into the clean room, and the level of automation varies from line-to-line and site-to-site.

13.2.2 Electron Beam Tunnel

After being unwrapped, outside surface decontamination occurs by introducing the naked tubs into an electron beam tunnel. The dose for sterilization is often set at 25 kilogray (kGy), which for most applications is more than needed given the very low bioburden on the outside surface of the tubs. An exhaust system removes the ozone generated by the electron beam. Using this method of outside decontamination requires demonstrating that the tub/nest configuration is compatible with remaining impurities or process contaminants. The tunnel unloads directly into the filling area.

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13.2.3 Hydrogen Peroxide Pass-through Locks

Hydrogen peroxide is a sterilizing agent suitable for externally decontaminating the tubs before introduction into the filling area. However, the hydrogen peroxide must not ingress into the tubs as this could have negative effects on the product.

13.2.4 Low-Pressure Plasma

The low-pressure plasma (LPP) unit is a process that decontaminates the outside surface of the syringe tub after removal of the protective bag and before entering the filling line. However, the process only removes low levels of incidental bioburden that may contact the tub during bag removal and transfer to the LPP unit. Because these operations are accomplished under laminar flow units producing Grade A quality air, any incidental contamination is low. The purpose of the LPP unit is to protect the environment within the filling line, not to sterilize the components.

The decontamination mechanism is UV light produced by generation of the plasma under low pressure. Plasma is a partly ionized gas. Using LPP to generate the UV light allows the plasma to be generated around the object, thereby directly contacting the surface of the object and eliminating any shadow areas. Measurement of cycle parameters and UV light output (by spectrometer) enables automatic confirmation of a satisfactory cycle.

UV light is also a sterilization method for surface decontamination. Neither the UV light nor the plasma can penetrate the object undergoing sterilization. When using the UV light as a decontamination method, the target microbiological reduction is log 4 (as opposed to log 6 for use as a sterilization method) ***(99)***.

13.2.5 Alcohol Wipe Disinfection Followed by Automatic Bag Opening and Transfer

In less automated processes, the bag with the tub is sprayed or wiped with alcohol before introducing it into higher classified areas. The disinfectant is commonly isopropanol, n-propanol, ethanol, or mixtures thereof. For a double-bagged tub, removal of the first bag might occur after being sprayed as it moves from a controlled but unclassified area to a Grade C area. The second bag can then be removed as it moves from a Grade C to a Grade A area. Ideally, a debagging unit should provide the automation needed for this critical introduction. In some recent cases, an automated spray tunnel replaces the manual spray. **Table 13.2.5-1** outlines a possible process for completing these tasks.

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Table 13.2.5-1 Process for Filling Presterilized Syringes

Process step	Description
1. Plunger stopper introduction	The three main plunger stopper insertion methods are the mechanical vent tube, vacuum insertion tub, and vacuum bell methods. Mechanical vent tube method A transfer arm moves the plunger stoppers over the vent tubes. A push rod then pushes and compresses each plunger stopper down to the bottom of the vent tube. The vent tube retracts out of the syringe barrel, and the push rod remains behind. The plunger stopper exits the vent tube and expands. The push rod retracts out of the syringe barrel, and the table moves to a new filled syringe position for the next plunger stopper insertion cycle. Due to differences in geometry and construction materials of the plunger stoppers, dedicated format parts (e.g., bowl or tracks) may be required. Vacuum insertion tub The plunger stopper is moved into a small enclosure, which seals the syringe, and a vacuum is pulled. Then, a push rod pushes the stopper a certain distance into the syringe, and the plunger stopper slides into the final position due to the vacuum. Vacuum bell method This is similar to the vacuum insertion tub, but does not include a push rod, meaning the stopper is pulled into the syringe by vacuum only. This method is mainly used for nested stoppers.
2. Tub delidding	The tub is moved into the fill area, and the Tyvek® seal sheet is peeled off and discarded. To make this operation easier, the rim of the tub might be heated to plasticize the glue.
3. Tub Inlay Sheet removal	The tub is conveyed to a second station, where the Tyvek® inlay sheet is removed.
4. Tub denesting	The tub is conveyed to the denester station, where the nest is lifted out of the tub and positioned on a coordinate table by a mechanical arm. The nest is lightly shaken to ensure that the syringe flange bottoms are in contact with the nest.
5. Syringe filling	The path of syringe filler consists of a holding tank and multiple pumps with filling needles. Different types of pumps are used in the process (e.g., rotary piston pumps, peristaltic pumps, membrane pumps, time-pressure systems). The user is responsible for evaluating and validating a method and process for the operation. The determination of the size of the filling needles is based on the required fill volume and solutions properties. Nitrogen overlay during filling/stoppering might be considered for oxygen-sensitive products. Discharge rate, fill needle movement, and other configurable settings can be adjusted to minimize undesirable filling characteristics (e.g., splashing, dripping). Filling volume accuracy must be well characterized to meet the end-user requirements.
6. Plunger stopper insertion	The three main plunger stopper insertion methods are the mechanical vent tube, vacuum insertion tub, and vacuum bell methods. In the mechanical vent tube method, the coordinate table moves filled syringe barrels under the mechanical vent tubes. The vent tubes insert the plunger stoppers into the syringes at the same time as the vibratory bowl conveys, sorts, and inverts sterile plunger stoppers down linear tracks for insertion. A transfer arm moves the plunger stoppers over the vent tubes. A push rod then pushes and compresses each plunger stopper down to the bottom of the vent tube. The vent tube retracts out of the syringe barrel, and the push rod remains behind. The plunger stopper exits the vent tube and expands. The push rod retracts out of the syringe barrel, and the table moves to a new filled syringe position for the next plunger stopper insertion cycle. Due to differences in geometry and construction materials of the plunger stoppers, dedicated format parts (e.g., bowl or tracks) may be required.
7. Tub re-nesting	Upon completion, the nest is removed from the coordinate table by the mechanical arm and returned to its original tub.

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13.3 Filling of Bulk Syringes

The processing approach for bulk syringes and the approach for presterilized syringes described in the previous section differ in that preparation of bulk syringes occurs at the location where filling occurs. Preparation includes washing, siliconization, and sterilization prior to filling. Although this provides direct process control, it also requires a high level of internal subject matter expertise. **Table 13.3-1** outlines a possible process for completing these tasks.

Table 13.3-1 Process for Filling Bulk Syringes

Process step	Description
1. Washing	Conducted with water to reduce particle levels, bioburden, lower endotoxin levels to acceptable levels, and lower levels of residue, such as tungsten (when required). Typically, washing, siliconization, and fitting of needle shields are performed on one machine, usually in a Grade C area under laminar flow. Consists of one or more cycles of rinsing with purified water (inside and outside of the barrel) followed by a final rinse with WFI. It is good practice to avoid recycling any of these rinses and to use only one pass. After washing, one or more drying steps with sterile filtered, pressurized air removes excess water. Process should result in a reduction of particles as well as endotoxin and bioburden levels because the syringes are subsequently steam sterilized.
2. Siliconization	Discussed in **Section 10.0**. For 1 mL long syringes with a staked needle, the process is usually performed with pure, medical-grade silicone oil of predefined viscosity. State-of-the-art approach is dynamic spraying of oil using a spraying nozzle. This technique allows application of silicone oil with uniform distribution and adequate quantities to ensure functionality of the syringes, without over-application of the oil. Also includes siliconization of the outside of the needle.
3. Fitting of Needle Shields	The final step before sterilization. Needle shields are provided via vibrator-bowl/conveyor-belt systems and mechanically pushed onto the syringes. Care must be taken not to pierce the needle shields because this would compromise the CCI of the system. Machines have units that check 100% of the integrity of the needle shields. The plastic caps on RNSs should not be damaged. Some operations fit the needle shields after separate sterilization of the syringes and needle shields.
4. Sterilization	It is good practice to transfer automatically the washed and siliconized syringes in holding systems to an autoclave under laminar flow in a Grade C environment (e.g., robotic systems). Steam sterilization in the autoclave is performed using a validated process. The validation needs to consider hard-to-reach areas in the C/C and the possibility that needle shields may become dislodged during the process.
5. Filling	Bulk syringes are usually fed into the filling machine in a single-lane mode, in contrast to the filling of presterilized syringes, which remain in the nest during filling. Filling is performed in a Grade A laminar flow environment with a Grade B background. Restricted access barrier systems have improved the classical aseptic process by separating operators from the critical fill areas. The use of isolators in a Grade C area has increased in recent years adding the advantage of fully automated decontamination. Several pump systems are used—often alternatively—on the same machine.
6. Stoppering	Two types of stopper placement technologies are available: placement tube/pin combinations and vacuum stoppering. Technology selection should be on a product-by-product basis. Placement tube/pin combinations allow precise placement but require a certain size of air bubble to avoid wet plunger stoppers. In contrast, vacuum stoppering can be used to fill syringes with a minimal air bubble, which can be very useful for the application of viscous solutions by avoiding the physical pressure-buffering effect of the air bubble.

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13.4 Visual Inspection

The purpose of the visual inspection process is to identify defects in filled syringes and reject those that do not meet established criteria. The visual inspection process is performed by trained technicians prior to assembly in specially designed rooms. The European Pharmacopeia in Section 2.9.20 describes visual inspection and provides details on both the apparatus consisting of a black and white background with a defined light source and the inspection method ***(85)***.

Syringe and plunger stopper inspection occurs during the visual inspection process. The process identifies and classifies defects, such as those described in **Table 13.4-1.** For each container, technicians document defects in a "defect library," which should include pictures and descriptions of each defect. For training purposes, operating procedures should provide documentation of defect classification and the handling of rejects. The classification of the defects must include an acceptance criterion for the number of rejected syringes per defect before further actions are initiated.

Table 13.4-1 Examples of post-filling visual inspection defects

Inspected Area	Examples of Defects	Inspected Area	Examples of Defects	Inspected Area	Examples of Defects
Syringe Barrel	Cracked	Plunger Stopper	Liquid in the ribs	Needle Shield	Bent
	Air line		Tilted		Missing
	High fill		Upside-down		Discolored
	Low fill		Soiled condition		Broken
	No fill		Foreign material		
	Foreign material		Scuffed		
	Damaged finger flange				

After the syringes are filled and closed in the syringe filling line, the syringes are taken to the visual inspection area or stored at the intended storage conditions for future inspection. If inspection occurs later, then technicians must provide adequate time to allow the syringes to reach ambient temperature before inspection begins. Otherwise, water condensation on the outer syringe surface could negatively affect inspection.

The visual inspection process starts once the syringes arrive at the inspection area. For manual inspections, each trained technician chooses a number of syringes and visually inspects each for the defects described in the defect library. Technicians identify syringes without defects as such and allocate accordingly; syringes with major and critical defects are rejected and segregated. Each trained technician documents the number of rejected syringes per defect. The inspection process continues until all syringes in a batch have been inspected.

13.4.1 Semiautomated Visual Inspection

In the semiautomated process, manufacturers may use denesters to remove the syringes from the tubs and automatically place the syringes on a moving track. The syringes move at a certain rate in front of the trained technician, who manually diverts any rejected syringes into a different track.

13.4.2 Automated Visual Inspection

Performing the visual inspection process may use a fully automated inspection machine on which syringes are denested, placed on a moving track, inspected by a vision system, and renested back into tubs or packed in rondo trays for subsequent assembly.

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The design of automated inspection equipment must consider factors that affect product quality. For example, an assessment of the effect of spinning, photo-exposure, and heat generated by equipment must ensure that the biologic product is suitable for automated inspection. Test runs with equipment or theoretical estimates leveraging stability data are recommended as part of the equipment specification and design.

Trays or tubs of filled syringes are fed into the inspection machine; syringes are detrayed/denested and automatically inspected for defects. Defective syringes are ejected from the machine and are commonly collated by defect category. Acceptable pieces are placed into trays or tubs for further processing. As part of the machine qualification, finished syringes undergo manual AQL inspection to ensure that the automated inspection machine is operating within its qualified state.

The automated inspection machine is qualified against a defect library. This library contains descriptions of C/C and product-specific defects and should remain current. Prior to inspecting batches of filled syringes, the inspection machine is challenged with the defect library to ensure that it is operating within the desired parameters. More details are available in PDA *Technical Report No. 43: Identification and Classification of Nonconformities in Molded and Tubular Glass Containers for Pharmaceutical Manufacturing* ***(33)***.

13.5 Final Drug Assembly and Product Secondary Packaging

Secondary packaging and final drug assembly are separate unit operations. Only high-throughput lines may combine filling, inspection, assembly, and final packaging operations. Additionally, final assembly sometimes has specific requirements for individual markets. Smaller packaging lots require frequent changeovers, so staging before final assembly makes sense.

Secondary packaging for syringes encompasses labeling, blistering, and cartoning. For this technical report, needlestick prevention devices and pen assemblies are out of scope.

13.5.1 Labeling, Plunger Rod, and Backstop Placement

In general, component suppliers label either the syringe or the surrounding needle-safety device. Labels might be transparent, semitransparent, or colored. Labels might also include graduations if not already printed on the glass barrel. Also included in the label are areas for variable data printed on line during production, such as the expiration date and batch number. Finally, there are coding areas, such as lumicodes, which check automatically the presence and position of labels.

This unit operation can also include the placement of backstops. Backstops are fitted onto the finger rest of the syringe and enlarge the grip area for injection while securing the plunger from accidentally being pulled out.

Plunger rod insertion for 1 mL long syringes includes the screwing-in motion of the threaded top of the rod into the threaded plunger stopper. The setting of the machine normally includes a torque that determines the final position. An important requirement is minimal movement of the plunger stopper during plunger rod placement. The risk is that the plunger stopper will move back into the nonsterile area of the syringe or the plunger stopper will move to the proximal end of the barrel, creating an overpressure inside that leads to splashing during removal of the needle shield.

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13.5.2 Blistering

Modern blister machines automatically place the syringes into thermoformed blister bodies. The sizes and shapes vary by use and size of the plunger rod and backstop. The manufacturer usually places low cost, frequent-dose products, such as heparin, into multisyringe blisters (i.e., blisters of five syringes), and pack less frequently dosed and higher-cost products into single-syringe blisters. The most common blister material is polyvinylchloride; polyethylene terephthalate and polypropylene are also blister materials.

Manufacturers can seal the blister with plastic foil, but paper or Tyvek® foils are also common. Balancing the need for complete sealing of the blisters and easy opening must be considered and carefully managed. Most blistering operations are performed under clean (i.e., controlled, not classified), but not aseptic conditions. Gassing the sealed blisters with EO, or in rare cases by gamma irradiation, can achieve external surface sterility of the syringe. The design of the blister ensures that the syringe remains undamaged during transport. Ideally, the plunger rod is fixed in the blister to avoid plunger movement during air shipment and ensure that the plunger stopper does not move into the nonsterile areas of the syringe. Sometimes manufacturers pack alcohol swabs into the blisters for the user's convenience. There are other packaging configurations in use as well that pack the syringe or syringes directly into a specially designed carton without blistering (Dividella-type packaging).

13.5.3 Cartoning

Automated lines can also perform cartoning. Component producers must add variable data, such as batch number and expiration date, and bar codes or data matrix codes to the boxes online using inkjet or laser printing.

Based on the blister size, the carton size and number of syringes will vary per box. Additionally, the blister inserts the folded patient information leaflet or booklet into the boxes.

13.5.4 Bundling and Palletizing

After cartoning of the syringe blisters, the cartons might be bundled and palletized into suitable packaging configurations for storage and shipping.

13.5.5 Special Considerations for Serialization

In an effort to stop the introduction of falsified medication into supply chains, an increasing number of authorities mandate individual serialization of package units. Initially, this was performed directly on the individual syringes with printing, etching, or other techniques. Today, manufacturers mark each package unit with a unique identifier. A central worldwide computer network generates and tracks the identifier. The healthcare provider or pharmacist at the point-of-sale can use this code to verify the identity of the product.

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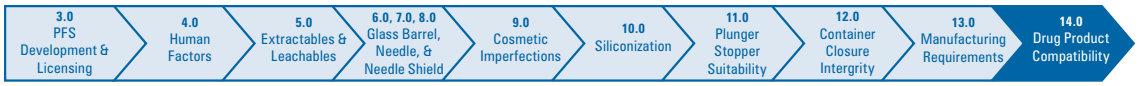
14.0 Drug Product Compatibility



This section focuses on the compatibility of a drug product with the various syringe components discussed in the previous sections, and the possible impact of surface interactions and stresses on drug product stability due to drug product manufacturing, shipping, storage, and handling.

14.1 Risk Assessment

Risk assessments are recommended to determine the sensitivity of the protein to various stress conditions and interfaces. It is important during the risk assessment to understand the molecular characteristics, including any known degradation pathways, to leverage this information during the risk assessment.

A risk assessment typically uses a risk-based approach involving FMEA to determine the criticality of process parameters and their effects on critical quality attributes, quality attributes, or manufacturing attributes. In the area of compatibility, an assessment of all interfaces of the product during manufacturing, and ultimately filling in the PFS, must be performed to determine their impact on the product. Any risk-mitigation activities applied must minimize the risk to the product. The risks are identified, scored, and characterized as being high, medium, or low to determine the mitigation strategy. The risk assessment makes it possible to obtain a complete understanding of the processes that the product will be subjected to, and helps to determine any incompatibilities that may result from the drug product manufacturing process.

The risk assessment should involve members of the research and development unit, and the manufacturing unit to ensure the characteristics of the product and manufacturing process are understood. Once actions that are associated with risk-mitigation activities are complete, the FMEA is reevaluated, rescored, and relevant documents are updated accordingly. **Table 14.1-1** presents only an illustrative example of the output of a risk assessment exercise, including process steps, parameters, risk rationale, criticality based on scoring, and recommended actions. For further information on conducting formal risk assessments, refer to *PDA Technical Report No. 54: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations* ***(36)***.

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Table 14.1-1 Example of Risk Assessment Output

Process Step	Process Parameter	Rationale	Criticality	Action
Filter into Filling System	Filter specification	Noncompatibility of filter with formulation that could affect the product	**High**	Use adequate filter and conduct bacterial retention study over a designated time and range integrity
Filling/ Stoppering	Residual vapor hydrogen peroxide (VHP) concentration	Possibility that residual VHP in stopper bowl, chutes, etc., will affect product; Demonstrated effect on product at specified levels	**High**	Monitor residual levels of VHP after the cycle and the profile of concentration due to air changes
Filter into Filling System	Filter contact time	Restricted time limit from bacterial retention study	**Medium**	Bacterial retention study over a designated time range
Filling	Fill speed	Need to ensure that appropriate filling speed range is defined for the product to prevent product impact	**Medium**	Shear level assessment of product for filling process
Filling	UV light exposure level from LPP	Possible effect on product of exposure to UV light	**Low**	Light level reading during LPP cycle
Stoppering	Stoppering speed	Possible effect on product and process of speed of stoppering	**Low**	Assessment of stoppering speed during operational qualification runs on isolator
Inspection	Syringe specification	Possible effect on product of specification of the product contact parts	**Low**	Confirmation that syringe is not damaged during loading and unloading from inspection equipment

14.2 Key Product and Manufacturing Attributes

14.2.1 Stability Studies

Stability studies that monitor the product in the syringe are critical to understanding the impact of the PFS on product characteristics. The stability studies should include studies in the expected/recommended storage conditions, as well as in stress temperatures to assess thermal stability in support of the intended usage and excursions. For example, if the recommended storage temperature is 2–8°C, a study may evaluate the effects of an accelerated temperature of 25°C or 30°C (depending on the stability of the drug product) and stress storage condition of 30° or 40°C. Care should be taken to choose a stress temperature based on the specific properties (e.g., melting temperature) of the product. Manufacturers should monitor photostability according to ICH guidelines to determine the effects during exposure to light ***(106)*.**

Other studies to consider are discussed in **Section 14.8,** including freeze and thaw cycling to understand the impact on protein stability in case of shipping excursions; CCI and shipping simulations via shear studies; or actual transit trials depending on the clinical phase of the product. In addition, studies to evaluate silicone and tungsten may be useful as discussed in **Section 14.5.** Manufacturers should also evaluate exposure to vapor hydrogen peroxide or other sterilizing agents used prior to filling the syringes. Finally, if the syringe will be used in conjunction with a device, the final drug product presentation can include stability studies as suggested by ICH Q1A(R2), Stability Testing of New Drug

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Substances and Products *(107)*. **Table 14.2.1-1** presents an outline of the stability studies that may be used to assess a typical PFS program.

Table 14.2.1-1 Outline of Typical Stability Studies Used to Assess PFSs

Study	Purpose	Considerations
Product Compatibility	Assess impact of product in siliconized PFS	If siliconization is an issue, process optimization should be considered. Options without silicone, such as polymer syringes, may also be considered if process optimization does not adequately address compatibility issues.
Comparability	Compare stability to previous drug product presentation in vials, if available	Leveraging of drug product data in vials may result in an advantage for the shelf-life setting if comparability of dosage forms can be established.
Long-Term Stability	Assess impact of product when stored at recommended temperature and in accelerated and stress conditions over time	The impact on the product and the ability of the syringe to deliver the product after stability testing must be assessed. Potential problems should be noted (e.g., needle clogging, changes in forces). Long-term stability tests should include the volumes used in clinical studies. May include a bracketing approach if the exact dose is not known at the time of the study ***(58).***
Freezing and Thawing	Assess impact of freezing and thawing on product when it is stored in PFS	After the cycling, assuming that the syringes maintain stability, stability testing of the syringes should be considered to assess any reductions in the expected stability based on the initial freeze/thaw cycling of the product in the PFS.
Shear	Assess impact of shear on product in PFS	Shear may be introduced during the drug product manufacturing process and subsequent distribution steps. Shear impact may be different in benchtop testing and commercial processes.
Silicone Sensitivity	Assess impact of silicone levels on product	A PFS with appropriate control of silicone levels must be chosen.
Tungsten Sensitivity	Assess impact of tungsten on product	Select a PFS with minimal tungsten that is compatible with the drug product
Photostability	Required as per ICH guidelines ***(107)***	Provide secondary packaging and include labelling text. Additional studies might be required in typical light conditions if photostability proves to be detrimental to the product.
Device Stability	Required as per ICH guidelines	This study determines whether an ancillary device will be used and must take into account the stability of the device when it is stored in nonrecommended conditions.

14.2.2 Analysis of Stability

Analysis of stability should include physical, biochemical, and potency/activity measures as outlined in **Table 14.2.2-1**. The tests in the table are examples and include characterization assays, which are not tested for each batch. Characterization can be conducted only on selected batches until sufficient knowledge for this parameter is obtained. Regulatory agencies have requested to consider particulate analysis as an additional assessment of the stability of the drug product ***(82,108).***

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Table 14.2.2-1 Stability analysis for PFS contents

Category	Parameter	Test Methods Examples
Physical	Appearance/Particulates	Light obscuration, microflow imaging
	Viscosity	Cone or plate viscometer/rheometer, Force measurement
Biochemical	Aggregation	Size exclusion chromatography, dynamic light scattering
	Deamidation/Oxidation	Reverse-phase HPLC, capillary isoelectric focusing gel, peptide mapping
	Conformation	Fluorescence, circular dichroism (CD)
	Potency/Activity	Biacore, ELISA, cell-based assays

During all analyses, manufacturers must give special consideration to the presence of the silicone coating in glass PFSs and the potential for the product to interact with the silicone, as well as the movement of the silicone over time or when the product is delivered (109; 110). Beyond having a direct product impact, the presence of the silicone affects the analytical techniques employed. For particulate analysis in particular, silicone must be distinguishable from other product-related particles as mentioned in **Section 10.8** ***(108).***

Depending on the fill volume of the syringe, there may be challenges for the analysis of the volumes required for visual inspection and particulate testing according to USP <788>, Particulate Matter in Injections. Manufacturers and suppliers commonly use small amounts of tungsten in the tips of glass PFSs with staked needles. The presence of tungsten has been shown to be incompatible with some products. This should be a consideration when regarding product quality. Refer to **Section 14.6.2.1** ***(81)*** for further discussion.

14.2.2.1 Force Measurements

The ability to measure the forces required to inject the drug product by means of the PFS is an important component of the stability testing process. This parameter may have an impact on a patient's ability to self-inject, or a healthcare provider's ability to use the product, as well as the reliability of the injection if the delivery is via a device. Monitoring the forces for delivery initially upon release is important. It is equally as important to monitor the forces for delivery over the intended shelf life of the drug product. This is to confirm that the potential physical changes in the drug product or in the silicone distribution in the syringe do not affect the ability of the PFS (or of the PFS in conjunction with a device) to deliver the required dose to the patient. The siliconization of the PFS, the properties of the drug product, and the needle gauge and its variability affect the forces required for delivery of the drug product. All of these issues should be a consideration when evaluating the force measurements for the drug product. Finally, syringeability, which measures the force required for delivery at a given injection rate via a needle of a predetermined gauge and length, will enable the optimal injection conditions to be determined for the drug product, taking into account the product's physical properties, flow rate, injection force, and needle gauge ***(111–114).***

14.2.2.2 Special Considerations: the Effect of Silicone on Analytical Testing, including Particle Analysis

To evaluate the effect of silicone on analytical testing and particle analysis (including light obscuration and other orthogonal analytical techniques), pooling practices are often required to overcome the challenges of low volumes per container. In addition to having a potential impact on drug product stability, the presence of silicone has a significant impact on particulate analysis via light obscuration and other orthogonal techniques used for particulate testing, such as microflow imaging. The amount of silicone oil droplets

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generated is dependent on the sampling method (removal from the barrel or injection via a needle) used to obtain the sample from the PFS for analysis. As a result, the sampling method that is used for the drug product should be evaluated and consistently used to analyze effectively the results of the particulate analysis. In addition, degassing and/or dilution may be needed to assist in providing consistent results. USP <788> and European Pharmacopeia, Section 2.9.19 require a volume of at least 25 mL, which is challenging to obtain when the volume in a PFS is generally 1 mL or less. When considering small-volume light-obscuration methods to overcome the volume requirement, the sampling plan must be properly qualified and validated for the specific drug product. Finally, because of the impact of the silicone, using both light obscuration and orthogonal techniques capable of identifying silicone within the sample matrix during development will assist in the interpretation of the results generated during particulate analysis.

14.3 Compatibility of the Drug Product with the Plunger Stopper and Needle Shield

Drug products may come in contact with rubber material in the plunger stopper and needle shield during stability testing. For some bioproducts, incompatibility between the drug product and the plunger stopper or needle shield can lead to aggregation, increased chemical degradation, and loss of drug product potency.

14.3.1 Impact of Interaction between the Drug Product and Rubber in the Needle Shield

Incompatibility between the drug product and needle shield can lead to increased polymer formation resulting in partial or complete needle clogging. The increased risk of clogging can lead to obstruction of liquid flow, and therefore, to the requirement for higher forces to expel the product. The higher forces required to push liquid out of the syringe can cause great discomfort, especially to patients with dexterity limitations. An increase in force to expel the product could also lead to the patient not receiving the dose. There is also a risk that injecting the patient with higher-order aggregates and polymer can potentially lead to an immune response ***(115).*** Thus, a comprehensive characterization of incompatibilities between the needle shield and drug product is essential.

Experiments conducted to evaluate incompatibility of the needle shield with the drug product need to account for the surface area to volume ratio between the product on the tip of the needle and the contact surface of the needle shield material. During product stability studies in syringes, the product solution may or may not come into contact with the needle shield. Thus, to evaluate the impact of the needle shield on product quality and stability more rigorously, extractables and leachables evaluations, such as those described in **Section 5,** may be useful.

14.3.2 Impact of Interaction between the Plunger Stopper and Drug Product

Section 5 discusses further considerations of extractables and leachables related to the plunger stopper. To evaluate the impact of the plunger stopper on product quality and stability, some or all of the following studies may be performed.

- **Impact of leachables from the plunger stopper on the product —** Plunger stoppers can have metals that are leachable. It is critical to evaluate the impact of metals on drug products in stability tests.
- **Impact of silicone oil on the product —** Plunger stoppers may be coated with silicone oil that may or may not be cross-linked. **Section 14.5** provides the evaluation of interactions between silicone oil and the product.

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- **Plunger stopper discoloration —** Some plunger stoppers may become significantly discolored when they come in contact with the product. This may lead to a loss of the visual aesthetics of the product. An evaluation of the interaction between buffer components, excipients, and surfactants on the plunger stopper is needed. Some plunger stoppers may preferentially adsorb certain excipients or surfactants, and some excipients and buffer salts may also promote greater water adsorption into the plunger stopper and result in greater dimensional distortion. This can potentially lead to plunger stopper discoloration over time and changes in the functionality of the syringe.

14.3.3 Needle Shield Permeability

Drying of the product in the needle may lead to needle clogging, increasing discomfort for the patient trying to inject the drug product through a clogged needle. Some products may have a greater propensity to become dry. For example, high-concentration formulations may become dry or form a gel more quickly than low-concentration formulations because high-concentration formulations are closer to their solubility limit and very little solvent is available for evaporation. In addition, drug product formulations may contain excipients that can precipitate, crystallize, or form a clog if solvent evaporation occurs.

Needle-shield permeability may be a factor (in addition to needle-shield incompatibility) in product clogging in the needle. Moisture loss through the needle shield could lead to evaporation of the solvent from the product on the needle tip leading to drying and clogging of the needle. The impact of relative humidity (5% to 70% relative humidity), and accelerated and stress temperatures (25°C and 40°C) on the rate of needle clogging in filled syringes should be considered when evaluating the overall risk of needle clogging. Product suppliers may perform moisture transmission studies through the needle shield to evaluate and compare needle shields.

Injection instructions should call for removal of the needle shield just prior to injection. Needle-shield removal usually leads to the release of a few drops of drug product due to initial vacuum generation followed by positive pressure in the syringes. In cases in which the drug is not injected immediately by the patient after needle-shield removal, there may be a risk that the drug product will dry out on the needle tip leading to clogging of the needle. This risk is higher in drug products that have a high concentration of proteins because these have a greater propensity to become dry. This phenomenon can lead to greater discomfort for the patient because more force to dislodge the clog (or partial clog) in the needle may be required.

For the above-mentioned reasons, it is necessary to characterize the clogging rate of the product as a function of environmental conditions and its impact for the delivery of appropriate patient instructions and training.

14.4 Needle Passage Shear

Protein therapeutics may experience multiple sources of shear stress during manufacturing processes, such as mixing, filtration, and filling. Manufacturing processes should consider the shear generated by fluid flow through the needle as it could yield sufficient torque to perturb, partially unfold, or even denature protein drug products ***(114,116).***

14.4.1 Experimental Design

Theoretical calculation is necessary to evaluate shear as the product passes through the filling and injection needles unless clinical study data has already demonstrated product efficacy with the chosen product design. The diameter and length of the needle, the flow rate (filling speed and injection speed), and the physical properties of the product (viscosity) can all affect the calculated shear. Therefore, a study

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designed with a design space around the calculated shear to evaluate protein impact is recommended. The design space allows for some flexibility to optimize filling parameters in the future.

Several studies have shown that the loss of protein activity is proportional to the exposure time and shear rate ***(114,117)***. However, other studies have found no evidence that shear significantly destabilizes proteins, but rather, show accelerated destabilization due to the turnover of interfacial interactions ***(100)***.

The experimental design should also isolate the air-liquid interface from the stress due to shear. The stability study should include recommended and accelerated storage to evaluate the critical quality attributes through stability-indicating assays; these attributes include pH, concentration, aggregation, clipping, charge profile, potency, oxidation, and visible and subvisible particle formation.

14.5 Compatibility with Process Intermediates and Additives

A typical evaluation includes the effects of extractables and silicone—individually and in combination. As mentioned in **Section 5**, product suppliers should evaluate silicone oil compatibility for product impact with different levels and distributions of silicone oil in the product contact area. An additional study to consider is testing whether material surfaces (such as the surfaces of the container and particulate matter in the formulation) can interact with the drug product in terms of extractables and leachables.

14.6 Types of Silicone

Commonly used silicone types for siliconization include silicone fluids, silicone emulsion, and silicone dispersion. Silicone fluids are identified in major international and national pharmacopeias. **Table 14.6-1** provides the classification of silicone fluids based on their kinematic viscosities (in cSt or mm^2 per second at 25°C).

Table 14.6-1 Types of Silicone

Pharmacopeia	Classification	Viscosity Range at 25°C (cSt)
USP	Dimethicone	20 – 30,000
European Pharmacopeia	Dimethicone	20 – 1,300
European Pharmacopeia	Silicone oil used as a lubricant	1,000 – 30,000
Japanese Pharmacopeia	General	2 – 5,000

14.6.1 Silicone Compliance with Monographs

To verify the compatibility with healthcare application, **Table 14.6.1-1** shows common characteristics tested as called for in pharmacopeia monographs. The silicone oil supplier typically tests these characteristics to meet compendial requirements for medical use.

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Table 14.6.1-1 Typical Silicone Oils Tests

Tests to Perform
Identification (infrared absorbance)
Viscosity at 2°C
Colorimetric (violet test)
Reaction of silicates
Acidity (potassium hydroxide titration)
Mineral oils
Phenylated compounds
Refractive index (at 20°C)
Heavy metals
Volatile matter or loss on heating

14.6.2 Effects of Extractables on Products (spiking controls)

14.6.2.1 Tungsten

Interactions between tungsten and some therapeutic proteins have been previously documented ***(49)***. Tungsten salts and tungsten oxides—as well as metallic tungsten particles—can remain in the syringe after the formation process. Thus, there is a need to assess the sensitivity of the drug product to tungsten. To determine the effects of (spiked) tungsten pin extracts, manufacturers should evaluate the stability of the drug product. The scope of the characterization can be dependent on the level of sensitivity observed.

14.7 Process-related Compatibility and Stability

The drug product manufacturing process can involve equipment that might affect the stability of the protein. Some of the key process steps for PFS manufacturing to evaluate include surface decontamination of syringe tubs, isolator technology for aseptic filling, automated visual inspection for defects, labeling, and packaging. A risk evaluation during manufacturing through relevant product impact studies is also needed, but is not covered in detail in this technical report.

14.8 Shock, Vibration, and Shipping Study

Agitating samples after filling should provide a preliminary understanding of how shipping and handling may affect the contact surface area at the silicone/air/solution interface and the corresponding factor under evaluation.

For biopharmaceutical products, it is important to demonstrate the stability of the protein product in both static and dynamic conditions. A common method of filling and stoppering a PFS involves placing the plunger stopper into the syringe barrel using an insertion tube or vacuum. This results in a headspace (air or inert gas) between the top of the plunger stopper and the meniscus of the drug solution. Proteins have the potential to aggregate in the presence of air-liquid interfaces. Thus, it is critical to evaluate the stability of the product under dynamic conditions (agitation) that are comparable or worse than the stress applied in a pharmaceutical supply chain.

The effect of agitation on the stability of the product should be evaluated as part of formulation development ***(67)***.

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Following selection of the final formulation composition, the manufacturer or supplier should perform a representative shipping study, either by actual transport or applying shipping conditions as recommended by ASTM, D4169-14, Standard Practice for Performance Testing of Shipping Containers and Systems ***(6)***. Simulation should include filling the drug solution into the PFS with the desired fill volume and plunger stopper position, then shipping the syringes under worst-case representative conditions ***(117)***. Worst-case conditions vary from product-to-product but should include those for fill volume, plunger stopper placement, temperature, excipient levels, pressure, and mode of transportation (ground and air). Incorporate shipping simulation if validation can show that it is representative of the shipping routes and generates stresses higher than stresses generated by real transport ***(6,8,117)***.

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15.0 Appendix I: Electronic Common Technical Document (ECTD) Application Submission Overview

The following table is a description of elements related to prefilled syringes to consider for inclusion in each module of the CTD.

Module	Overview PROPOSED SECTIONS
1	Administrative information Cover letter and forms Letters of authorization Device manufacturer Medical Device Master File (MAF) or 510(k) Labeling Device index—description of where device content can be found
2	Introduction and overall summaries
3	Quality
5	Clinical List of clinical studies (module **Section 5.2**) Comparative bioequivalence study reports (module **Section 5.3.1.2**) Reports of postmarketing experience (module **Section 5.3.6**) References (module **Section 5.4**)

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16.0 Appendix II: Module 3 for Electronic Common Technical Document (ECTD) Submission Overview

The following table is a description of elements related to prefilled syringes to consider for inclusion in module 3 of the CTD.

Section	PROPOSED BLA/NDA
3.2.P.1	Description and Composition of the Drug Product Describe the device constituent/component(s) (e.g., injector presentation or kit components) in a brief descriptive sentence and include in component table.
3.2.P.2.1	Components of the Drug Product No content applicable to devices
3.2.P.2.2	Pharmaceutical Development – Drug Product Describe the device constituent/component(s) (e.g., injector presentation) briefly and generally.
3.2.P.2.3	Manufacturing Process Development Describe the device constituent/component(s) (e.g., injector presentation) briefly and explain how the manufacturing process or assembly was developed. Cross-reference 3.2.P.3.3.
3.2.P.2.4	Container Closure System Provide "suitability" information for the device constituent/component(s). For delivery devices, include a high-level overview of drug-specific data showing that delivery is accurate, 510(k) history, summary of human factors studies of target population, shear studies, component aging, ISO 10993 data, etc. Link to 3.2.R (medical devices). Cross-reference to device master files.
3.2.P.2.5	Microbiological Attributes Provide information regarding testing (e.g., development, stability, and shipping studies) to confirm that CCI is maintained.
3.2.P.2.6	Compatibility Describe the compatibility of the device with the drug product if there is contact, and provide summaries of compatibility studies if conducted. Include a human factor engineering summary report with a cross-reference to the studies in Section 3.2.P.2.
3.2.P.3.1	Manufacturers Provide names and addresses of the sites performing assembly, kitting, or packaging operations.
3.2.P.3.3	Description of Manufacturing Process and Process Controls Include a description of the final combination product assembly steps and testing, but not detailed component manufacturing (parts molding or machining). Include hold times, process flow diagrams, assembly or packaging equipment, etc.
3.2.P.3.4	Control of Critical Steps and Intermediates For integral combination products, define intermediate checks, in-process controls, and critical process parameters of the manufacturing or assembly process. The content is dependent on product complexity. This section is not applicable to kit components.
3.2.P.3.5	Manufacturing Process Validation and/or Evaluation Provide evidence that installation qualification, operational qualification, production qualification (IQ/OQ/PQ) and process validations were completed (e.g., device release test data on multiple representative clinical/stability/commercial lots).

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Section	PROPOSED BLA/NDA
3.2.P.5.1	Specifications Provide a table of release specifications for the combination product with limited, key device quality inspection or performance test specifications (ranges). Do not include device component design specifications.
3.2.P.5.2	Analytical Procedures Provide an adequate summary of the release test method/protocol for each reported device component specification.
3.2.P.5.3	Validation of Analytical Procedures If noncompendial, provide summary validation information (e.g., gauge repeatability and reproducibility [R&R]) with data on measured parameters or cite standards. Descriptive device inspections may not require validation.
3.2.P.5.4	Batch Analysis Provide release test results for device characteristics for clinical and/or process validation lots as part of overall combination product batch data.
3.2.P.5.6	Justification of Specifications Provide overview of device design inputs that justify specifications (e.g., force to actuate). Supporting design verification data may be necessary.
3.2.P.7	Container Closure System Provide a high-level description of device and device components. Link to 3.2.R. Cross-reference to device master file.
3.2.P.8.1	Stability Summary and Conclusion Provide a summary of the device constituent aging test program and combination product functional stability test program, stability-indicating parameters, and proposed shelf life for the combination product. Present and discuss stability data.
3.2.P.8.2	Post-approval Stability Protocol and Commitment Provide commitments (e.g., annual stability lots) and protocol summaries.
3.2.P.8.3	Stability Data Provide functional stability testing data.
3.2.A.1	Facilities and Equipment Describe the combination product (final assembly/kitting) facility, flow diagrams, principal equipment, other products, and cross-contamination procedures.
3.2.R	Regional Information (medical devices) United States: 510(k)-like document for review by the Center for Devices and Radiological Health of FDA; European Union: Technical file-like document for review by notified body or competent authority

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17.0 APPENDIX III: Tests, Standards, and Guidance Documents for PFS Combination Products

The following table is a guide to the relevant tests, standards and guidance documents for PFS combination products by individual component and the assembled system.

Category	Descriptions of Tests, Standards, and Guidance Documents		Syringe Component				
			Syringe	Plunger stopper	Needle	Needle Shield	Assembled
Protection	The container closure system provides the dosage form with adequate protection from factors that can cause degradation in the quality of the drug product over its shelf life.						
Sterility	Microbial limits and bioburdens/endotoxins		•	•	•	•	
	21 CFR 610.12 USP <71> Sterility Tests	EU Annex 1 Pharmacopeia European 2.6.1					•
Sterilization	FDA Guidance for Industry - Submission Documentation for Sterilization Process Validation in Applications for Human and Veterinary Drug Products 1994 USP <1211> Sterilization and Sterility of Compendial Articles	ISO 11135-1 (2007) ISO 11137-1 (2006) ISO 11137-2 (2006, revised) ISO 11137-3 (2010, revised) ISO 11138 (2006)	•	•	•	•	•
Sterility	ISO 11737-1 (2006) ISO 17665-1 (2006)	ISO 14937 (2009) ISO 10993-7 (2008)	•	•	•	•	•
Container Closure Integrity	USP <1207> Sterile Product Packaging—Integrity Evaluation FDA Guidance for Industry, Container and Closure System Integrity Testing *in Lieu* of Sterility Testing as a Component of the Stability Protocol for Sterile Products FDA Guidance for Industry, Container Closure Systems for Packaging Human Drugs and Biologic ASTM F2338 ICH Q5C EMA, Note for Guidance on Pharmaceutical Development						•

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Category	Descriptions of Tests, Standards, and Guidance Documents			Syringe Component				
				Syringe	Plunger stopper	Needle	Needle Shield	Assembled
Safety	Packaging components should be constructed of materials that will not leach harmful or undesirable amounts of substance.							
Glass	Dimethicone NF USP <660> Containers—Glass	Ph.Eur. 2.9.17 Ph.Eur. 3.1.8 Ph.Eur. 3.2.1	ISO 8106 (2004) (118) ISO 11040-4 (2007)	•				
Elastomer (plunger stopper and needle shield)	Dimethicone NF USP <381> Elastomeric Closures for Injection USP <661> Containers—Plastics	ISO 8871-1 (2003) ISO 8871-2 (2003) ISO 8871-4 (2006) ISO 8871-5 (2005) ISO 11040-5 (2012)	Ph.Eur. 3.1.8 Ph.Eur. 3.1.9 Ph.Eur. 3.2.9		•		•	
Needle	ISO 23908 (2011) ISO 594-2	ISO 7864 ISO 9626 (1991)				•		
	AISI 304 DIN 13097 ISO 11608-2 (2012) U.S. Congress, Needlestick Safety and Prevention Act	FDA, Medical Devices with Sharps Injury Prevention Features EU, Framework Agreement on Prevention from Sharp Injuries in the Hospital and Health-care Sector				•	•	•
General Safety	USP <1031> The Biocompatibility of Materials Used in Drug Containers, Medical Devices and Implants	USP <87> Biological Reactivity Tests, In Vitro USP <88> Biological Reactivity Tests, In Vivo ISO 10993						•
Compatibility	Packaging components that are compatible with the drug product do not interact sufficiently to cause unacceptable changes in the quality of either the dosage form or the packaging component.							
Extractables and Leachables	USP <1663> Extractables Associated with Pharmaceutical Packaging Systems (in development) USP <1664> Leachables Threshold and Best Practices (in development) (ref to PF)			•	•	•	•	•

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Category	Descriptions of Tests, Standards, and Guidance Documents				Syringe Component				
					Syringe	Plunger stopper	Needle	Needle Shield	Assembled
Performance	The container closure system must be able to function in the manner for which it was designed. Drug delivery refers to ability of the packaging system to deliver the dosage form in the amount stated in the label claims.								
Performance as Combination Product	USP <1> Injections			ISO 7864					•
	Ph.Eur. 2.9.17. Test for Extractable Volume of Parenteral Preparations			ISO 9626					
	FDA Guidance documents: Medical Device Use-Safety - incorporating HFE into Risk Management Draft Guidance for Industry and Food and Drug Administration Staff—Applying Human Factors and Usability Engineering to Optimize Medical Device Design Technical Considerations for Pen, Jet, and Related Injectors Intended for Use with Drugs and Biological Products (draft guidance) Glass Syringes for Delivering Drug and Biologic Products Submissions for Post-approval Modifications to a Combination Product Approved Under a BLA, NDA, or PMA **ISO Standard:** ISO 7886-1:1993 Sterile hypodermic syringes for single use — Part 1: Syringes for manual use, Annex C - Method for determination of dead space								•
General Quality Systems	21 CFR 210	21 CFR 610	ICH Q8	ISO 15378 (2006)					•
	21 CFR 211	21 CFR 820	ICH Q9	EN ISO 13485 (2003)					
			ICH Q10	EN ISO 14971 (2007)					
Packaging/ Shipping	ISO 11607	ASTM F1929	ASTM D4728	ASTM D6653					•
	ISO 14644-1	ASTM F1980	ASTM D5276	ISTA 3 E					
	ASTM F88/F88M-09	ASTM D4169	ASTM D5487						

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18.0 Appendix IV: Compound Classification with the Associated Threshold of Toxicological Concern (TTC)

The following table reflects the compound classifications used in the threshold evaluation of extractables and leachables for a PFS system.

Compound Class PQRI - PODP	Compound Class Details	Suggested TTC
Class I	Cramer Classification – Class I "***Low Risk***" Substances of simple chemical structure with known metabolic pathways and innocuous end products which suggest a low order of oral toxicity	150 µg/day
Class II	Cramer Classification – Class II "***Medium Risk***" Substances that are intermediate in toxicity	45 µg/day
Class III	Cramer Classification – Class III "***High Risk***" Substances with chemical structures that permit no strong initial impression of safety and may even suggest a significant toxicity (Given that no Toxicity alerts are generated for the following toxicological end points: sensitization, irritation, genotoxicity, mutagenicity, carcinogenicity, teratogenicity)	7.5 µg/day
Class IV	Sensitizer – irritant – teratogen: compounds suspected of endocrine disruptor activity	5 µg/day
Class V	Genotoxin, mutagen, carcinogen + Substances of very high concern (e.g., DEHP, bisphenol A)	1.5 µg/day

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19.0 References

1. *The Design and Construction of Prefilled Syringes.* **Eakins, M.** 2006, Am. Pharm. Rev. , Vol. 10, pp. 47-51.
2. *Technical Considerations in the Development of Prefilled Syringes for Protein Products.* **Overcashier, D., et al.** 2006, Am. Pharm. Rev., Vol. 9, pp. 77-83.
3. *Best Practices for Extractables and Leachables in Orally Inhaled and Nasal Drug Products: An Overview of the PQRI Recommendations.* **Norwood, D., et al.** 2008, Pharm. Res., Vol. 25(4), pp. 727-739.
4. *ISO 23908:2011. Sharps Injury Protection — Requirements and Test Methods — Sharps Protection Features for Single-Use Hypodermic Needles, Introducers for Catheters and Needles Used for Blood Sampling.* **International Organization for Standardization.** 2011. www.iso.org (accessed May 24, 2013).
5. *Guidance for Industry: Container Closure Systems for Packaging Human Drugs and Biologics.* **U.S. Food and Drug Administration.** 1999. http://www.fda.gov/downloads/Drugs/Guidances/ucm070551.pdf (accessed May 16, 2013).
6. *ASTM D4169-14. Standard Practice for Performance Testing of Shipping Containers and Systems.* **ASTM International.** 2014. www.astm.org (accessed September 14, 2014).
7. *ISTA 3 Series Protocols: General Simulation Performance Tests.* **International Safe Transit Association.** 2008. http://www.ista.org/pages/procedures/ista-protocols.php#3series (accessed September 14, 2014).
8. *ISO/CD 13355:2001 Packaging: Complete, Filled Transport Packages and Unit Loads -- Vertical Random Vibration Test.* **International Organization for Standardization.** 2001. www.iso.org (accessed May 24, 2013).
9. *U.S. Code of Federal Regulations Title 21 Part 3.4: Designated Agency Component.* U.S. Food and Drug Administration. 2014. http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/CFRSearch.cfm?CFRPart=3&showFR=1&subpartNode=21:1.0.1.1.3.1 (accessed March 2, 2015).
10. *U.S. Code of Federal Regulations Title 21 Part 3.2: Product Jurisdiction – Definitions.* **U.S. Food and Drug Administration.** 2014. http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/CFRSearch.cfm?fr=3.2 (accessed December 11, 2014).
11. *U.S. Code of Federal Regulations Title 21 Part 820.30: Design Controls.* **U.S. Food and Drug Administration.** 2014. http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/cfrsearch.cfm?fr=820.30 (accessed May 16, 2013).
12. *U.S. Code of Federal Regulations Title 21 Part 211: Current Good Manufacturing Practice for Finished Pharmaceuticals.* **U.S. Food and Drug Administration.** 2014. http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/CFRSearch.cfm?CFRPart=211 (accessed March 2, 2015).
13. *U.S. Code of Federal Regulations Title 21 Part 610: Biological Products Standards.* **U.S. Food and Drug Administration.** 2012. http://www.gpo.gov/fdsys/pkg/CFR-2012-title21-vol7/pdf/CFR-2012-title21-vol7-part610.pdf (accessed March 2, 2015).
14. *U.S. Code of Federal Regulations Title 21 Part 4: Current Good Manufacturing Practice for Combination Products.* **U.S. Food and Drug Administration.** 2013. http://www.gpo.gov/fdsys/pkg/CFR-2013-title21-vol1/pdf/CFR-2013-title21-vol1-part4.pdf (accessed December 11, 2014).
15. *Eudralex 2001/83/ED, Directive 2001/83/Ed of the European Parliament Relating to Medicinal Products for Human Use.* **European Commission.** 2012. http://ec.europa.eu/health/files/eudralex/vol-1/dir_2001_83_consol_2012/dir_2001_83_cons_2012_en.pdf (accessed March 2, 2015).
16. *EUC, C., Council Directive 93/42/EEC Concerning Medical Devices.* **The Council of European Communities.** 2007. http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CONSLEG:1993L0042:20071011:en:PDF (accessed March 2, 2015).
17. *Draft Guidance: Current Good Manufacturing Practice for Combination Products.* **U.S. Food and Drug Administration.** 2009. http://www.fda.gov/RegulatoryInformation/Guidances/ucm126198.htm (accessed December 11, 2014).
18. *ISO 13485:2003. Medical Devices--Quality Management Systems--Requirements for Regulatory Purposes.* **International Organization for Standardization.** 2003. www.iso.org (accessed May 16, 2013).
19. *ISO 14971:2007. Medical Devices --Application of Risk Management to Medical Devices.* **International Organization for Standardization.** 2007. www.iso.org (accessed May 28, 2013).
20. *Guideline on Active Substance Master File Procedure.* **European Medicines Agency.** 2011, CHMP/QWP/227/02 Rev 3/Corr. http://www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2012/07/WC500129994.pdf.
21. *General Chapter <381> Elastomeric Closures for Injections. USP 381/NF 376.* **U.S. Pharmacopeia.** 2009. www.usp.org.
22. *General Chapter <87> Biological Reactivity Tests, in Vitro. USP31/NF26.* **U.S. Pharmacopeia.** 2011. www.usp.org.
23. *General Chapter <88> Biological Reactivity Tests in Vivo. USP 29/NF 24.* **U.S. Pharmacopeia.** 1988. www.usp.org.
24. *CDC guideline for preventing transmission of infectious agents in healthcare settings.* **Centers for Disease Control and Prevention.** 2007. http://www.cdc.gov/hicpac/pdf/isolation/Isolation2007.pdf (accessed March 2, 2015) .

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25. *U.S. Code of Federal Regulations Title 29 Part 1910.1030: OSHA Blood-Borne Pathogen Standard.* **U.S. Department of Labor.** 2012. https://www.osha.gov/pls/oshaweb/owadisp.show_document?p_id=10051&p_table=STANDARDS (accessed March 2, 2015).

26. *U.S. Code of Federal Regulations Title 21 Part 3.2(e): Definitions: Combination Products.* **U.S. Food and Drug Administration.** 2014. http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfCFR/CFRSearch.cfm?fr=3.2 (accessed September 14, 2014).

27. *Draft Guidance for Industry: Appropriate Use of Voluntary Consensus Standards in Premarket Submissions for Medical Devices.* **U.S. Food and Drug Administration.** 2014. http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfstandards/search.cfm (accessed September 14, 2014).

28. *Technical Report No. 48: Moist Heat Sterilizer Systems: Design, Commissioning, Operation, Qualification and Maintenance.* **Parenteral Drug Association.** 2010. www.pda.org/bookstore (accessed May 17, 2013).

29. *Technical Report No. 1 (Revised 2007): Validation of Moist Heat Sterilization Processes: Cycle Design, Development, Qualification and Ongoing Control.* **Parenteral Drug Association.** 2007. www.pda.org/bookstore (accessed May 17, 2013).

30. *Technical Report No. 3 (Revised 2013): Validation of Dry Heat Processed Used for Sterilization and Depyrogenation.* **Parenteral Drug Association.** 2013. www.pda.org/bookstore (accessed May 17, 2013).

31. *Technical Report No. 11: Sterilization of Parenterals by Gamma Irradiation.* **Parenteral Drug Association.** s.l. : 1988. www.pda.org/bookstore (accessed May 17, 2013).

32. *Technical Report No. 27: Pharmaceutical Package Integrity.* **Parenteral Drug Association.** 1998. www.pda.org/bookstore (accessed May 16, 2013).

33. *Technical Report No. 43: Identification of Nonconformities in Molded and Tubular Glass Containers for Pharmaceutical Manufacturing (Revised 2013).* **Parenteral Drug Association.** 2013. www.pda.org/bookstore (accessed May 16, 2013).

34. *Guidance for Industry: Medical Device Use-Safety: Incorporating Human Factors Engineering into Risk Management Human Factors Engineering Process.* **U.S. Food and Drug Administration.** 2000. http://www.fda.gov/medicaldevices/deviceregulationandguidance/guidancedocuments/ucm094460.htm (accessed May 28, 2013).

35. *IEC 62366:2007. Medical Devices - Application of Usability Engineering to Medical Devices.* **International Organization for Standardization.** 2007. www.iso.org (accessed September 11, 2014).

36. *Technical Report No. 54: Implementation of Quality Risk Management For Pharmaceutical and Biotechnology Manufacturing Operations.* **Parenteral Drug Association.** 2012. www.pda.org/bookstore (accessed May 17, 2014).

37. *ISO 14971:2012. Medical Devices --Application of Risk Management to Medical Devices.* **International Organization for Standardization.** 2012. www.iso.org (accessed May 28, 2013).

38. *ANSI/AAMI HE75:2009. Human Factors Engineering: Design of Medical Devices.* **American National Standard.** 2009. http://www.aami.org (accessed May 28, 2013).

39. **Stanton, N., et al.** *Human Factors Methods: A Practical Guide for Engineering and Design, 2ed.* Burlington : Ashgate Publishing Company, 2005. ISBN-13: 978-1409457541.

40. *Human Factors Implications of the new GMP Rule Overall Requirements of the New Quality System Regulation.* **U.S. Food and Drug Administration.** 2011. http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/HumanFactors/ucm119215.htm (accessed March 2, 2015).

41. *Technical Report No. 44: Quality Risk Management for Aseptic Processes.* **Parenteral Drug Association.** 2008. www.pda.org/bookstore (accessed May 17, 2014).

42. **U.S. Food and Drug Administration.** Recognized Consensus Standards. *U.S. Food and Drug Administration.* [Online] [Cited: March 2, 2015.] http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfstandards/search.cfm.

43. *Quality Guideline Q9: Quality Risk Management.* **International Conference on Harmonisation.** 2006. www.ich.org (accessed May 7, 2014).

44. *Medical Devices; Current Good Manufacturing Practice (CGMP) Final Rule; Quality System Regulation.* **U.S. Food and Drug Administration.** 1996. http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/PostmarketRequirements/QualitySystemsRegulations/ucm230127.htm (accessed March 2, 2015).

45. *Beyond the five-user assumption: Benefits of increased sample sizes in usability testing.* **Faulkner, L.** 3, 2003, Behavior Research Methods, Instruments, and Computers, Vol. 35, pp. 379-383.

46. *AAMI TIR49:2013: Design of training and instructional materials for medical devices used in non-clinical environments.* **Association for the Advancement of Medical Instrumentation.** 2013. http://my.aami.org/store/detail.aspx?id=TIR49 (accessed September 15, 2014).

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47. *Guidance on Medical Device Patient Labeling; Final Guidance for Industry and FDA Reviewers.* **U.S. Food and Drug Administration.** 2001. http://www.gmp-compliance.org/eca_guideline_1317.html (accessed May 29, 2013).

48. *IEC 62366:2014 Annex k: Application of usability engineering to medical devices.* **International Electrotechnical Commission.** 2014. www.iec.ch (accessed September 15, 2014).

49. *General Chapter <1660> Evaluation of the Inner Surface Durability of Glass Containers. USP 37/NF 32.* **U.S. Pharmacopeia.** 2012. www.usp.org.

50. *Root Cause Analysis of Tungsten-Induced Protein Aggregation in Pre-Filled Syringes.* **Liu, W., et al.** 2010, PDA J. Pharm. Sci. Technol., Vol. 64, pp. 11-19.

51. *Precipitation of a Monoclonal Antibody by Soluble Tungsten.* **Bee, J., et al.** 2009, J. of Pharm. Sci. DOI 10.1002/jps.21707.

52. *Interactions between Therapeutic Proteins and Acrylic Acid Leachable.* **Liu, D., et al.** 2012, PDA J. of Pharm. Sci. and Technol., Vol. 66, pp. 12-19.

53. *Effects of Surfaces and Leachables on the Stability of Biopharmaceuticals.* **Bee, J., et al.** 10, 2011, J. of Pharm. Sci., Vol. 100, pp. 4158-70.

54. *ISO 10993-7:2008. Biological Evaluation of Medical Devices – Part 7: Ethylene Oxide Sterilization Residuals.* **International Organization for Standardization.** 2008. www.iso.org (accessed June 3, 2013)

55. *ISO 11135-1:2007. Sterilization of Health Care Products -- Ethylene Oxide -- Part 1: Requirements for Development, Validation and Routine Control of a Sterilization Process for Medical Devices.* **International Organization for Standardization.** 2007. www.iso.org (accessed June 3, 2013).

56. *EMA CPMP/QWP 159/01: Limitations to the Use of Ethylene Oxide in the Manufacture of Medicinal Products.* **European Agency for the Evaluation of Medicinal Products.** 2001. http://www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2009/09/WC500002915.pdf (accessed March 2, 2015).

57. *ISO 8871-2:2004. Elastomeric Parts for Parenterals and for Devices for Pharmaceutical Use. Identification and Characterization.* **International Organization for Standardization.** 2004. www.iso.org (accessed June 3, 2013).

58. *Quality Guideline Q2(R1): Validation of Analytical Procedures: Text and Methodology.* **International Conference on Harmonisation.** 2005. www.ich.org.

59. *Guidance for Industry: Immunogenicity Assessment for Therapeutic Protein Products.* **U.S. Food and Drug Administration.** 2014. http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/UCM338856.pdf (accessed September 14, 2014).

60. *Safety Thresholds and Best Practices for Extractables and Leachables in Orally Inhaled and Nasal Drug Products.* **Product Quality and Research Institute.** 2006. http://www.pqri.org/pdfs/LE_Recommendations_to_FDA_09-29-06.pdf (accessed September 15, 2014).

61. *Biopharmaceutical Quality: Factoring Extractables into the Risk Equation.* **Product Quality and Research Institute.** D. Paskiet Presentation, CPhI Pharma Extractables & Leachables Conference, November 29, 2012, Hyderabad, India.

62. *The Product Quality Research Institute (PQRI) leachables and extractables work group initiatives for parenteral and ophthalmic drug product (PODP).* **Paskiet, D., et al.** 2013, PDA J. Pharm. Sci. Technol., Vol. 67, p. 430.

63. **Norwood, D., et al.** The Analytical Evaluation Threshold (AET) and its Relationship to Safety Thresholds. [ed.] D. Ball, D. Norwood and et al. *Leachables and Extractables Handbook.* Hoboken : John Wiley and Sons, 2012, pp. 59-78.

64. *ISO 11040-4:2007. Prefilled Syringes--Part 4: Glass Barrels for Injectables and Ready-to-Use Prefillable Syringes.* **International Organization for Standardization.** 2007. www.iso.org (accessed May 16, 2013).

65. *EP 8.0 Chapter 2.9.17 Test for extractable volume of parenteral preparations.* **European Pharmacopeia 8.0.** 2014. www.edqm.eu (accessed September 15, 2014).

66. *ISO 7864:1993. Sterile Hypodermic Needles for Single Use.* **International Organization for Standardization.** 1993. www.iso.org (accessed June 3, 2013).

67. *ISO 11608-2:2012. Needle-Based Injection Systems for Medical Use -- Requirements and Test Methods -- Part 2: Needles.* **International Organization for Standardization.** 2012. www.iso.org (accessed June 3, 2013).

68. *ISO 9626:1991. Stainless Steel Needle Tubing for the Manufacture of Medical Devices.* **International Organization for Standardization.** 1991. www.iso.org (accessed June 3, 2013).

69. *Variability in syringe components and its impact on functionality of delivery systems.* **Rathore N., et al.** 5, 2011, PDA J Pharm Sci Technol., Vol. 65, pp. 468-80. PubMed PMID: 22293836.

70. *29 Gauge Needles Improve Patient Satisfaction Over 27 Gauge Needles for Daily Glatiramer Acetate Injections.* **Glenski, S and Conner, J.** 2009, Drug Healthcare Patient Safety, Vol. 1, pp. 81-86. Epub 2009 Dec 9. PubMed PMID: 21701612.

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71. *Improving needle-point sharpness in prefillable syringes.* **Vedrine, L and et al.** 4, 2003, Med Device Technol., Vol. 14, pp. 32-35. PubMed PMID: 12774577.

72. *Impact of a modified needle tip geometry on penetration force as well as acceptability, preference, and perceived pain in subjects with diabetes.* **Hirsch, L and et al.** 2, 2012, J Diabetes Sci Technol., Vol. 6, pp. 328-335. PubMed PMID: 22538142.

73. *Improving patient acceptance of insulin therapy by improving needle design.* **Norman, J and Prausnitz, M.** 2, 2012, Diabetes Sci Technol., Vol. 6, pp. 336-338. PMID: 22538143.

74. *Design Control Guidance for Medical Device Manufacturers.* **U.S. Food and Drug Administration.** 1997. http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/GuidanceDocuments/ucm070627.htm (accessed December 11, 2014).

75. Shire, S and et al. *High-Concentration Antibody Formulations, in Formulation and Process Development Strategies for Manufacturing Biopharmaceuticals.* [ed.] **F Jameel and S Hershenson.** Hoboken : John Wiley & Sons, Inc., 2010. Online ISBN: 9780470595886.

76. *Reduced Flow resistance in insulin pen needles designed with thin-wall technology.* **Molin, A and et al.** 2002, Diabetes, Vol. 51 (suppl 1), p. A475.

77. *EP 3.2.9 Rubber Closures for Containers for Aqueous Parenteral Preparations, for Powders and for Freeze-Dried Powders.* **European Pharmacopeia 5.0.** 2005. www.edqm.eu (accessed September 15, 2014).

78. *Dimethicone Monograph. USP 37/NF 32.* **U.S. Pharmacopeia.** 2012. www.usp.org .

79. *EP 8.0 Dimeticone Monograph 5.12 Reference Standards.* **European Pharmacopeia 8.0.** 2014. www.edqm.eu (accessed September 15, 2014).

80. *Silicone Resin Monograph.* **Japanese Pharmacopoeia 16.** 2011. http://www.ffcr.or.jp/zaidan/ffcrhome.nsf/7bd44c20b0dc562649256502001b65e9/916cae3da5a8a11b49256f320018877f/$FILE/D327.pdf (accessed March 2, 2015).

81. *Syringe siliconization process investigation and optimization.* **Chan, E and et al.** 2, 2012, PDA J Pharm Sci Technol., Vol. 66, pp. 136-150.

82. *General Chapter <788> Particulate Matter in Injections. USP 37/NF 32.* **U.S. Pharmacopeia.** 2012. www.usp.org .

83. *General Chapter <789> Particulate Matter in Ophthalmic Solutions. USP 37/NF 32.* **U.S. Pharmacopeia.** 2012. www.usp.org.

84. *EP 2.9.19 Particulate Contamination: Sub-Visible Particles.* **European Pharmacopeia 5.0.** 2005. www.edqm.eu (accessed September 15, 2014).

85. *EP 2.9.20 Particulate Contamination: Visible Particles. European Pharmacopeia 5.0.* **European Pharmacopeia. 2005.** www.edqm.eu (accessed September 15, 2014).

86. *ISO 11137-2:2013. Sterilization of Health Care Products—Radiation—Part 2: Establishing the Sterilization Dose.* **International Organization for Standardization.** 2012. www.iso.org (accessed June 3, 2013).

87. *ISO 11040-5:2012. Prefilled syringes -- Part 5: Plungers for Injectables.* **International Organization for Standardization.** 2012. www.iso.org (accessed September 15, 2014).

88. *ANSI/ASQ Z1.4–2003 (R2013): Sampling Procedures and Tables for Inspection by Attributes.* **American Society for Quality.** 2013. www.asq.org (accessed September 15, 2014).

89. *ISO 2859-1:1999. Sampling Procedures for Inspection by Attributes -- Part 1: Sampling Schemes Indexed by Acceptance Quality Limit (AQL) for Lot-by-Lot Inspection.* **International Organization for Standardization.** 2012. www.iso.org (accessed September 15, 2014).

90. *ISO 3951-1:2013. Sampling Procedures for Inspection by Variables - Part 1: Specification for Single Sampling Plans Indexed by Acceptance Quality Limit (AQL) for Lot-by-Lot Inspection for a Single Quality Characteristic and a Single AQL.* **International Organization for Standardization.** 2012. www.iso.org (accessed September 15, 2014).

91. *ISO 11607-1:2006. Packaging for terminally sterilized medical devices -- Part 1: Requirements for materials, sterile barrier systems and packaging systems.* **International Organization for Standardization.** 2006. http://www.iso.org/ (accessed March 2, 2015).

92. *ASTM F 1980-07. Standard Guide for Accelerated Aging of Sterile Barrier Systems for Medical Devices.* **ASTM International.** 2011. http://www.astm.org (accessed March 2, 2015).

93. *ISO 11040-6:2012. Prefilled syringes - Part 6: Plastic Barrels for Injectables.* **International Organization for Standardization.** 2012. www.iso.org (accessed September 15, 2014).

94. *General Chapter <1207> Sterile Product Packaging--Integrity Evaluation. USP 37/NF 32.* **U.S. Pharmacopeia.** 2012. www.iso.org (accessed September 15, 2014).

95. *Technical Report No. 60 Process Validation: A Lifecycle Approach.* **Parenteral Drug Association.** 2013. www.pda.org/bookstore (accessed September 11, 2013).

96. *Technical Report No. 57 Analytical Method Validation.* **Parenteral Drug Association.** 2012. www.pda.org/bookstore (accessed September 11, 2013).

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97. *Technical Report No. 56: Phase Appropriate Application of cGMPs.* **Parenteral Drug Association.** 2012. www.pda.org/bookstore (accessed May 17, 2013).

98. *EU Guide to Good Manufacture Practice: Annex 1, Manufacture of Sterile Medicinal Products.* **European Commission.** 2009.

99. *Guidance for Industry. Sterile Drug Products Produced By Aseptic Processing – Current Good Manufacturing Practice.* **U.S. Food and Drug Administration.** 2004.

100. *Technical Report No. 39 (Revised 2007): Guidance for Temperature-Controlled Medicinal Products: Maintaining the Quality of Temperature-Sensitive Medicinal Products through the Transportation Environment.* **Parenteral Drug Association.** 2007. www.pda.org/bookstore (accessed September 14, 2014).

101. *Non-destructive vacuum decay method for pre-filled syringe closure integrity testing compared with dye ingress testing and high-voltage leak detection.* **Simonetti, A and Amari, F.** 2015, PDA J Pharm Sci Technol., Vol. 69, pp. 108-122. doi: 10.5731/pdajpst.2015.01004.

102. *ASTM F2338-09. Standard Test Method for Nondestructive Detection of Leaks in Packages by Vacuum Decay Method.* **ASTM International.** 2013. http://www.astm.org (accessed March 2, 2015).

103. *ISO 11040-07: Prefilled syringes -- Part 7: Packaging systems for sterilized subassembled syringes ready for filling.* **International Organization for Standardization.** (Under development). www.iso.org (accessed March 2, 2015).

104. *ISO 17665-1:2006. Sterilization of Health Care Products -- Moist Heat -- Part 1: Requirements for the Development, Validation and Routine Control of a Sterilization Process for Medical Devices.* **International Organization for Standardization. 2006.** www.iso.org (accessed June 3, 2013).

105. *ISO 14937:2009 Sterilization of Health Care Products -- General Requirements for Characterization of a Sterilizing Agent and the Development, Validation and Routine Control of a Sterilization Process for Medical Devices.* **International Organization for Standardization.** 2009. www.iso.org (accessed June 3, 2013).

106. *Quality Guideline Q1B: Stability Testing: Photostability Testing of New Drug Substances and Products.* **International Conference on Harmonisation.** 1996. www.ich.org (accessed September 14, 2014).

107. *Quality Guideline Q1A(R2): Stability Testing of New Drug Substances and Products.* **International Conference on Harmonisation.** 2003. www.ich.org (accessed September 14, 2014).

108. *Silicone-Oil-Based Sub-Visible Particles: Their Detection, Interaction and Regulation in Prefilled Container Closure Systems for Biopharmaceuticals.* **Felsovalyi, F and et al.** 2010, J. Pharm. Sci., Vol. 101, pp. 4569-4583. doi 10.1002/jps.23328.

109. *Rheological and Syringeability Properties of Highly Concentrated Human Polyclonal Immunoglobulin Solutions.* **Burckbuchler, V and et al.** 2010, Eur. J. Pharm. Biopharm., Vol. 76, pp. 351-356. doi 10.1016/j.ejpb.2010.08.002.

110. *Quality Guideline Q5C: Stability Testing of Biotechnological/Biological Products.* **International Conference on Harmonization.** 1996. www.ich.org (accessed May 20, 2013).

111. *Enzyme Inactivation with Shearing.* **Charm, S and Wong, L.** 6, 1970, Biotechnol. Bioeng., Vol. 12, pp. 1103-1109.

112. *Effect of High Shear Rate on Stability of Proteins: Kinetic Study.* **Olivia, A and et al.** 2, 2003, J. Pharm. Biomed. Anal., Vol. 33, pp. 145-155.

113. *Do Protein Molecules Unfold in a Simple Shear Flow?* **Jaspe, J and Hagen, S.** 9, 2006, Biophys. J., Vol. 91, pp. 3415-3424.

114. *Protein Denaturation by Combined Effect of Shear and Air-Liquid Interface.* **Maa, Y and Hsu, C.** 6, 1997, Biotechnol. Bioeng., Vol. 54, pp. 503-512.

115. *Immunogenicity of therapeutic proteins: Influence of aggregation.* **Ratanji, Kirsty and et al.** 2, 2014, J Immunotoxicol., Vol. 11, pp. 99-109.

116. *Determining antibody stability: Creation of solid-liquid interfacial effects within a high shear environment.* **Biddlecombe, J and et al.** 2007, Biotechnol. Prog., Vol. 23, pp. 1218-1222.

117. *ISTA 7 Series Protocols. 7D: Development Test to Evaluate the Effects of External Temperature Exposures of Individual Packaged-Products.* **International Safe Transit Association.** 2014. http://www.ista.org/forms/7Doverview.pdf (accessed September 14, 2014).

118. *ISO 8106:2004. Glass containers -- Determination of capacity by gravimetric method -- Test method.* **International Organization for Standardization.** 2004. www.iso.org (accessed September 15, 2014).

119. *General Chapter <1207> Sterile Product Packaging--Integrity Evaluation. USP 37/NF 32.* **U.S. Pharmacopeia.** 2012. www.iso.org (accessed September 15, 2014).

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